

**Critical Care and Emergency Nursing  
Department**

**Lecture 1**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**Pre-Hospital Emergency Care**  
**Chain of Survival**  
**First Aid**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**



# Objectives

**By the end of this lecture the student will be able to:**

## **Chain of survival**

- Indicate links of chain of survival according to AHA. (American Heart Association)

## **First aid**

- Define first aid.
- Mention purpose of first aid
- Indicate meaning of first Aid .
- Explain general principles of first aid
- Enumerate roles and responsibilities of rescuer
- Discuss the victim assessment



# Outlines

- Adult chain of survival
- Definition of first aid
- Purpose of first aid
- General principles of first aid
- Role & responsibilities of rescuer
- Victim assessment



# Definition of Adult chain of survival

- The chain of survival refers to a series of actions that, properly executed, reduce the mortality associated with cardiac arrest.

# Adult chain of survival



## Out-of-hospital Chain of Survival

رقم الإسعاف الموحد لجمهورية مصر العربية هو 123 يعمل 24 ساعة على مدار الأسبوع لتقديم الخدمات الإسعافية لجميع المواطنين



## The 6 links in the adult out-of-hospital Chain of Survival are:

- **Recognition** of cardiac arrest and **activation** of the emergency response system.
- Early **cardiopulmonary resuscitation (CPR)** with an emphasis on chest compressions.
- Rapid **defibrillation** automated external defibrillator (**AED**).
- **Advanced resuscitation** by Emergency Medical Services (EMS).
- **Post-cardiac arrest care.**
- **Recovery** (including additional treatment, observation, rehabilitation, and psychological support).

# Definition of First Aid

First aid is the immediate, temporary care provided to a person who is injured or suddenly ill before professional medical help arrives.

First aid involves simple, often life-saving techniques, such as wound care or performing CPR, and requires only basic training and minimal equipment.



# Purpose of first aid (The 3 P's)

- **Preserve Life:** To prevent death, for example, by managing airway, breathing, and circulation (ABC).
- **Prevent Worsening:** To stop the injury or illness from becoming more severe.
- **Promote Recovery:** To help the person feel better and support their recovery.

# Meaning of First Aid

- Following the systematic approach
- Identifying the nature of injuries or illness
- Responding quickly and Reassurance
- Safety promoting
- Treatment in a timely manner
- Ask for help
- Intervening victim(s) (follow algorithms or protocols)
- Do not further harm the victim(s)



# Assess the Victim's Condition (DRSABCD):

**D:** danger: Check for any dangers to yourself or others.

**R:** response: Check if the person is responsive or conscious.

**S:** send for help: Call emergency services.

**A:** airway: Check if the airway is clear.

**B:** breathing: Check for normal breathing.

**C:** CPR: If they are not breathing, start cardiopulmonary resuscitation (CPR).

**D:** defibrillation: If an AED (Automated External Defibrillator) is available, use it according to its instructions.

# General principles of first aid

- Keep the victim away from danger sources.
- Apply victim assessment quickly and competently.
- Keeping the victim quite, calm, and warm.
- Sure of your safety.
- Be calm and act wisely.
- Seeking EMS as rapidly as possible.
- Collect all information.



# Roles and responsibilities of rescuer

- Ensure the continuing safety of themselves.
- Assess victims and find out the nature & cause of their injuries.
- Prioritize victims according to their medical needs.
- Move the victim safely, minimize pain and stabilize his/her condition.
- Avoid any drinks for unconscious or semiconscious victim.
- Provide reassurance and guidance to the victims.
- Follow the first aid protocols and workplace guidelines.

# Victim Assessment

## Component of Victim Assessment

- Primary survey.
- Secondary survey



# Primary Assessment

- It is an initial rapid assessment of victim. It should be completed within 30 seconds to 2 minutes.
- The priorities at the scene are Airway, Breathing, Circulation, and Disability (A, B, C, and D).



# Aim of primary assessment

- To assess level of consciousness
- To establish patent **A**irway
- To provide adequate **B**reathing (ventilation and oxygenation).
- To evaluate **C**irculation (HR, pulse, & BP) & to control hemorrhage
- To establish the mechanism of injury (**D**isability)

# Secondary Assessment

- After the victim's condition has been stabilized during the primary assessment.

## Components of secondary assessment:

- **Medical History:** gathering information about what happened and the victim's medical background.
- **Vital Signs:** check of vital signs.
- **Physical Exam:** a head-to-toe examination to find other injuries or problems.

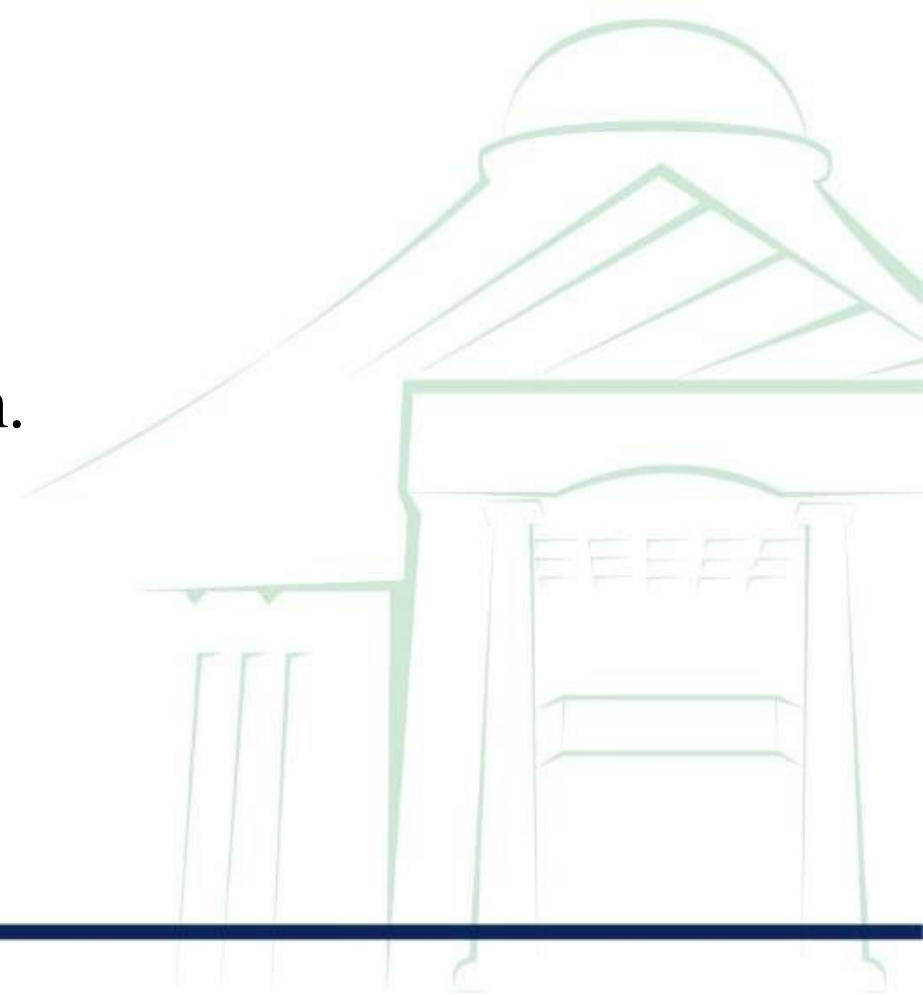
# First aid measures to victim with Fainting



# Objectives

**By the end of this lecture the student will be able to:**

- Define fainting.
- Recognize the main causes of fainting.
- Summarize signs and symptoms of fainting.
- Discuss the first aid intervention for fainting victim.
- Enumerate the preventive measures of fainting.



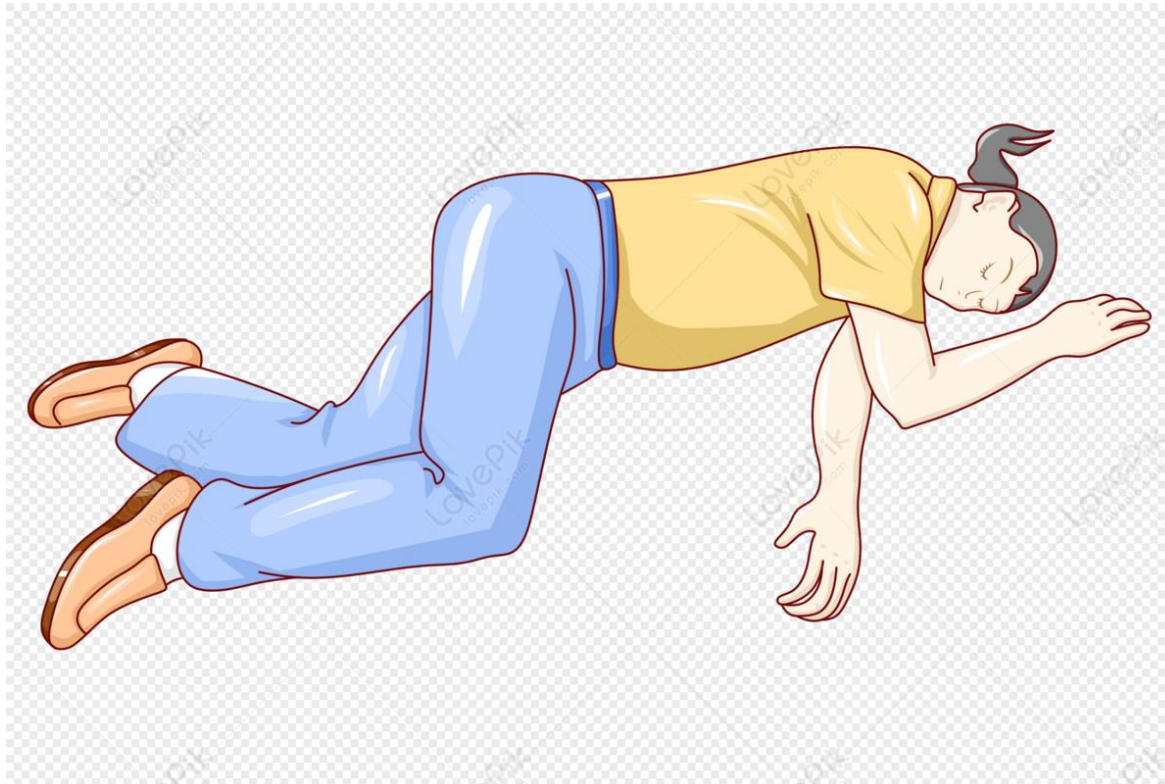
# Outlines

- Definition of fainting.
- Causes of fainting.
- Signs and symptoms of fainting.
- First aid for fainting victim.
- Cautions during providing fainting first aid.



# Definition of Fainting

**Fainting** is a temporary loss of consciousness that can be partial or complete due to decrease blood supply and oxygen to the brain.



# Fainting

- Sudden drop in glucose level
- Anemia
- A side effect of some medicines
- Being in hot, humid weather
- Standing a long time in one position
- Standing up too fast
- Anxiety, stress and emotional upset
- Severe pain
- Low blood pressure

# Causes



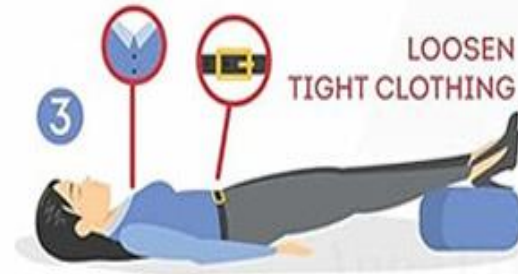
# Signs and symptoms of fainting

- Nausea
- Drowsiness
- Excessive sweating
- Vision changes "double vision".
- Numbness and tingling of the hands and feet
- Rapid heartbeat or palpitations
- losing consciousness



# First Aid for Fainting

## FAINTING: FIRST AID



DO  
NOT



GIVE PERSON  
ANY FOOD OR DRINK



SLAP FACE OR SPLASH WATER  
TO TRY TO RETRIEVE PERSON



LEAVE PERSON ALONE

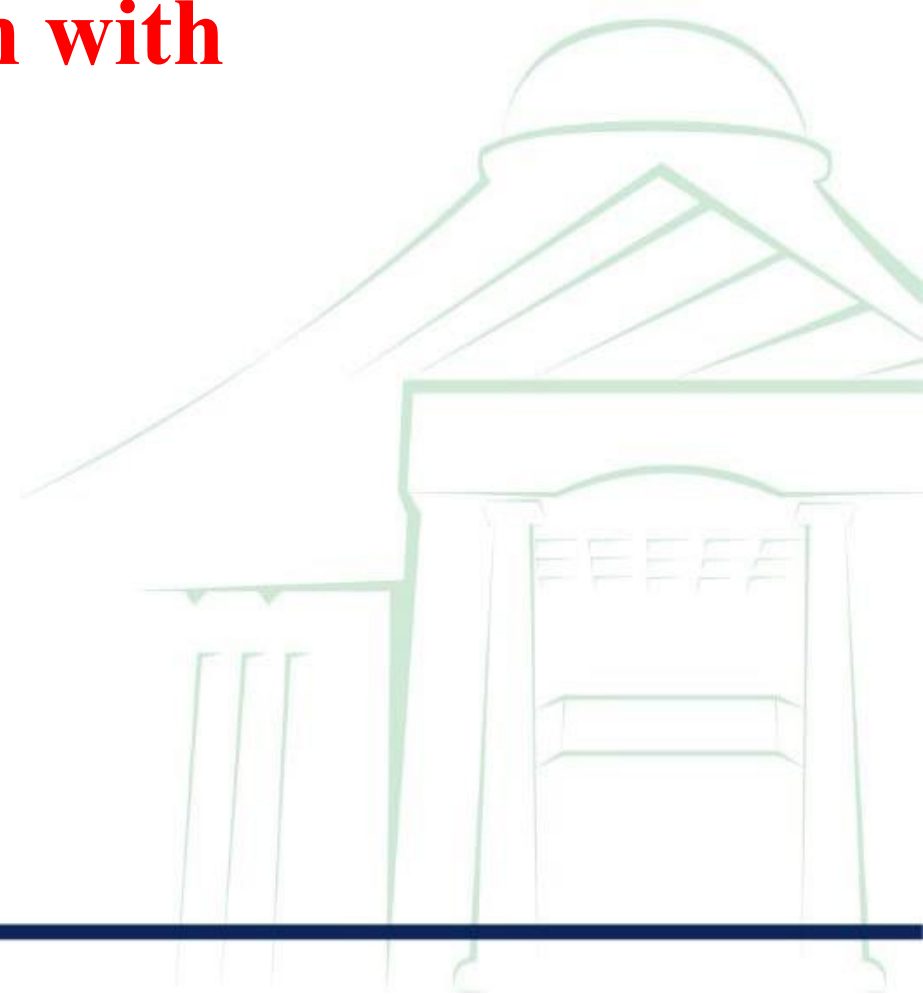
# First Aid for Fainting

**(DO NOT)**



- Splash water over the victim's face
- Give any fluids unless the victim become conscious.
- Give the victim any perfumes or ammonia to smell.
- Hit the victim's face

# First aid measures to victim with Convulsion



# Objectives

**By the end of this lecture the student will be able to:**

- Define convulsion
- Enumerate causes, signs and symptoms of convulsions.
- Provide first aid for victims with convulsions.
- List cautions during providing convulsions first aid.

# Outlines

- Definition of convulsion
- Causes, signs and symptoms of convulsions.
- First aid for victims with convulsions.
- Cautions during providing convulsions first aid



# Definition of convulsion

It is a sudden, abnormal electrical activity in the brain. It causes temporary, involuntary changes in body movement, function, sensation, awareness or behavior.



# Causes of convulsion

- Heatstroke
- Poisoning
- Electric shock
- Hypoglycemia
- High fever in children
- Brain injury, tumor, or stroke
- Alcohol withdrawal, drug abuse/overdose

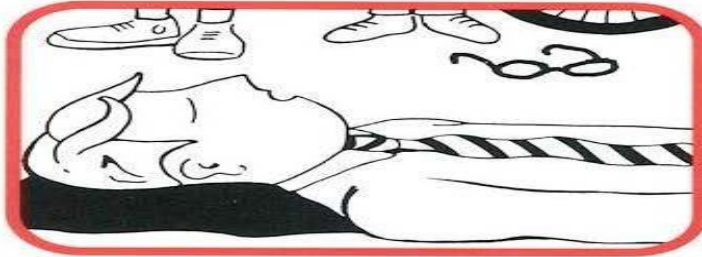


# Signs and symptoms of convulsion

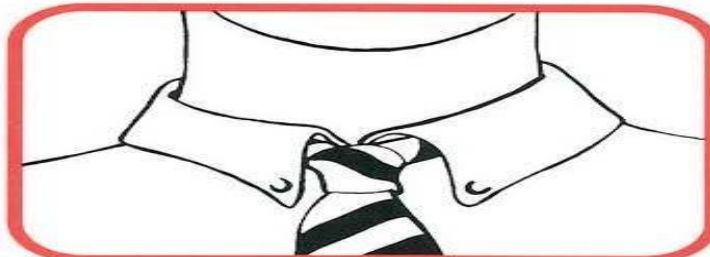
- Uncontrolled muscular contractions
- Change in consciousness
- Rhythmic jerking of the head and limbs
- Loss of bladder or bowel control
- Drowsiness, Confusion
- Presence of tears
- Becoming rigid
- Holding breath
- Upward rolling of the eyes



# First Aid for Convulsion



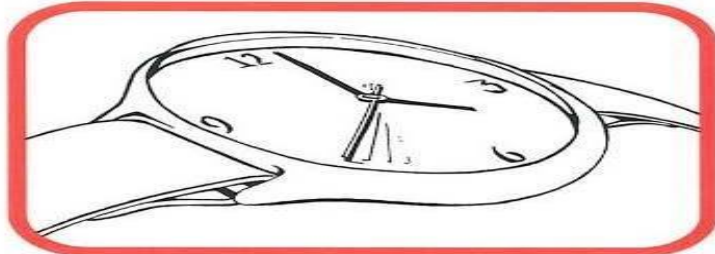
Cushion head,  
remove glasses



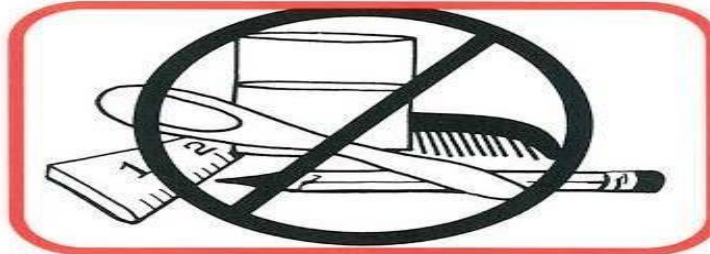
Loosen tight clothing



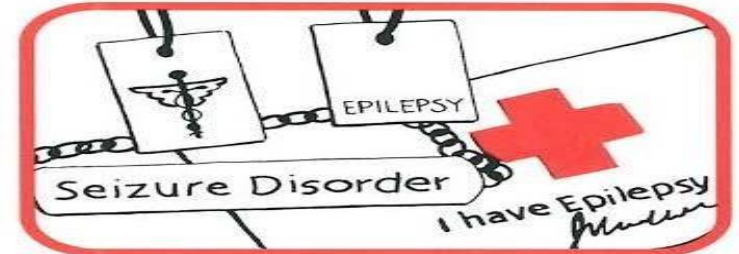
Turn on side



Time the seizure with  
a watch



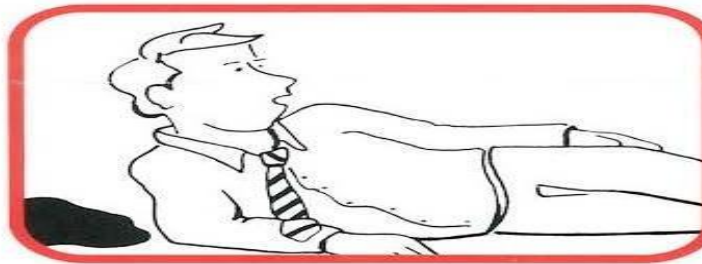
Don't put anything  
in mouth



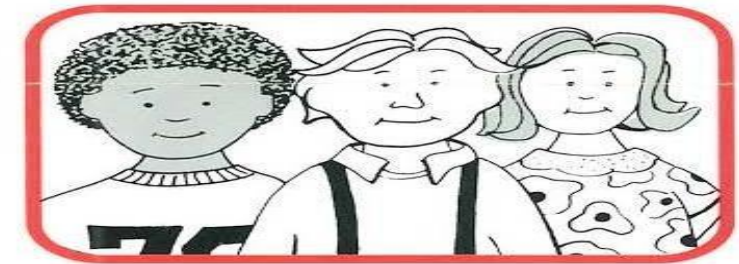
Look for I.D.



Don't hold down



As seizure ends...



...offer help

# First Aid for Convulsion (DO NOT)



- Restrain the victim.
- Put anything between the victim's teeth during the convulsion.
- Give the victim anything to eat or drink.
- Splash or pour any liquid on the victim's face.
- Move the victim to another place.
- Leave a victim until he or she is fully alert and recovered

# Question 1

**Which of the following considered first step in the Chain of Survival?**

- A. Start CPR immediately
- B. Give rescue breaths
- C. Early recognition and call for emergency help**
- D. Use an AED



# Question 2

**2. Which steps comes immediately after calling emergency services in the Chain of Survival?**

A. Place the victim in recovery position

**B. Start high-quality CPR**

C. Give water to the victim

D. Wait for paramedics



## Question 3

**3. What is the purpose of using an AED (Automated External Defibrillator) in the Chain of Survival?**

- A. To measure blood pressure
- B. To restart the heart with an electric shock if needed**
- C. To provide oxygen
- D. To stop external bleeding



**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture 2**



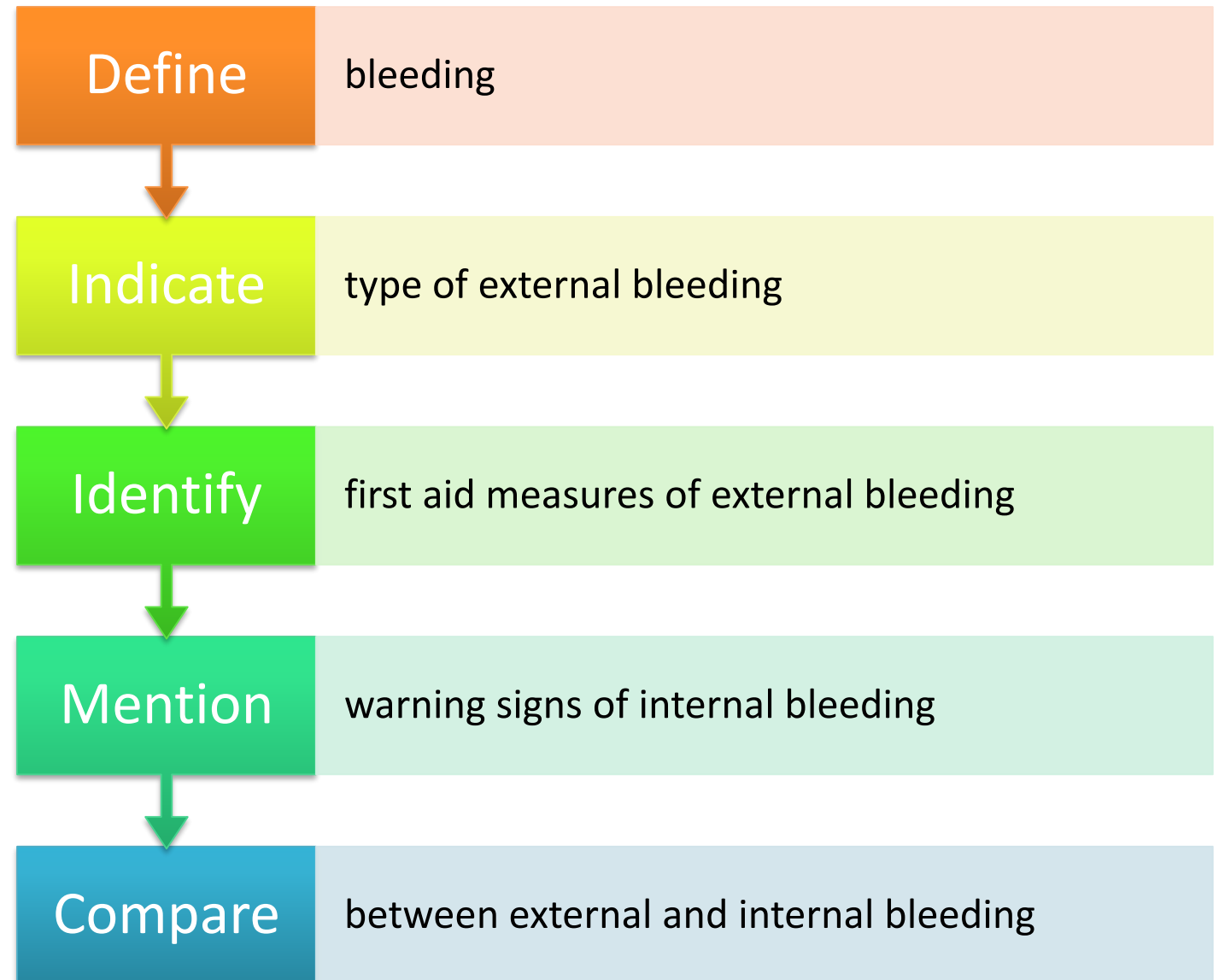
**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**First aid measures to victim with  
Bleeding**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**

## Objectives

By the end of this lecture the student will be able to:



# Outlines

- Definition of bleeding
- Types of bleeding
- First aid measures of external bleeding
- Epitaxies (external bleeding)
- Internal bleeding

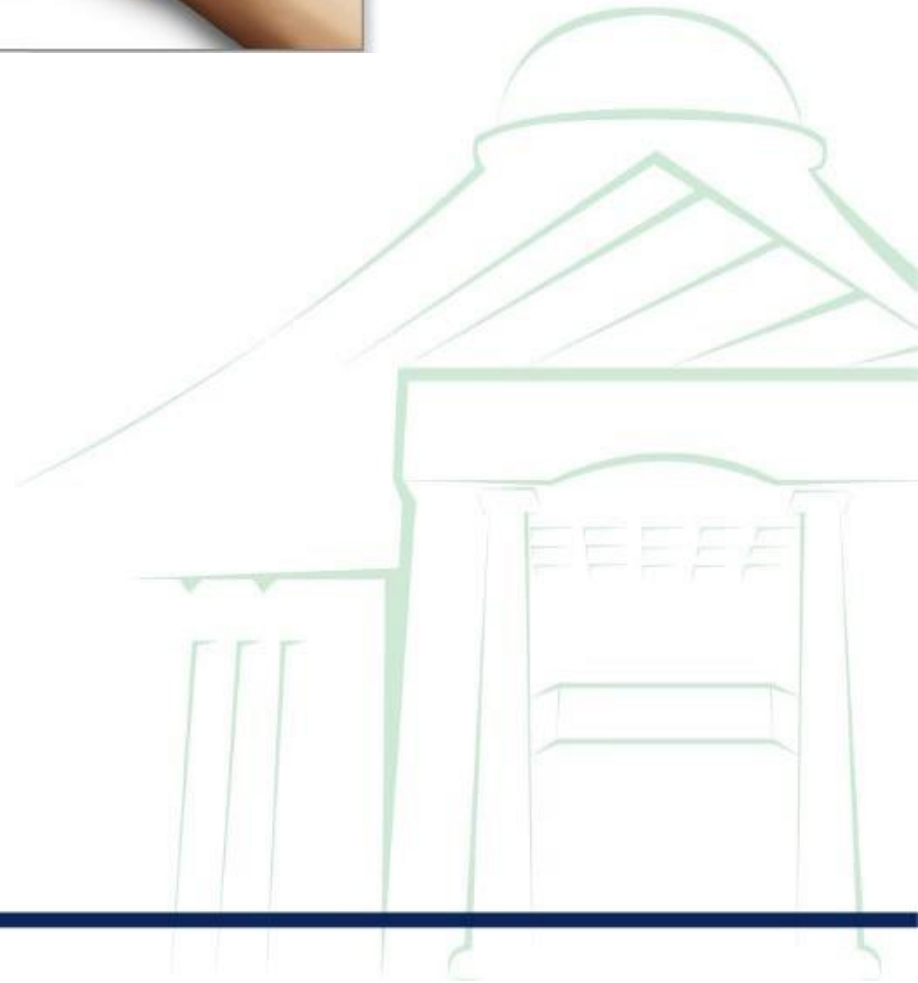


# Definition of Bleeding

- Bleeding is the escape or loss of blood from the circulatory system.
- This can occur externally, visible outside the body, such as from cuts and abrasions, or internally, where blood leaks from blood vessels or organs inside the body.

# Types of Bleeding

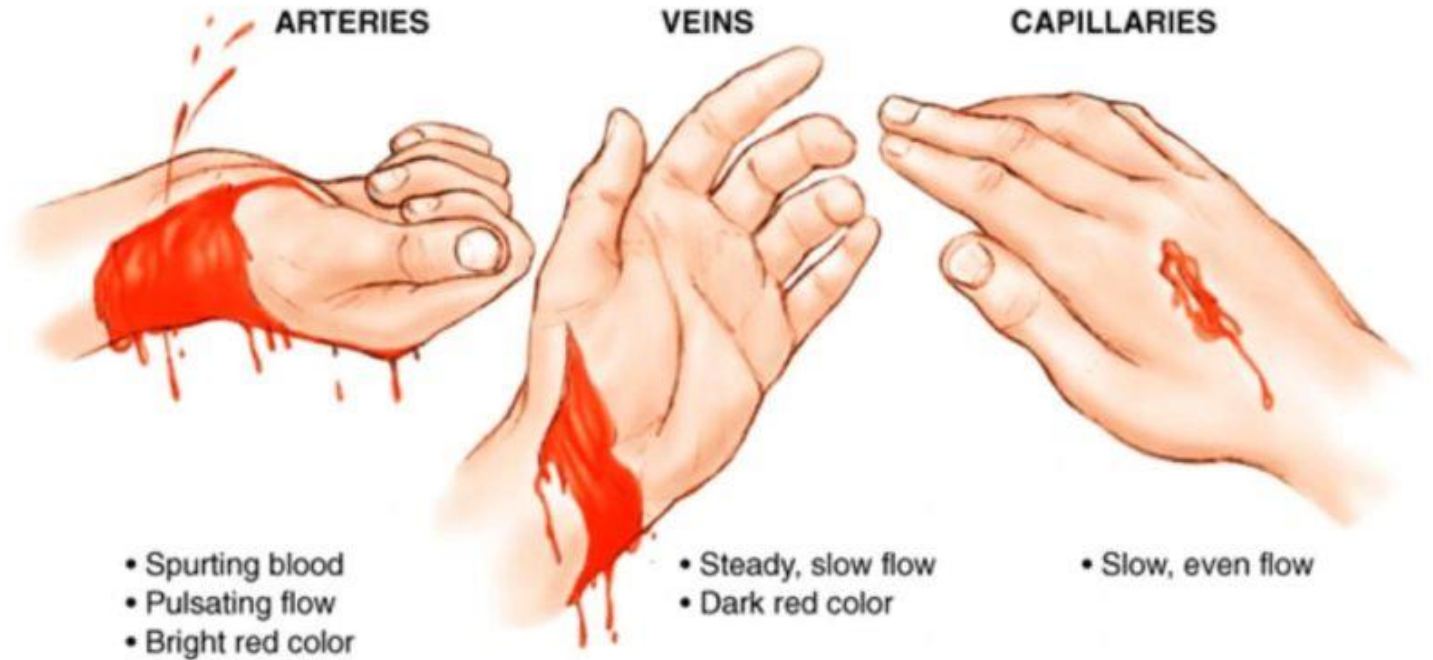
- Internal bleeding
- External bleeding



# Types of External Bleeding

---

1. Arterial bleeding
2. Venous bleeding
3. Capillary bleeding



# Types of External Bleeding

1. Arterial bleeding it is the most serious type of bleeding. characterized by bright red blood which can spurt with each heartbeat.
2. Venous bleeding it is easier to control and characterized by dark red blood that has steady or gushes flow.
3. Capillary bleeding it is the most common types of bleeding and characterized by blood oozes from capillaries.

- **Definition of Hypo perfusion:** Inadequate circulation of blood through an organ.
- Uncontrolled bleeding or significant blood loss leads to Shock (hypo perfusion) and possibly death.

# Sudden loss of blood (Serious conditions )

1liter of blood in an adult



Half liter of blood in an child



100 – 200cc of blood in an infant



**FIRST AID  
GUIDE**

**EXTERNAL  
BLEEDING**

**ALSCO**

**1 Use Protection**



**2 Identify and Examine**

**5 Dress the Wound**



**4 Immobilise the Wound**

**3 Use Direct Pressure**



**6 Clean Up**

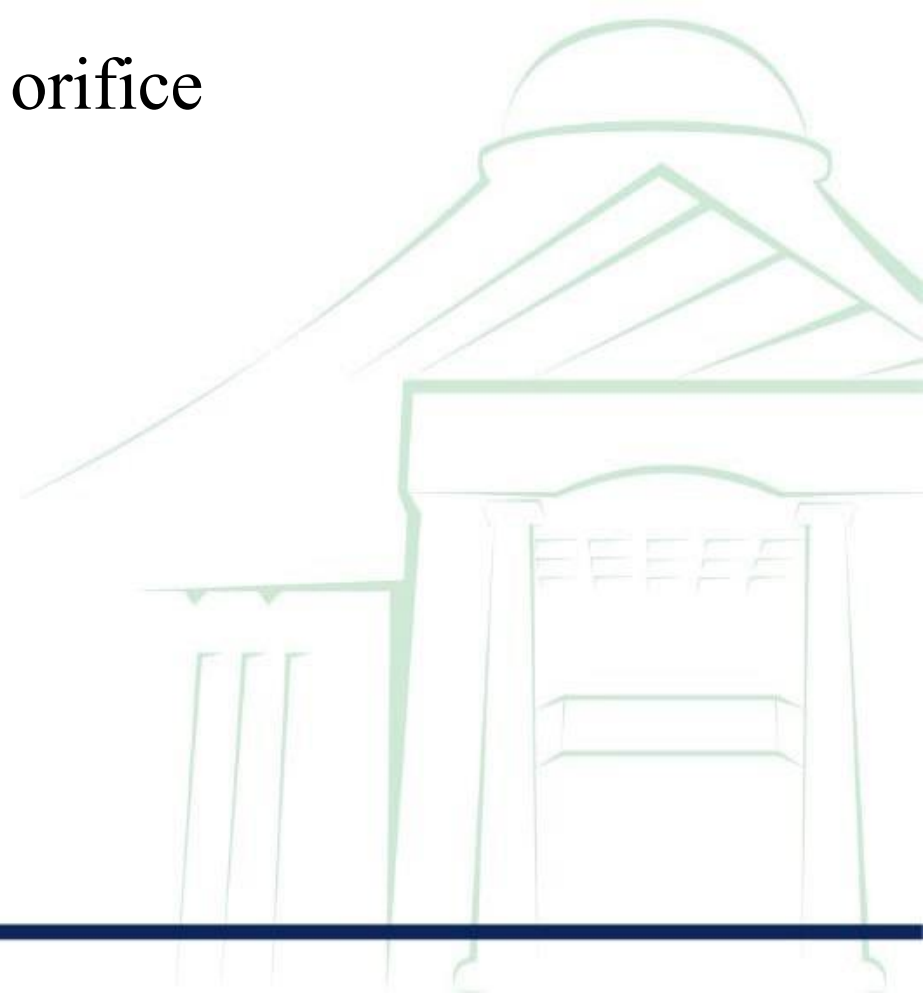


# First aid for external bleeding

1. Put the victim in lying position
2. Protect your self
3. Cut any clothes above the bleeding area
4. Apply direct pressure over bleeding area and make a bandage
5. If the dressing soaked with blood, apply another one
6. Reassess the bleeding area if the bleeding continued, elevate the bleeding area above the level of the heart
7. If the bleeding continued apply indirect pressure in combination with direct pressure and elevation

# Signs and symptoms of internal bleeding

1. Pain, tenderness, deformity, swelling, discoloration
2. Bleeding from the mouth, rectum, vagina, or other orifice
3. Tender, rigid, and/or distended abdomen
4. Vomiting blood
  - Bright red
  - Coffee-ground color and consistency



# Cont.....

6. Blood in stool

- Bright or dark red

7. Capillary refill time  $> 2$  seconds

8. Nausea and vomiting

9. Decreasing blood pressure

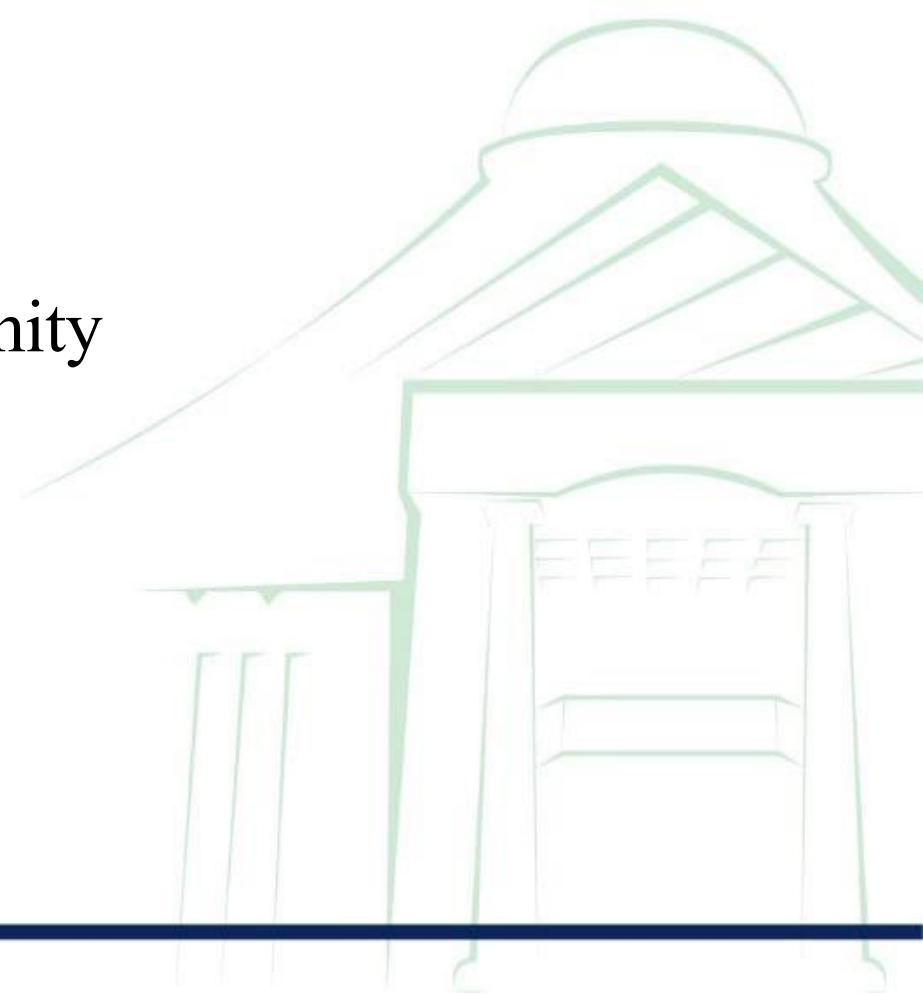
10. Dilated pupils (*late sign*)



# First aid of Internal Bleeding

## Emergency Care Steps

1. Airway care and oxygen
2. Immediate transport to appropriate facility
3. Apply direct pressure if injured area is on an extremity
4. Splint extremity



# Epistaxis



## EPISTAXIS



## Bleeding from the nose

# Causes of Epistaxis

- Facial trauma.
- Cardiovascular disorders as hypertension.
- Blood disorders.
- Nasal tumors.
- A foreign body in the nose.
- Vigorous nose blowing.
- Low humidity.



# First aid for Epistaxis

1. Put the victim in sitting position with the head forward and slightly flexed
2. Remove any tight clothes around the neck
3. Apply pressure by pinching the nostril ( 5- 10 m)
4. Ask the victim to spit out any blood accumulated in his mouth but shouldn't blow the nose.

# Tilt Forward



**Do Not Tilt  
Back**

# Reassessment

- If bleeding didn't stop repeat the pressure for another 10 minutes with ice.
- After stopping the bleeding, advise the victim to rest for few hours.
- If the bleeding isn't stopping, ask for medical aids

# First aid measures to victim with Fracture



# Objectives

By the end of this lecture the student will be able to:

|           |                                |
|-----------|--------------------------------|
| Define    | fracture                       |
| Enumerate | causes of fracture             |
| Indicate  | general fracture symptoms      |
| Mention   | first aid measures of fracture |

# Outlines

- Definition of fracture
- Causes of fractures
- Signs and symptoms of fractures
- First aid measures of fracture



# Definition of Fracture

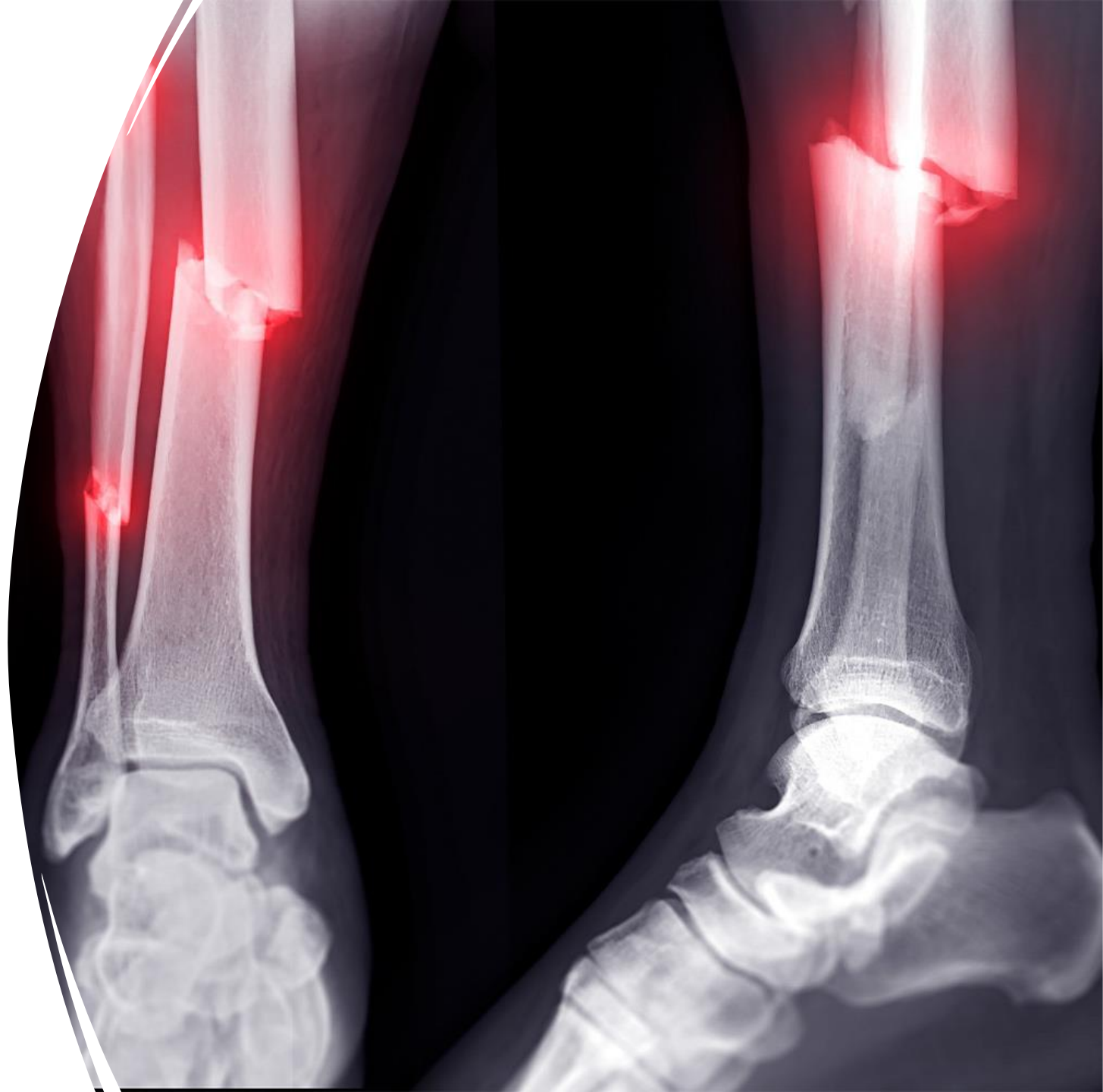
A fracture is any break in the continuity of a bone



# Causes of fracture

---

- Fall from a height
- Motor vehicle accidents
- Direct blow
- Injuries from sports



# Signs and symptoms of fracture

- Severe pain at the site of the fracture
- Deformity
- Tenderness
- Swelling
- Inability to move the injured part.
- The patient may report that he or she heard the sound of the bone breaking.
- Associated wound, and bleeding.

# Emergency action plan for fracture

1. Treat the victim for **shock** if present. If the person feels faint or is breathing in short, rapid breaths, lay the person down with the head slightly lower than the trunk and, if possible, elevate the leg

2. Determine **what happen** and the location of injury.

3. Listen to the **victims complain**.



# First Aid

## for BROKEN BONE

1

**Control the bleeding** by keeping sterile, clothes/ cotton pad with a firm pressure.



2

**Don't try to move the person** only if necessary to avoid further injury.

3

**Apply ice packs** to limit swelling. Never apply ice directly to the fracture. **Wrap in a towel** before applying.



4

**A Splint helps to immobilize the bone** before professional medical attention can take over.

5

**A Sling can help to stabilize an arm** that is broken.



6

If the fracture is severe, call **Blood For Sure Ambulance** 080 67335555 for medical help.



[www.bloodforsure.com](http://www.bloodforsure.com)

# First aid measures for fracture

1. Wash hand and protect yourself
2. Carefully, **remove or cut clothing** covering the injured area without moving the injured part.
3. **Look and feel** the injured part to determine if there is tenderness, swelling or deformities.
4. **Control any external bleeding** using direct pressure, being careful not to press over the site of the fracture or over any wound in the case of an open fracture.

## Cont....

5. Assess circulation, sensory and motor function before and after splint.
6. Support the fractured limb in the most comfortable position.
7. **DO NOT** attempt to straighten a bent or deformed limb as this may result in considerable damage to adjacent soft tissue.
8. Cover any broken skin with a clean or sterile cloth.
9. Splint the limb as it lies.
10. Splint is secured in place by bandages above and below the fracture site.

## Cont....

11. Bandaging should be firm enough to secure the splint but not so tight as to restrict blood flow to and from the limb.
12. Fractures of the upper limbs should be supported in a sling and extra padding may be applied at the fracture site.





# Question 1

**What is the immediate first aid response to control severe external bleeding?**

A. Wash the wound with soap and water

**B. Apply direct pressure to the wound**

C. Elevate the injured area only

D. Put ice directly on the wound



# Question 2

**What is the recommended duration for pinching the soft part of the nose in case of epistaxis ?**

- A. 1–2 minutes
- B. 5 minutes
- C. 10–15 minute**
- D. 30 minutes



## Question 3

**Which of the following is considered a common sign of a fracture?**

- A. Mild itching
- B. Abnormal shape of the limb**
- C. Sneezing
- D. Red eyes





**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture 3**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**Basic Life Support  
Automated External Defibrillation  
(AED)**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**

# Objectives:-

**By the end of this lecture the student will be able to:**

- Define cardiopulmonary resuscitation
- List purposes of cardiopulmonary resuscitation
- Mention indications of CPR
- Indicate signs and symptoms of cardiac arrest
- Define Automated External Defibrillator(AED)
- Mention the principles of AED application



# Outlines

- Definition of cardiopulmonary resuscitation(CPR)
- Purposes of cardiopulmonary resuscitation
- Indications of CPR resuscitation
- Signs and symptoms of cardiac arrest
- Definition of Automated External Defibrillator(AED)
- Principles of AED application



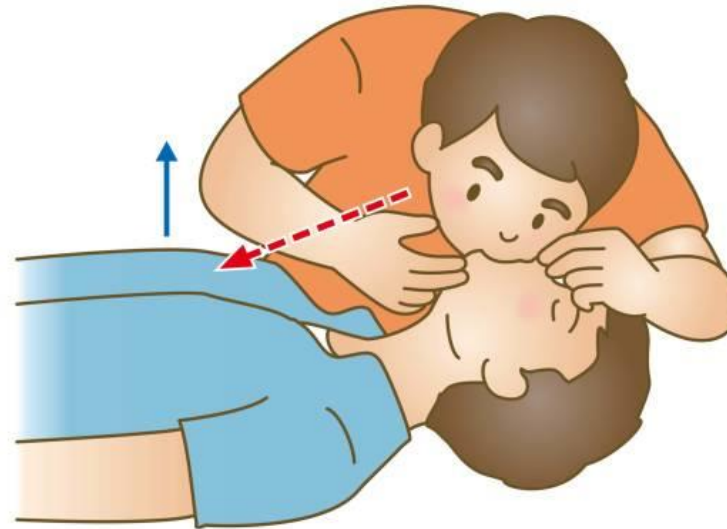
# CPR

Cardio  
Pulmonary  
Resuscitation



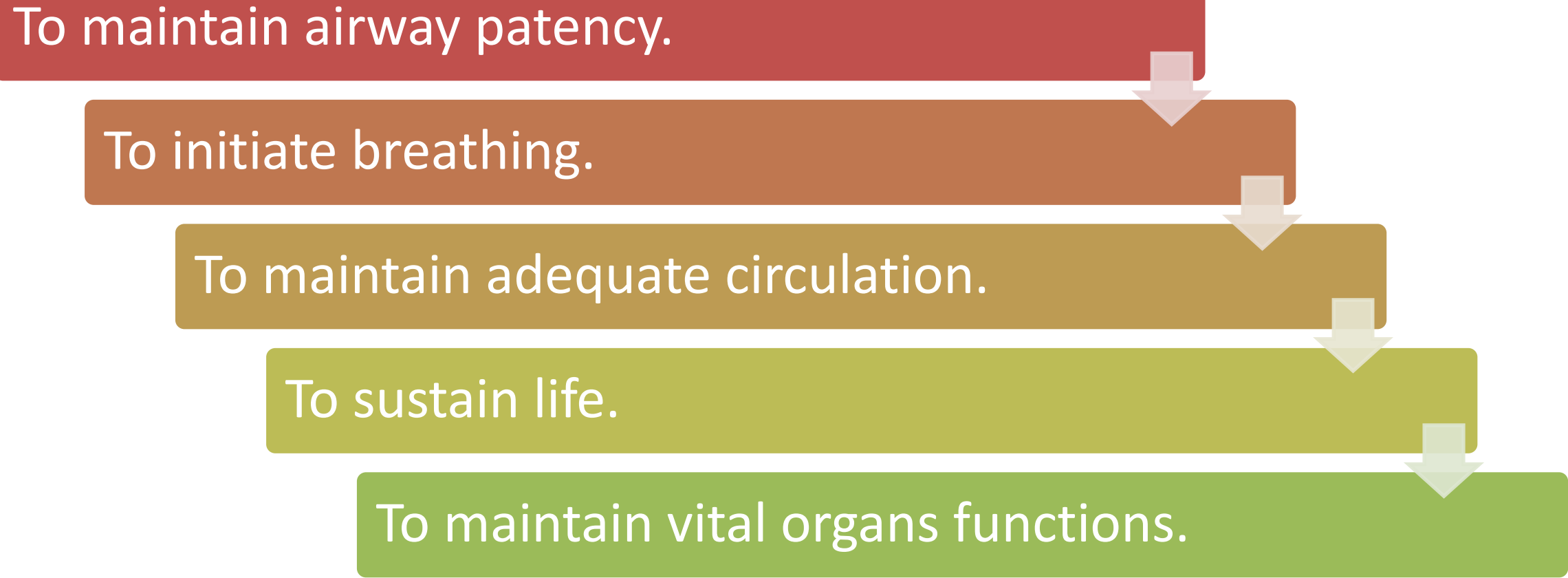
# Definition of Cardiopulmonary Resuscitation(CPR)

It is an emergency procedure for life support, starts with the immediate delivery of **chest compressions** followed by **ventilations**.



# Purposes of CPR

To maintain airway patency.



```
graph TD; A[To maintain airway patency.] --> B[To initiate breathing.]; B --> C[To maintain adequate circulation.]; C --> D[To sustain life.]; D --> E[To maintain vital organs functions.]
```

To initiate breathing.

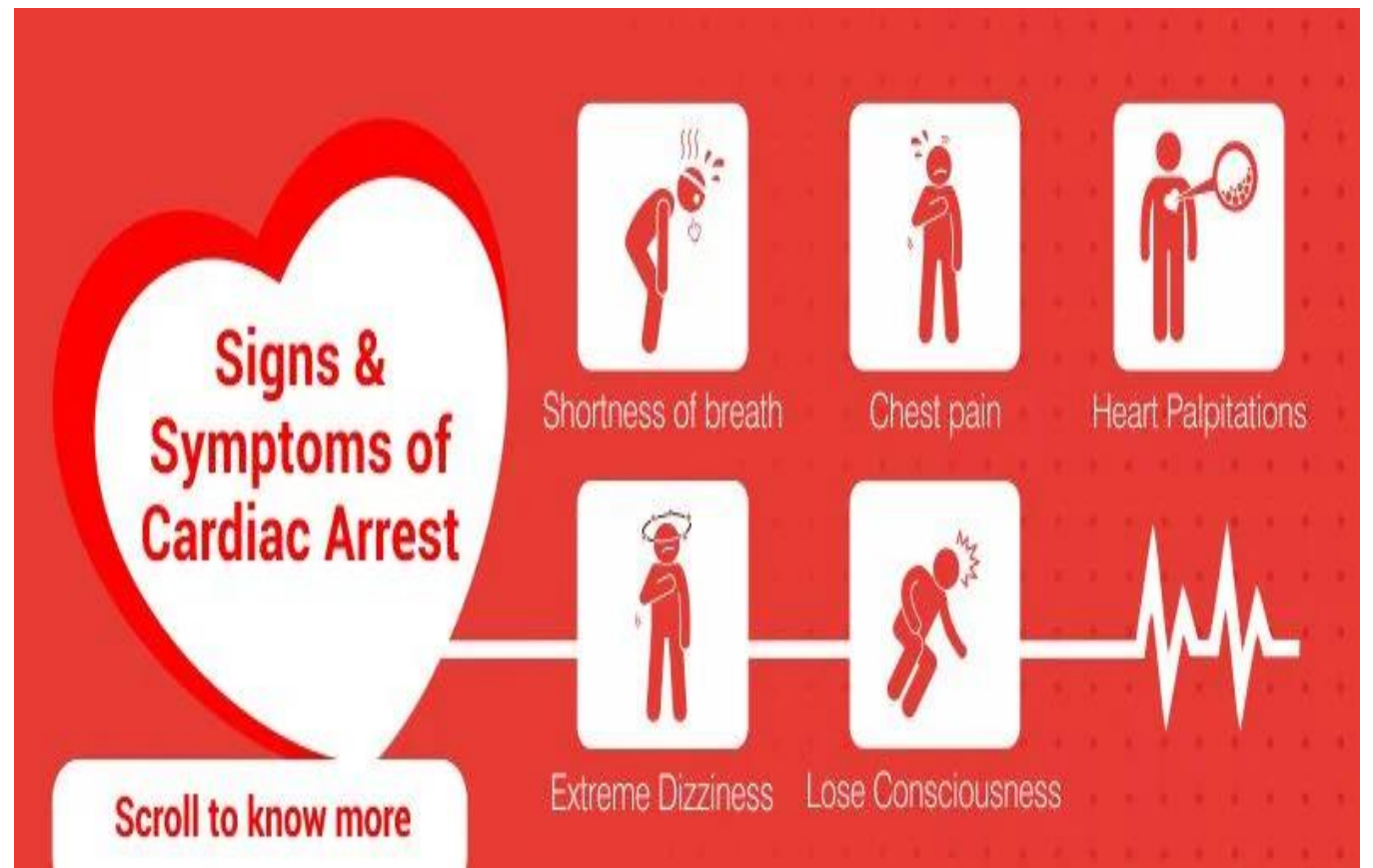
To maintain adequate circulation.

To sustain life.

To maintain vital organs functions.

# Signs and Symptoms of Cardiac Arrest

- Loss of consciousness.
- Gasping or absent respiration.
- Pulselessness in large arteries (carotid and femoral).
- Dilated pupils.



# Types of CPR

## 1. Basic life support (BLS):

It consists of **Circulation**, **Airway** management & **Breathing**.

## 2. Advanced Cardiac Life Support (ACLS) :

It includes the use of equipment (e.g. **ECG**, **Defibrillation** & **I.V access**) and **drug therapy**.

## 2. Prolonged Life Support (PLS) :

**Post- resuscitation care** to maintain brain orientation in the intensive care unit (brain protection).

# Basic life support (BLS) It consists of the following components

**Circulation.**



**Airway management.**



**Breathing.**



# CPR is as easy as **C - A - B**



**C**ompressions  
Push hard and fast  
on the center of  
the victim's chest



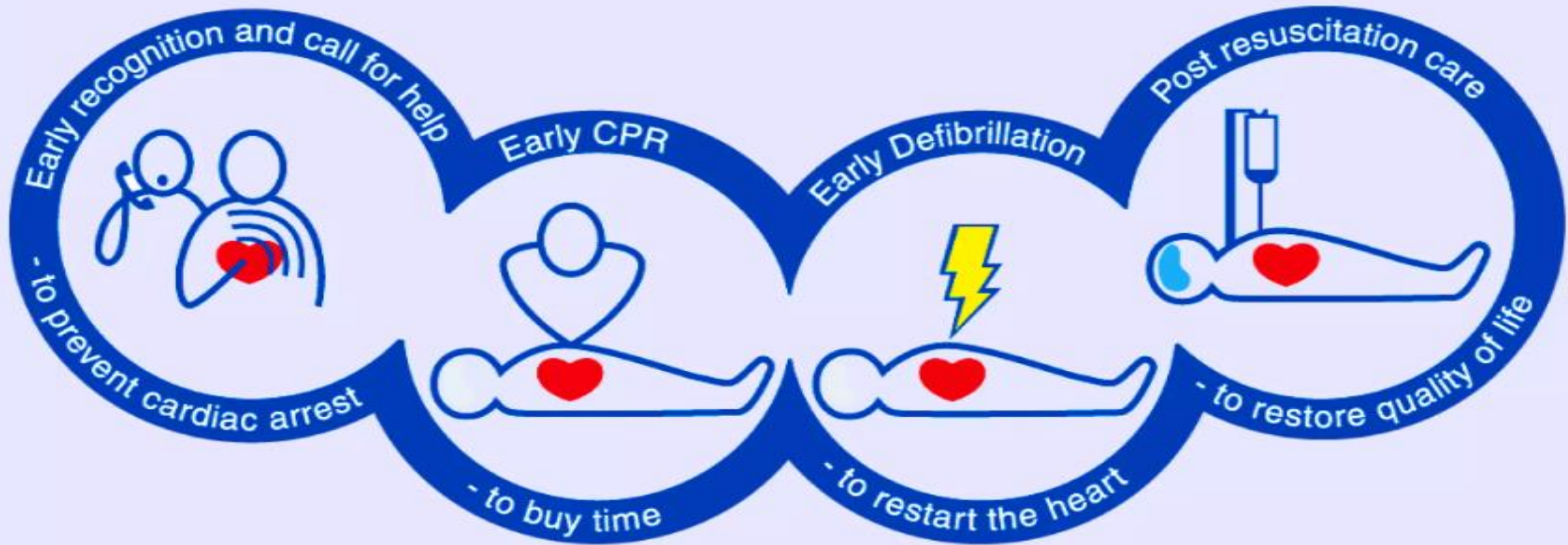
**A**irway  
Tilt the victim's head  
back and lift the chin  
to open the airway



**B**reathing  
Give mouth-to-mouth  
rescue breaths

Early chest compression can immediately circulate oxygen that is still in the bloodstream. By changing the sequence, chest compressions are initiated sooner and the delay in ventilation should be minimal.

# ”CHAIN OF SURVIVAL“



# Level of Consciousness

- **Response**
  - **Yes**
    - **Check for injuries**
    - **Reassess and get help**



# Response

## ?How to check the victim's response



- **Shake** the shoulders gently.
- **Shout** loudly "Are you alright?"

# Response Yes

## If the victim responds

1. Leave him/her as it is.
2. Call EMS (123)
3. Try to find out what is wrong.
4. Reassess regularly.

# Circulation

Check the patient for a carotid pulse for 5-10 seconds.

(Do not check for more than 10 seconds.)

## **If the patient has a pulse:**

Move to the airway and rescue breathing :

- Provide 10 rescue breaths per minute (1 breath every 6 seconds).
- Recheck pulse every 2 minutes.



shutterstock.com • 289553630

## If the patient doesn't have a pulse:

- Begin 5 cycles of CPR (lasts approximately 2 minutes).
- Start with chest compressions:
  - Provide 100 to 120 compressions per minute. This is 30 compressions every 15 to 18 seconds.
  - Place your palms midline, one over the other, on the lower 1/3 of the patient's sternum between the nipples.
  - lock your arms.

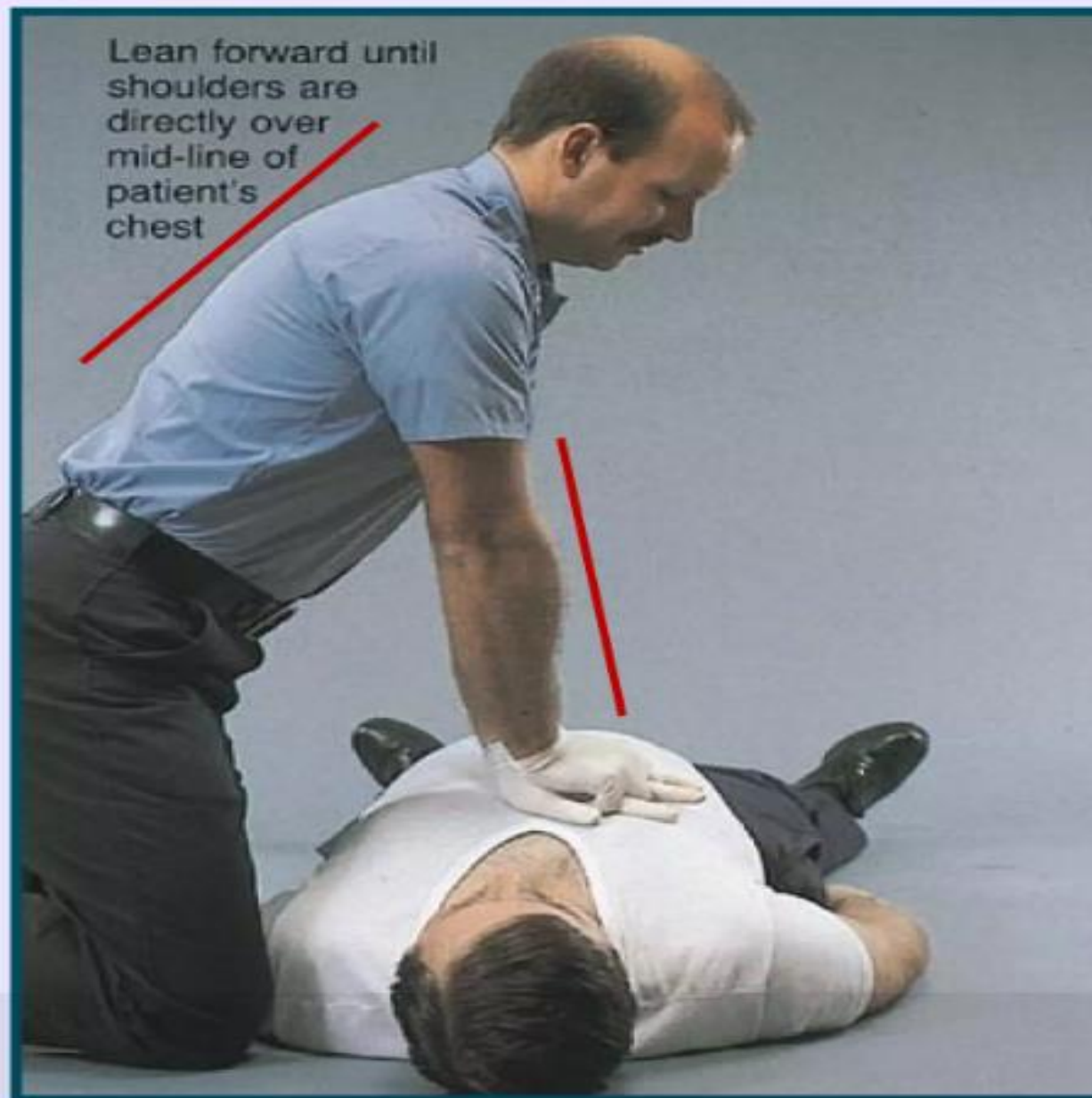
# How to give Chest compressions?

✗



✓



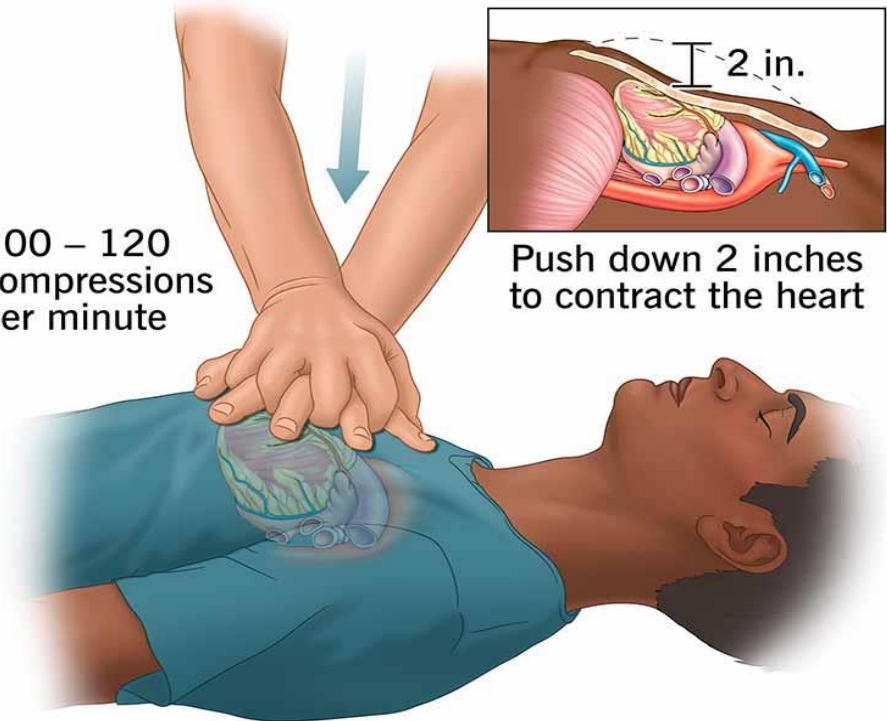


# Give 30 effective chest compressions



CPR (cardiopulmonary resuscitation)

100 – 120  
compressions  
per minute



Push down 2 inches  
to contract the heart

Cleveland Clinic © 2025

# How to perform effective chest ?compressions



- Place the **heel** of one hand on the lower half of sternum.
- Place **other hand on top**.
- **Interlock** your fingers.
- keep your **arms straight** and **lock your elbows** so you can compress the chest fully while conserving your energy.
- Compress the chest:
  - ☐ Rate **at least 100/min.**
  - ☐ Depth **5-6 cm**
  - ☐ Allow chest recoil.
- When possible change CPR operator/**5** cycles.

# How to give Chest compressions?

- Using two arms press to a depth of 2 to 2.4 inches (5- 6cm) or more on the patient's chest.
- Press hard and fast.
- Allow for full chest recoil with each compression.
- 1 cycle of adult CPR is 30 chest compressions to 2 rescue breaths.
- If two providers are present: switch rolls between compressor and rescue breather every 5 cycles.

# Continue Chest compressions & Rescue :breaths at a rate of



30



2

**If you confirm Cardiac arrest**

**Call 123 & Start CPR**



123



# Open Airway



- Open airway by **head tilt chin lift**
- If you suspect head or neck injury use **Jaw thrust**



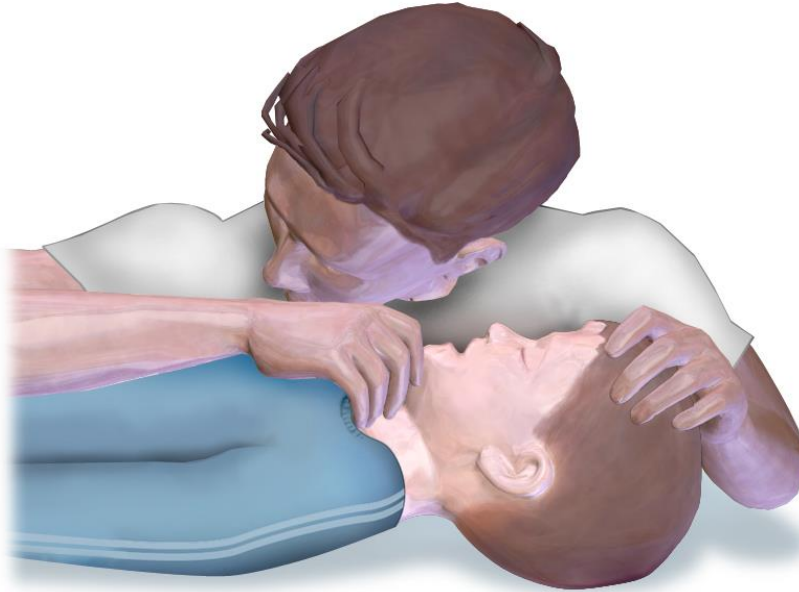
# How to Open The ?Airway

- **Head tilt & Chin lift**
- If cervical spine injury suspected: **jaw thrust**



# Check for breathing

- **Look**
- **Listen**
- **Feel**



**Look, Listen, and Feel  
for Breathing**

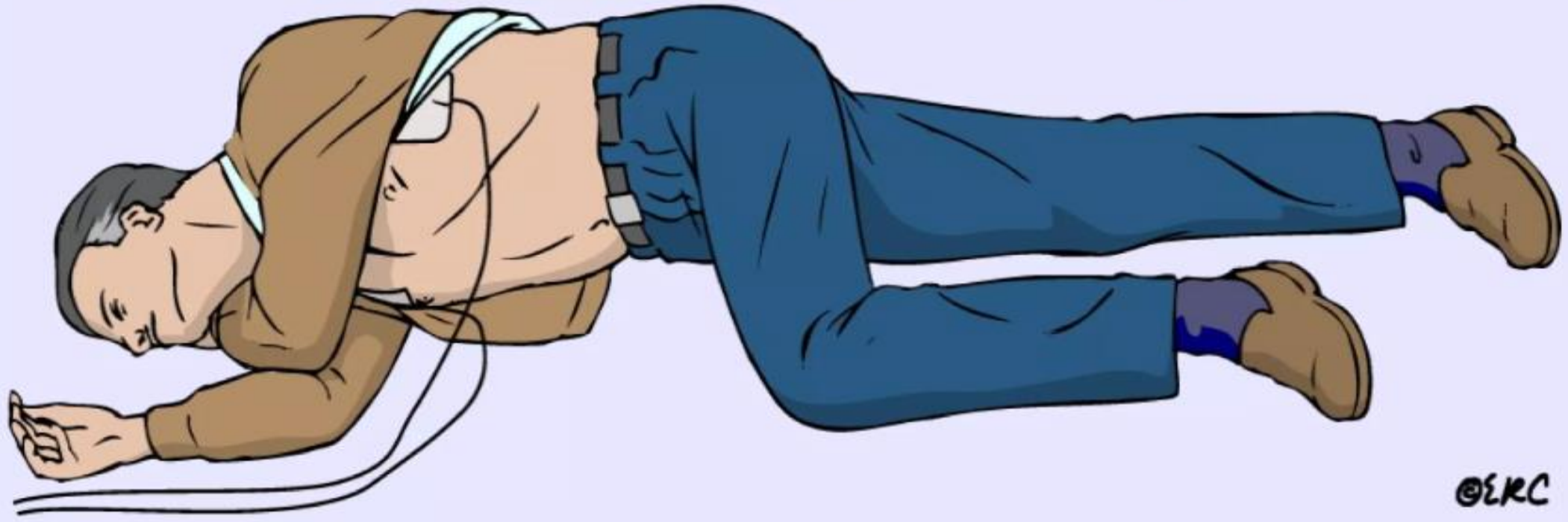
# Is the victim breathing ?

**Yes**

- **Recovery position**
- **Reassess**



If the victim starts to breathe normally, Put him/her in the  
**Recovery Position**



©ERC

# Steps of positioning the victim in the Recovery position

1



2



3



4





## ☐ Aim of the Recovery Position.

1. Gravity assistance to the clearance of **physical obstruction** of the airway by the tongue.
2. Gives a clear route by which **fluid (e.g. vomitus)** can drain from the airway.
3. There is a room for the **chest to rise and fall freely**.

## ☐ All forms of the recovery position share basic principles:

- 1) **The mouth is downward** so that fluid can drain from the patient's airway.
- 2) **The chin is upward** to keep the epiglottis opened.
- 3) **Arms and legs are locked** to stabilize the position of the patient.

**If the patient is not breathing or is breathing inadequately:**

**If the patient has a pulse:**

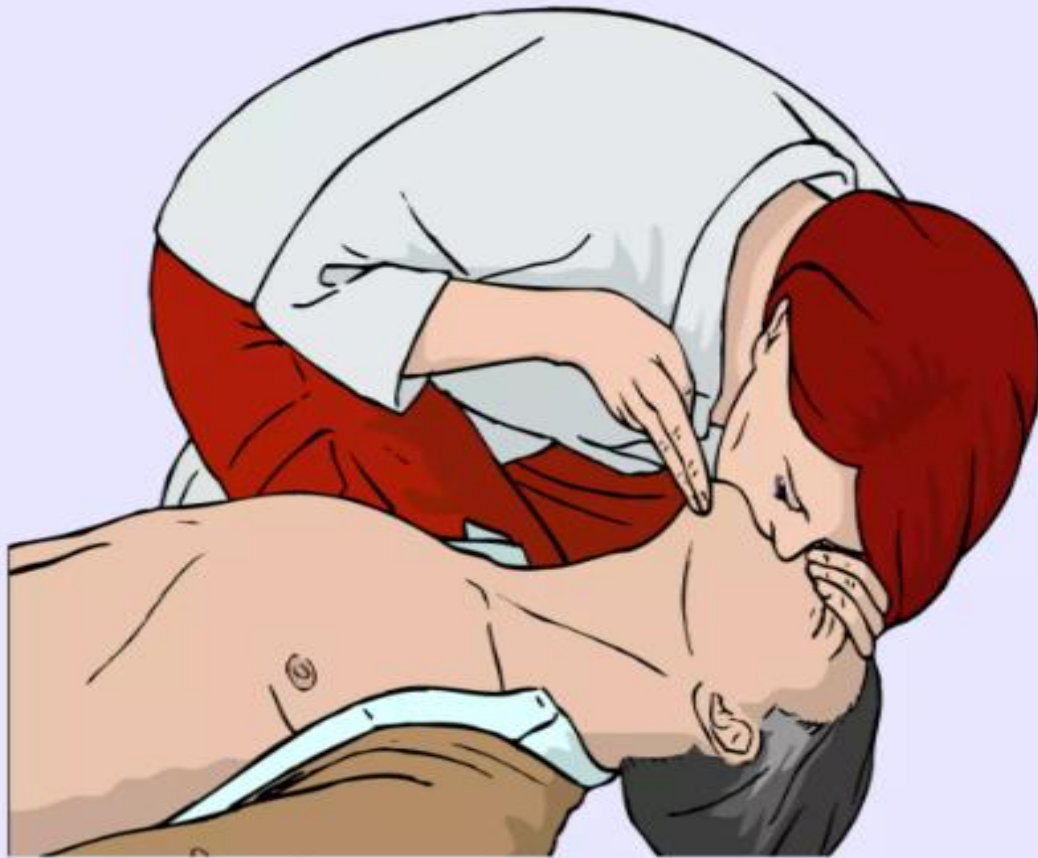
- Commence rescue breaths immediately.
- Give two slow breaths (every one breath over 1 seconds)

**If the patient has no pulse:**

- Begin CPR.



## ? How to deliver effective rescue breaths



- ❖ Open the airway.
- ❖ Close the victim's **nose**.
- ❖ Take a **normal breath**.
- ❖ Seal your lips over the victim's mouth.
- ❖ Blow **until the chest rises**.
- ❖ Pause for 1 second.
- ❖ **Allow chest to fall**.
- ❖ Repeat one more time.

**Simply place a towel or a piece of clothing over victim mouth and nose.**

**Do not perform  
mouth-to-mouth**

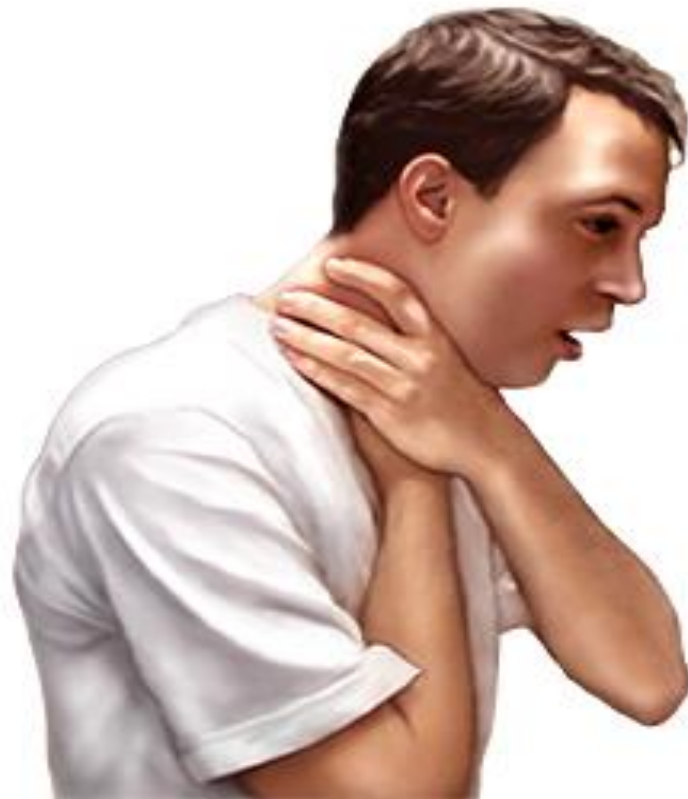


# If no air entry

- Retile head and try 2 more breaths.
- If still unsuccessful suspect choking



# Choking (Airway Obstruction)



Universal sign  
of choking

 ADAM.



# Heimlich Maneuver Technique (Conscious victim)

- Press fist into abdomen quick upward thrust
- Apply 5 abdominal thrust

Five and five

Give five back blows

Give five abdominal thrusts



# Heimlich Maneuver Technique (unconscious victim)

**Begin the steps of CPR- Starting with compressions.**

**The only difference between CPR & choking is:**

That each time you open the airway – look for the obstructing object before giving each breath. then remove the object.

Heimlich maneuver for an unconscious adult



# Finger Sweep Technique

- If foreign body comes within reach, grasp and remove it, do not force object deeper.



# Automated External Defibrillators (AED)

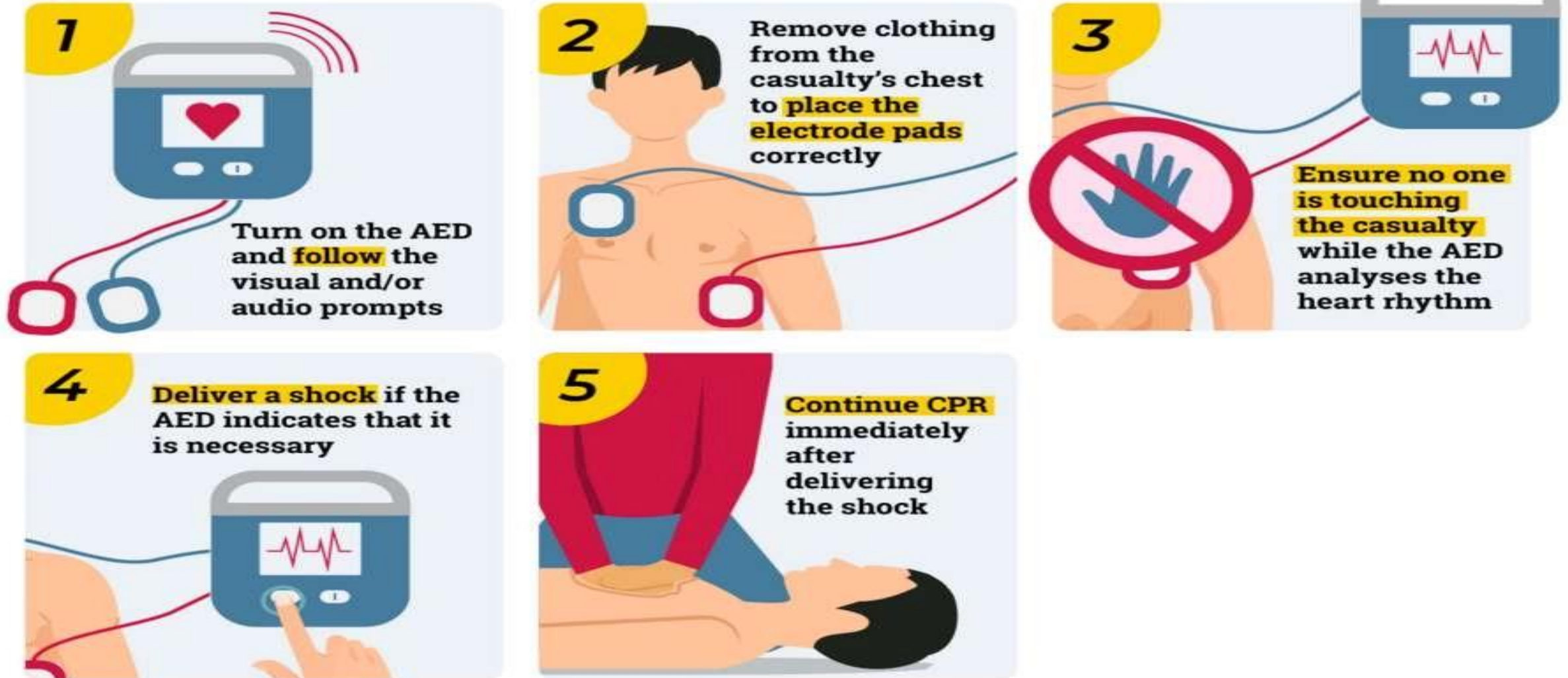
## Definition:-

**AED** is a small, lightweight, portable device used to give electric shock to restore a stopped heartbeat.



# Use of Automated External Defibrillator (AED)

## ✚ Steps to use AED



# Purposes of (AED)

- To rapidly identify the victim of sudden cardiac arrest.
- To provide early defibrillation to the victim of sudden cardiac arrest.



# Principles of AED Application

- **Do not** use alcohol to wipe the victim's chest .
- **Do not** use an AED and/or pads designed for adults on a child younger than 8 years, unless pediatric pads specific to the device are not available.
- **Do not** touch the victim while the AED is analyzing as it affects on the rhythm.
- **Do not** defibrillate someone when around flammable materials, such as gasoline.
- **Do not** use an AED in a moving vehicle.
- **Do not** place the victim on a pool or puddle of water when the rescuer is operating an AED.
- **Do not** use a mobile phone or radio within 6 feet of the AED, as infrared interference, generated by radio signals can disrupt analysis.
- Bore delivering shock to the victim with an AED, make sure that the **area is clear**

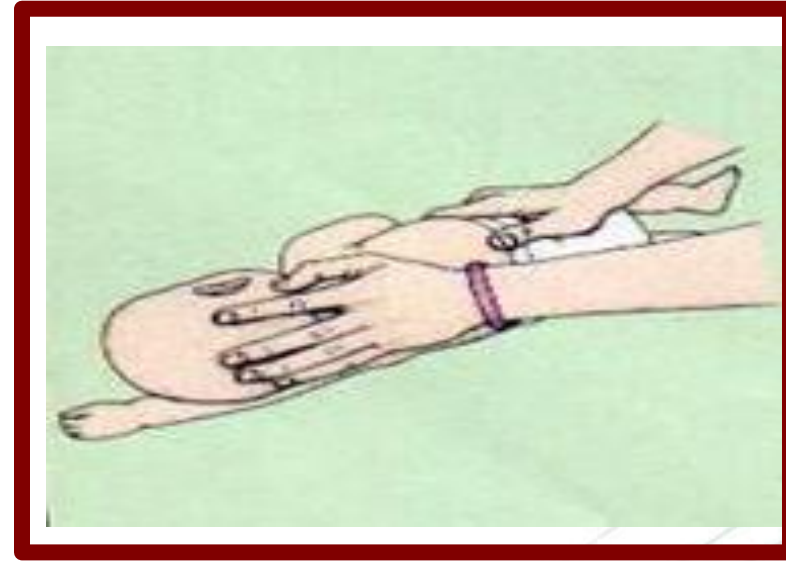
# Infant CPR

## Pediatric chain of survival



# Infant CPR

- Check responsiveness
- Shout for help
- Roll infant onto back

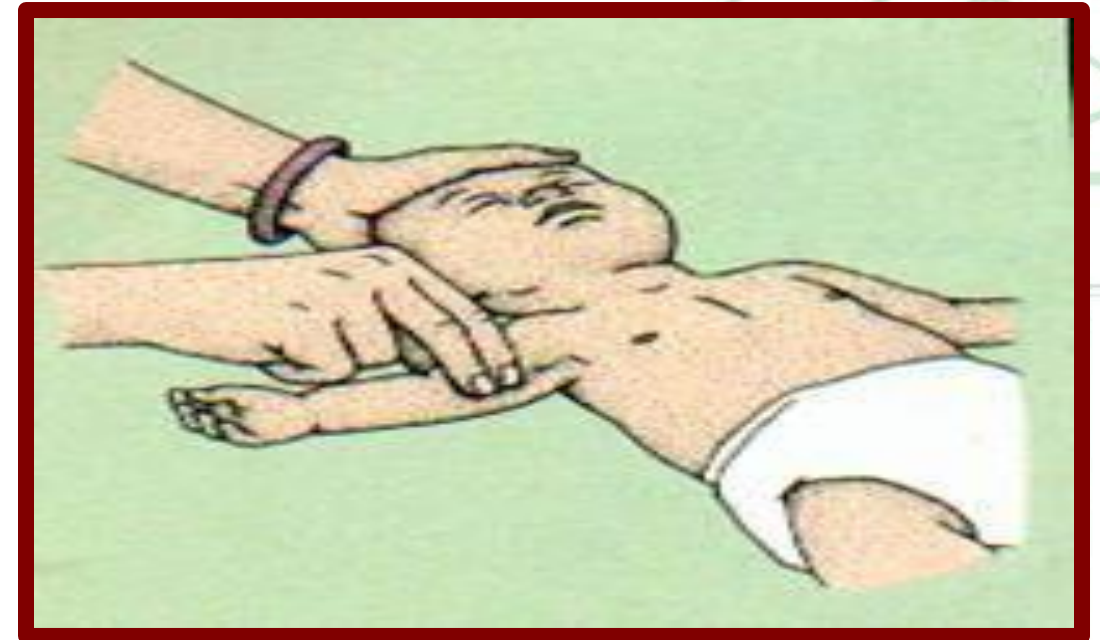


## ■ Open airway

- Check for breathing (3 – 5 seconds)
- Give 2 slow breaths for 1 – 1.5 seconds
- Check for pulse
- If there is pulse give rescue breathing every 3 seconds



- **Check for pulse**
- **If there is pulse give rescue breathing every 3 seconds**
- **Give 5 compressions and 1 breath (repeat for 1 minute) then check pulse**
- **At a rate of 100 per minute**



# Infant CPR Technique



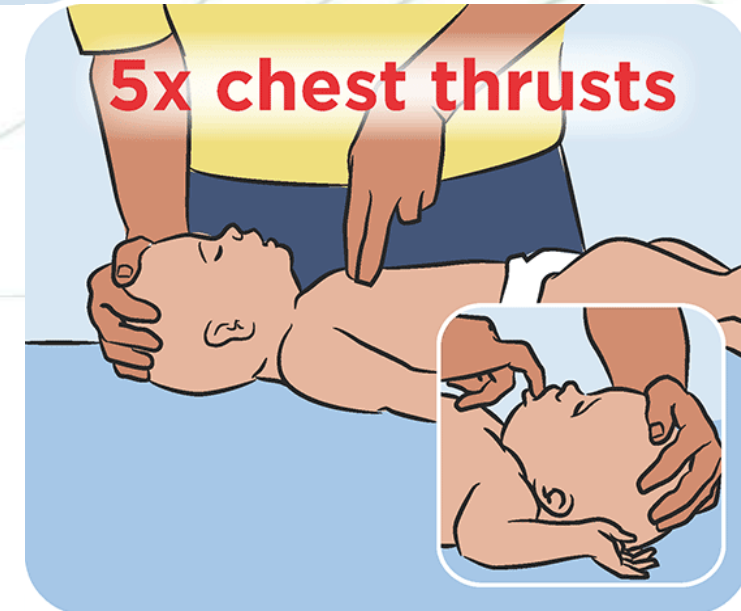
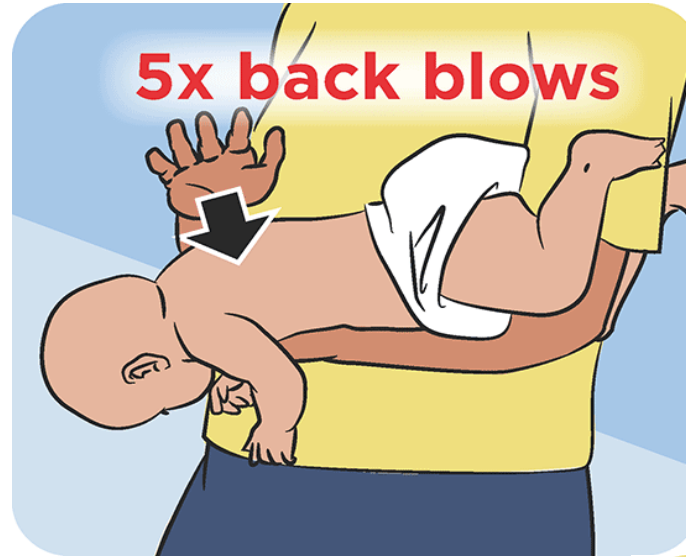
# Conscious infant with obstructed airway

- Give 5 back blows
- Give 5 chest thrusts
- Repeat up to 5 blows then 5 chest thrusts until infant is unconscious or object expelled



# Unconscious infant with obstructed airway

- Give 2 slow breaths
- Give 5 back blows
- Give 5 chest thrusts
- Check mouth for foreign object



# How long do we keep Giving CPR ?

- Victim revives
- Trained help arrives
- You are completely exhausted
- The scene becomes unsafe for you





# Question 1

**45 Year old man collapse in a shopping mall he is unresponsive and not breathing normally. What is first action?**

- A. Give two rescue breaths
- B. Check for pulse for 30 seconds
- C. Ensure safety and check responsiveness**
- D. Put him in recovery position



# Question 2

**A drowning victim is pulled from a pool she is unresponsive and not breathing. What is first action?**

A. Start chest compressions immediately

**B. Give 5 initial rescue breaths**

C. Place her in recovery position

D. Wait for emergency services



# Question 3

**A road traffic accident and the victim is unconscious but breathing normally. What is first action?**

- A. Start chest compressions
- B. Give rescue breaths
- C. Place the victim in the recovery position**
- D. Offer water



# Question 4

**What compression-to-ventilation ratio should be use for victim has no pulse and breathing?**

A. 15:2

**B. 30:2**

C. 5:1

D. 10:1



## Question 5

**While performing CPR on a victim in a public park, an AED becomes available. What should you do?**

- A. Continue CPR without interruption
- B. Use the AED only after 10 minutes
- C. Attach the AED immediately and follow instructions**
- D. Stop CPR completely



**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

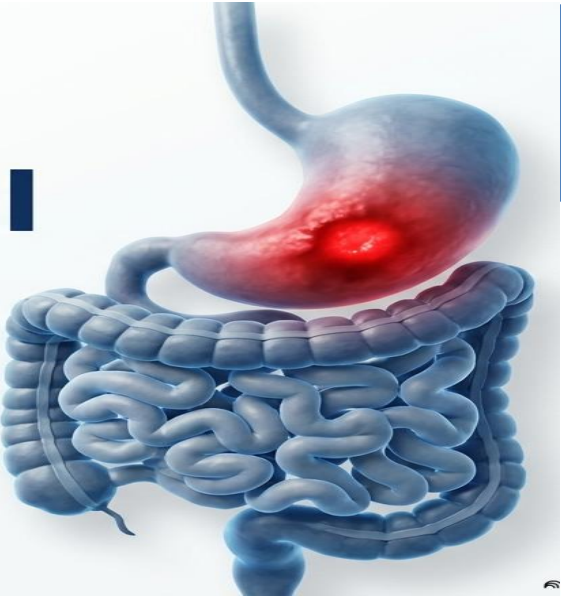
**Lecture 8&9**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

# **Acute Gastroenteritis Hemorrhage**

**Prepared by : Dr. Islam Sabry**  
**lecturer of Critical Care Nursing**





# LECTURE OUTLINE

01



Definition & Overview

02



Epidemiology & Incidence

03



Classification: Upper vs Lower GI Bleeding

04



Etiology & Risk Factors

05



Pathophysiology

06



Clinical Presentation & Assessment

07



Diagnostic Investigations

08



Emergency Nursing Management

09



Medical & Surgical Interventions

10



Complications & Prognosis

# Definition of Acute Gastroenteritis Hemorrhage



Acute, significant blood loss from any part of the gastrointestinal tract (esophagus to rectum) requiring immediate medical attention.



**~100/100K**

Annual Incidence.  
Accounts for ~300,000  
U.S. hospital admissions  
annually.



**5-10%**

Mortality Rate for Upper  
GI bleeds. The 2<sup>nd</sup>  
most common GI  
emergency.



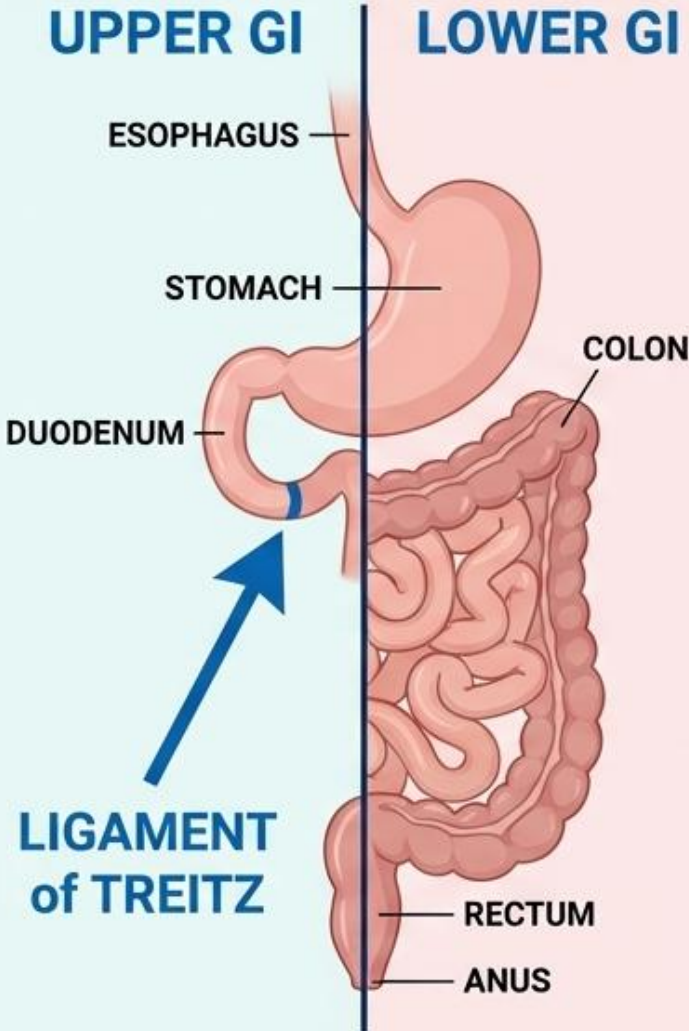
**3:1**

Male to Female ratio.

# Type of Acute Gastroenteritis Hemorrhage

## Upper GI - Proximal to Treitz

- Peptic Ulcer Disease (~50% - Most Common)
- Esophageal Varices
- Erosive Esophagitis / Gastritis
- Gastric / Duodenal Ulcers



## Lower GI - Distal to Treitz

- Diverticular Disease (~40% - Most Common)
- Colorectal Cancer / Polyps
- Inflammatory Bowel Disease (IBD) & Ischemic Colitis
- Hemorrhoids, Anal Fissures & Meckel's Diverticulum

# Causes And Risk Factor



## NSAIDs & Aspirin

Inhibits prostaglandins, causing mucosal injury. Increases bleeding risk 3-4x.



## H. pylori Infection

The most common cause of Peptic Ulcer Disease. Present in 90% of duodenal ulcers.



## Anticoagulants

Warfarin, heparin, DOACs. Danger escalates with a supratherapeutic INR.



## Liver Cirrhosis & Alcohol

Portal hypertension leads to varices. Coagulopathy and direct mucosal damage (gastritis).



## Advanced Age

Reduced mucosal defense paired with a higher comorbidity burden.

# The Pathophysiological Cascade

**Step 1:** Mucosal Injury or Vascular Disruption occurs in the GI tract.

**Step 2:** Active Blood Loss pours directly into the GI Lumen.

**Step 3:** Decreased Circulating Volume triggers systemic alarms.

**Step 4:** Compensatory Mechanisms activate to maintain perfusion.

**Step 5 (Critical): Hypovolemic Shock** ensues if severe blood loss is unmanaged.

# The Hallmarks Of GI Hemorrhage

## Visualizing the Output



### **Hematemesis (Bright Red or Coffee-Ground)**

Vomited blood. Indicates an Upper GI Bleed.



### **Melena (Black, Tarry, Foul-Smelling)**

Digested blood in stool. Indicates Upper GI or high Lower GI bleed.



### **Hematochezia (Bright Red per Rectum)**

Fresh blood. Indicates Lower GI or massive Upper GI rapid bleed.



### **Systemic Shock Alerts**

Hemodynamic instability (Hypotension, tachycardia, syncope) and Shock Signs (Cool clammy skin, altered consciousness, oliguria).

**Key Nursing Assessment:** ABCs first → Vital signs → IV access → Blood type & crossmatch.

# Diagnostic procedures



## Endoscopy (The Gold Standard)

Upper GI Endoscopy (EGD) for proximal, Colonoscopy for stable lower GI, Capsule for obscure bleeds.



## Blood Panels

CBC, BMP, Coagulation (PT/INR).

Note: BUN:Creatinine ratio  $>36$  strongly suggests Upper GI bleeding.



## Advanced Imaging

CXR (checks for free air/perforation), CT Angiography (detects active bleeding  $\geq 0.3$  mL/min), Tagged RBC scan.



## Tubes & Catheters

Pulmonary Artery Catheter (PAC) to guide fluid replacement. Large Nasogastric (NG) Tube for decompression.

# Diagnostic Intervention: Gastric Lavage



## The Protocol

Used to confirm diagnosis, assess bleeding volume, and prepare the GI tract for endoscopy.

Lavage with water or normal saline until clear.



**NEVER USE ICED SALINE.**

**Why?** Iced saline decreases oxygen delivery to the tissues, prolongs Prothrombin Time (PT), and can actively aggravate the hemorrhage.

# Nursing Management

## Immediate Resuscitation

A

### Module A: Airway & Breathing

Ensure patent airway. Administer O<sub>2</sub> 100% via non-rebreather mask. Position semi-recumbent. Prepare for intubation if LOC drops.



B

### Module B: IV Access & Fluids

Establish TWO large-bore IVs ( $\geq 16G$ ). Rapid administration of Normal Saline or Ringer's Lactate. Target MAP  $\geq 65$  mmHg. Avoid over-resuscitation.



C

### Module C: Blood Transfusion

Transfuse if Hgb  $< 7$  g/dL (or  $< 8$  g/dL with cardiac disease). Administer FFP for coagulopathy (INR  $> 1.5$ ). Platelets if  $< 50,000$ .



# Nursing Management

## Stabilization & Monitoring

D

### Module D: Monitoring

Continuous cardiac monitoring. Hourly urine output via Foley. Vital signs q15-30 min. Gastric pH monitoring (goal > 5).



E

### Module E: Medications

IV PPI (pantoprazole 80mg bolus). Octreotide / Vasopressin for variceal bleeding. Prophylactic antibiotics.



F

### Module F: Documentation

Strict I&O charting. Exact stool and emesis characteristics.



## Advanced Interventions

- Endoscopic Thermal/Injection therapy.



- Sengstaken-Blakemore tube (balloon tamponade).



- TIPS (Transjugular intrahepatic portosystemic shunting).



# The Hypovolemia Threat

## Level: Systemic Complications



**Systemic Threat:** Hypovolemic Shock & Rebleeding (25-30% risk within 72 hours - highest predictor of death).



### Brain (Hepatic Encephalopathy):

Triggered by the breakdown of blood proteins in the gut (critical for cirrhotic patients).

### Lungs (Aspiration Pneumonia):

Risk of blood aspiration into lungs during hematemesis.

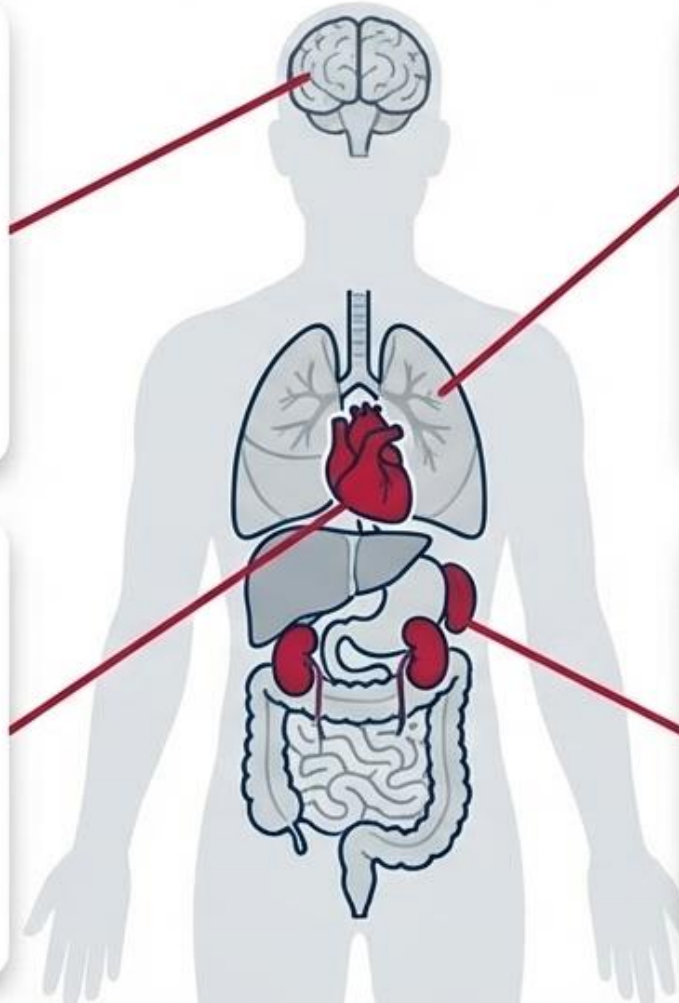


### Heart (Acute MI / ACS):

Induced by severe decrease in oxygen delivery, especially in the elderly.

### Kidneys (Renal Failure):

Decreased renal perfusion leads to Pre-renal azotemia and Acute Tubular Necrosis (ATN).



# Patient Education and Discharge Teaching



**Medications:** Strict compliance. Completely avoid NSAIDs unless explicitly directed.



**Dietary Modifications:** Avoid spicy and acidic foods. Consume small, frequent meals to reduce gastric irritation.



**Lifestyle:** Maintain moderate exercise and healthy sleep patterns.



**Vigilance:** Report any recurrence of symptoms (melena, pain) to the physician immediately.



**Stress Management:** High stress increases acid secretion. Practice relaxation techniques.



**Cessation:** Stop Alcohol & Smoking immediately—both worsen mucosal damage and drastically increase re-bleeding risk.

# NURSING DIAGNOSES & CARE PLANNING

|                                                                                                                                                 |                                                                                                                                      |                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Deficient Fluid Volume</b><br><i>R/T: Active blood loss</i>                                                                                  |  Stable vital signs, Hgb $\geq 7$ g/dL, adequate UO | IV fluid resuscitation • Transfuse as ordered • Hourly UO via Foley • Monitor CVP/MAP         |
| <b>Risk for Ineffective Tissue Perfusion</b><br><i>R/T: Decreased cardiac output</i>                                                            |  Alert & oriented, SpO <sub>2</sub> $\geq 95\%$     | O <sub>2</sub> therapy • Continuous monitoring • Elevate HOB 30° • Report changes immediately |
|  <b>Acute Pain</b><br><i>R/T: Mucosal erosion / ulceration</i> |  Pain $\leq 3/10$ within 1 hour                     | IV PPI administration • NPO status • Position of comfort • Non-pharmacologic measures         |
|  <b>Anxiety</b><br><i>R/T: Acute illness threat</i>          |  Patient verbalizes reduced anxiety               | Explain all procedures • Calm reassurance • Allow family presence • Provide information       |

# Remember and Summary

1

**Immediate Threat:** This is a life-threatening emergency requiring systematic assessment.

2

**The 3 Hallmarks:**  
Memorize Hematemesis, Hematochezia, and Melena.

3

**Safe Diagnostics:** Utilize PAC, NG Tube, and Gastric Lavage (Never use iced saline).

4

**Resuscitate First:** Secure two large-bore IVs, administer crystalloids, then blood products.

5

**Vigilant Monitoring:**  
Continuously observe and report any bleeding signs immediately.

6

**Prevent Recurrence:**  
Discharge education on meds, diet, and lifestyle is a life-saving intervention.

**Any question**





# THANK YOU

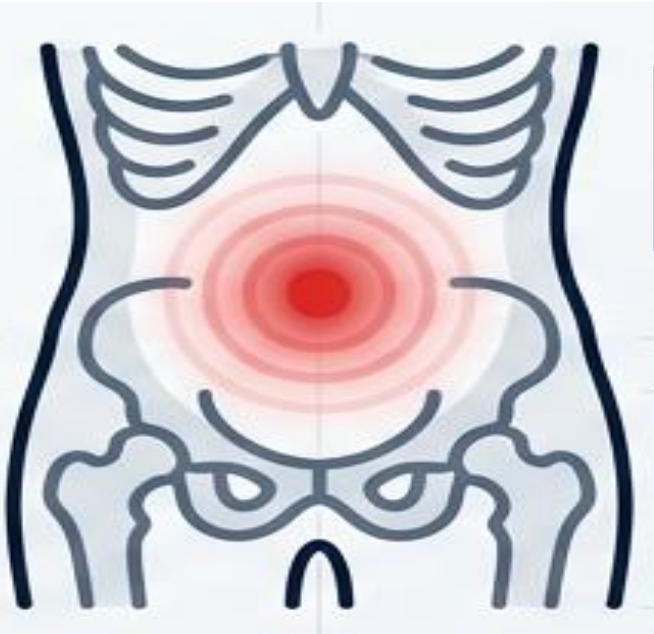
[www.must.edu.eg](http://www.must.edu.eg)



MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY

## Acute Abdominal Pain

**Prepared by : Dr. Islam Sabry Abd Elgwad**  
**Lecturer of Critical Care Nursing**



# Learning Objectives:



By the end of this lecture, students will be able to:

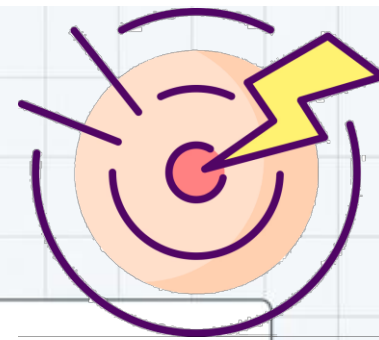
- Define and classify acute abdominal pain
- Identify key clinical signs and symptoms
- Apply systematic nursing assessment (OPQRST + IAPP)
- Interpret diagnostic investigations
- Implement evidence-based nursing interventions
- Recognize special populations with atypical presentations



# What is Acute Abdomen?



# The Urgency of the Acute Abdomen



Sudden onset of severe pain in the abdomen lasting less than 5 days, potentially indicating a life-threatening condition requiring immediate evaluation.



## Frequency

5–10% of all  
Emergency Department  
visits worldwide.



## Urgency

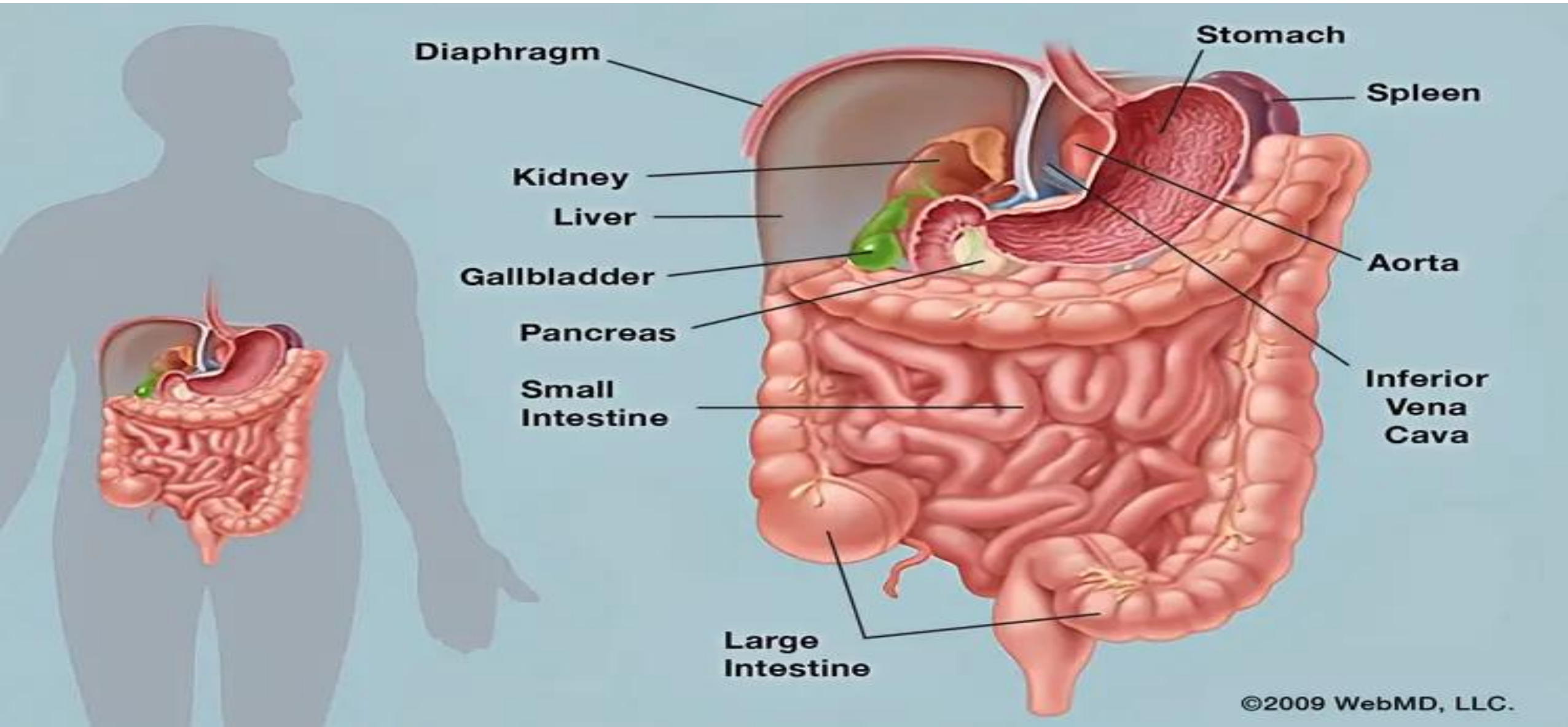
25% of cases require  
immediate surgical  
intervention.  
Time-critical decisions  
are mandatory.



## Mortality Risk

Delayed diagnosis  
significantly increases  
morbidity and mortality.  
Rapid systematic  
assessment is the  
primary defense.

# ABDOMINAL ANATOMY



# ABDOMINAL ANATOMY & QUADRANTS

## RUQ

Liver  
Gallbladder  
Duodenum  
Right kidney

## LUQ

Stomach  
Spleen  
Pancreas  
Left kidney

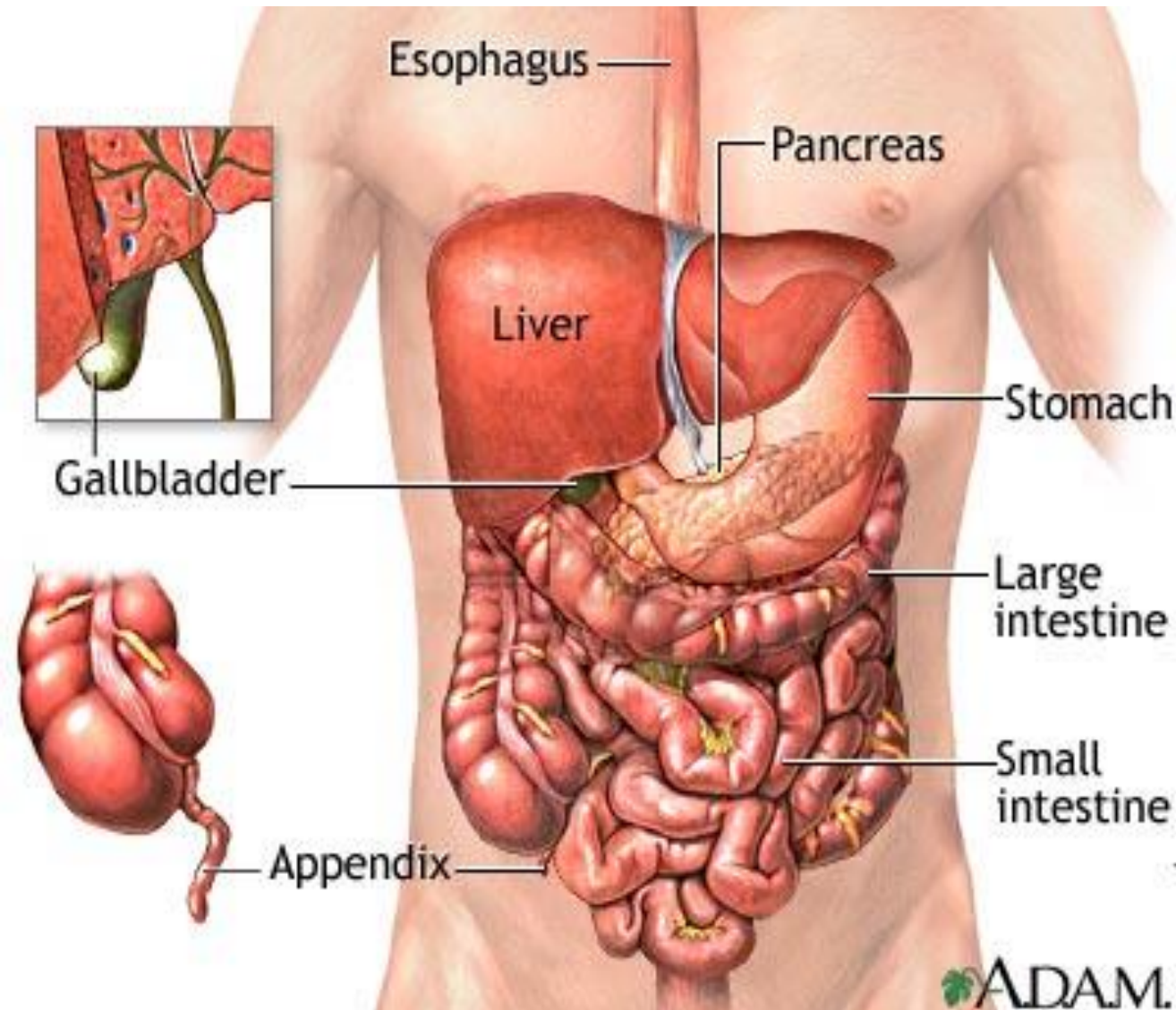
U

## RLQ

Appendix  
Right ovary  
Cecum  
Right ureter

## LLQ

Sigmoid colon  
Left ovary  
Descending colon  
Left ureter



# Classification Causes



## Surgical Causes

- Appendicitis
- Perforated peptic ulcer
- Bowel obstruction
- Mesenteric ischemia
- Ruptured ectopic pregnancy



## Medical/Non-Surgical Causes

- Gastroenteritis
- Pancreatitis (often managed medically)
- Renal/biliary colic
- Diabetic ketoacidosis
- Pelvic inflammatory disease

# Pathophysiology of Abdominal Pain



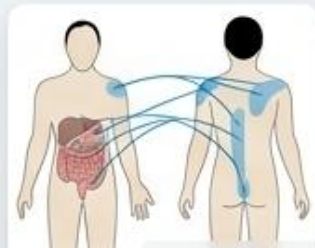
Understanding this cascade explains why pain character changes as inflammation spreads from the organ to the peritoneum.

# Source of Abdominal Pain



# The Pain Typology Matrix

|                          | Visceral Pain 📍                          | Somatic Pain 🩺                              | Referred Pain ↗                                      |
|--------------------------|------------------------------------------|---------------------------------------------|------------------------------------------------------|
| <b>Origin:</b>           | Hollow organ distension/ischemia         | Parietal peritoneum irritation              | Shared neural pathways (organ to skin)               |
| <b>Character:</b>        | Dull, crampy, poorly localized (colicky) | Sharp, well-localized, worsened by movement | Felt distant from actual source                      |
| <b>Location:</b>         | Periumbilical or midline                 | Directly over affected area                 | Shoulder, back, or groin                             |
| <b>Clinical Example:</b> | Early appendicitis, intestinal colic     | Peritonitis, late appendicitis, perforation | Diaphragm irritation → shoulder; Renal colic → groin |



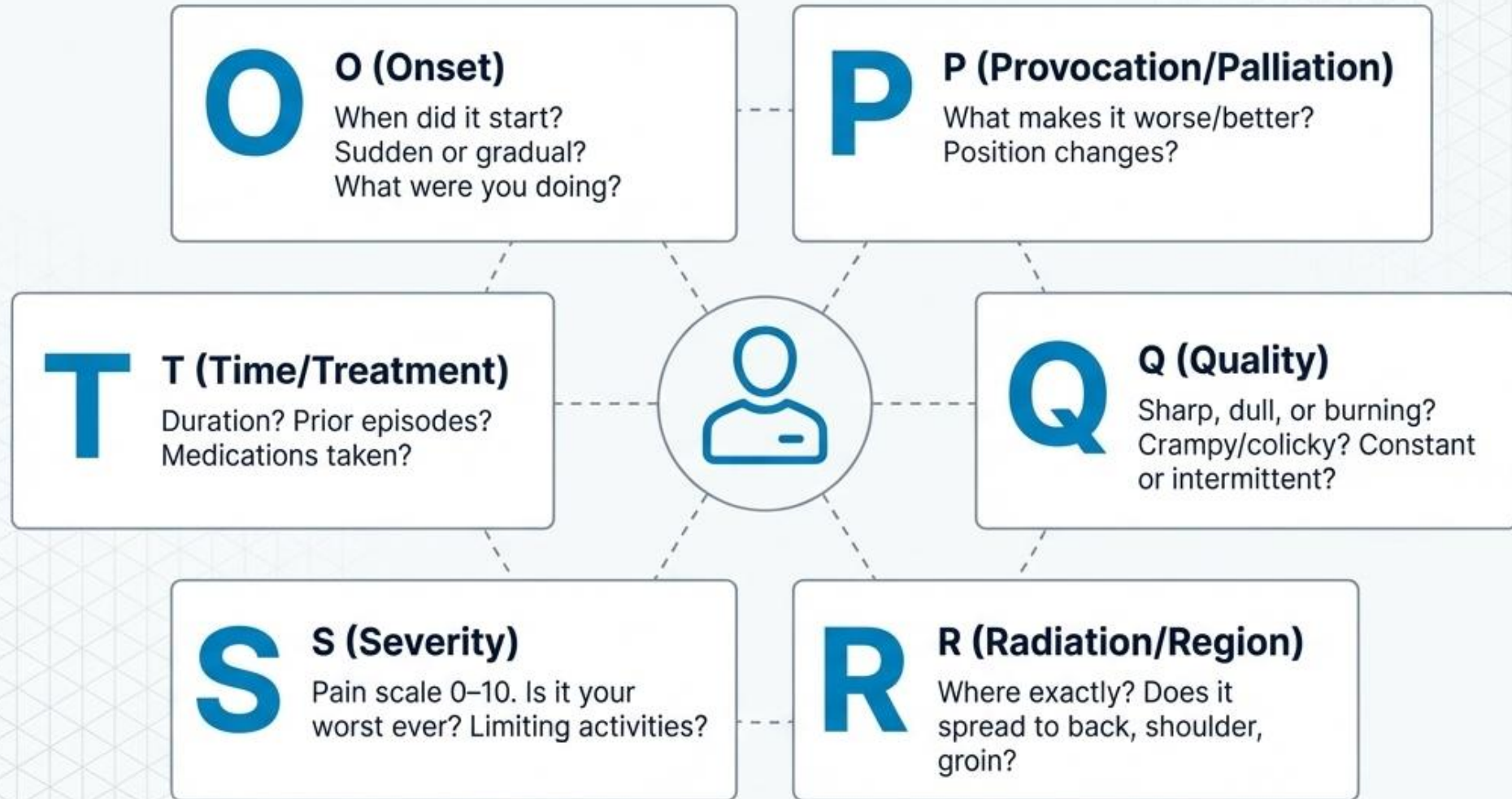
# OPQRST



- Onset
- Palliation/ Provocation
- Quality
- Radiating/Referred
- Severity
- Time



# Subjective Investigation: The OPQRST Dashboard



# Objective Evidence: The Physical Exam Pipeline

**⚠ Always follow the order. Never palpate before auscultating!**



## INSPECTION 🧐

Look for distension, asymmetry, visible peristalsis, skin color/bruising, surgical scars, patient posture.



## AUSCULTATION 🔊

Listen for bowel sounds: normal 5-35/min, hypoactive/ileus, hyperactive/obstruction, absent/peritonitis, bruits.



## PERCUSSION 🥁

Assess for tympany/gas, dullness/fluid, shifting dullness/ascites.

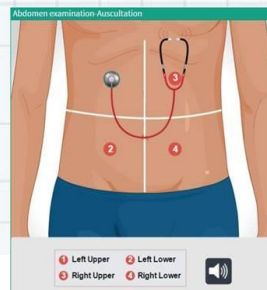
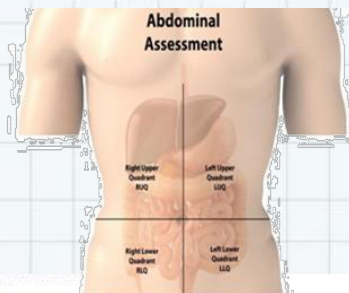


## PALPATION 🤏

Assess for guarding, rigidity, rebound tenderness, specific eponymous signs.

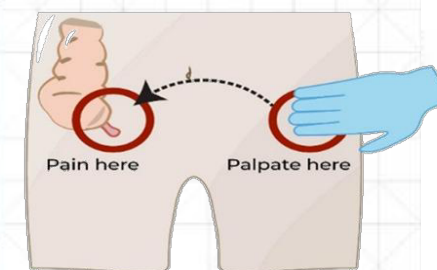


**Palpation alters bowel sounds.  
Stop and listen first.**



# The “Signs & Suspects” Diagnostic Grid

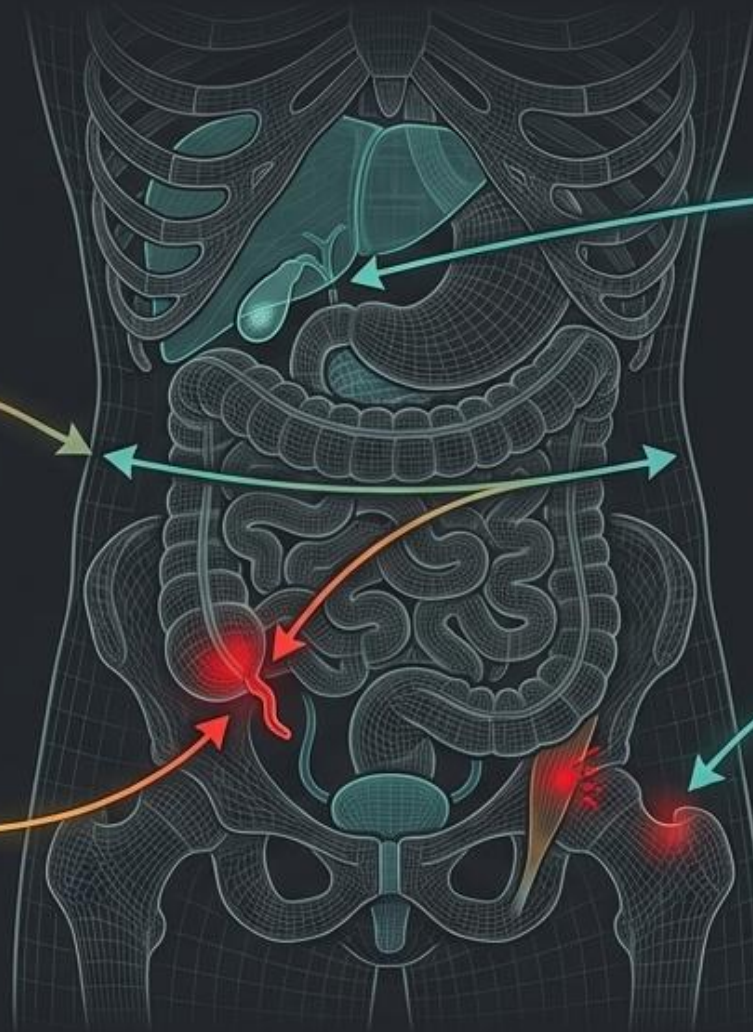
| Clinical Sign                       | Action (How to Elicit)                                        | Finding                          | Clinical Suspicion                                                                              |
|-------------------------------------|---------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Cullen's &amp; Grey Turner's</b> | Inspect periumbilical / flanks                                | Bruising or discoloration        | Retroperitoneal hemorrhage (e.g., AAA, severe pancreatitis)<br><b>Abdominal Aortic Aneurysm</b> |
| <b>Murphy's Sign</b>                | Deep palpation RUQ during deep inspiration                    | Patient abruptly stops breathing | <b>Acute Cholecystitis</b>                                                                      |
| <b>Rovsing's Sign</b>               | Apply deep pressure to LLQ                                    | Referred pain felt in RLQ        | <b>Acute Appendicitis</b>                                                                       |
| <b>Psoas / Obturator Signs</b>      | Hip extension/flexion against resistance or internal rotation | Deep pelvic pain                 | <b>Retrocecal/Pelvic Appendicitis</b>                                                           |



Referred pain to RLQ with deep palpation of LLQ.  
Suggests Appendicitis



# The Diagnostic Anatomy Map



## Murphy's Sign

Halt in inspiration upon deep palpation = Acute cholecystitis.



## Rebound Tenderness

Pain worsens upon sudden release of pressure = Peritonitis.



## Rovsing's Sign

LLQ pressure causes RLQ pain = Acute appendicitis.



## Psoas / Obturator Signs

Pain on hip extension/rotation = Retrocecal appendicitis / pelvic abscess.



# Cross-Referencing Evidence: Labs & Imaging



## Laboratory Forensics

- **Amylase & Lipase:** **Lipase  $>3\times$  ULN** = Pancreatitis
- **$\beta$ -hCG:** **Mandatory** to rule out ruptured ectopic pregnancy
- **Lactate:** **Elevation indicates poor perfusion** / bowel ischemia
- **Routine:** CBC (WBC $\uparrow$ ), CMP (LFTs/Renal), Urinalysis, Blood Cultures



## Imaging Evidence

**FAST (First-line):** Ultrasound (biliary, renal, FAST exam for free fluid) & X-ray (free air / obstruction).

**GOLD STANDARD:** CT Abdomen/Pelvis (Appendix, diverticulitis, AAA).

**URGENT VASCULAR:** CT Angiography (Mesenteric ischemia).

# The Danger Zone: Vital Signs & Red Flags

|                                                           |                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------|
| <b>BP: Hypotension (&lt;90/60)</b>                        | → Septic/hemorrhagic shock, AAA rupture, ectopic rupture. |
| <b>HR: Tachycardia (&gt;100 bpm)</b>                      | → Hypovolemia, infection, internal bleeding.              |
| <b>Temp: Fever (&gt;38.5°C) or Hypothermia (&lt;36°C)</b> | → Infection, abscess, severe sepsis.                      |



## IMMEDIATE ACTION RED FLAGS

- Rigid, board-like abdomen
- Hypotension + tachycardia (Shock index > 1)
- Absent bowel sounds + rebound tenderness
- Severe pallor/diaphoresis
- Pain completely unresponsive to analgesia

# Nursing management of Abdominal Pain

## The Rescue Protocol: Action Timeline

0 min 10 min 30+ min

### Immediate Resuscitation (0–10 min)



- **Position:** Semi-Fowler's (reduces abdominal tension).



- **Access:** Large-bore IV access (2 × 18G) & fluid resuscitation (0.9% NaCl or LR).



- **Oxygenate:** O2 therapy if SpO2 <94%.



- **Control:** Keep patient NPO (prepare for potential surgery).



- **Pain:** Administer analgesia as ordered.  
\*Evidence note: Do NOT withhold pain meds; it does not mask the diagnosis.\*

### Ongoing Assessment (10–30 min)



- **Draw bloods** (CBC, CMP, Lipase, β-hCG, Lactate).



- **Perform serial** abdominal exams to detect deterioration.



- **Insert NG tube** if vomiting/obstruction is present.



- **Monitor hourly** urine output (target >0.5 mL/kg/hr).

# NURSING DIAGNOSES & CARE PLAN (NANDA)

## Nursing Diagnosis

**Acute Pain**

**Risk for Deficient Fluid Volume**

**Risk for Infection**

**Anxiety**

## Interventions

- |                                                                                                                                            |                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Administer analgesia as prescribed</li><li>• position semi-Fowler's</li></ul>                      | <ul style="list-style-type: none"><li>• reassess q30 min</li><li>• non-pharmacologic measures</li></ul>              |
| <ul style="list-style-type: none"><li>• IV fluid replacement</li><li>• monitor I&amp;O</li><li>• assess mucous</li></ul>                   | <ul style="list-style-type: none"><li>• membranes</li><li>• daily weights</li><li>• electrolyte monitoring</li></ul> |
| <ul style="list-style-type: none"><li>• Aseptic wound care</li><li>• monitor WBC temperature</li><li>• antibiotics as prescribed</li></ul> | <ul style="list-style-type: none"><li>• hand hygiene education</li></ul>                                             |
| <ul style="list-style-type: none"><li>• Therapeutic communication</li><li>• explain all procedures</li><li>• allow questions</li></ul>     | <ul style="list-style-type: none"><li>• include family in education</li><li>• comfort measures</li></ul>             |

**Any question**





**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

# **Critical Care Trauma Nursing**

## **Head, Chest, Spinal Injuries**

**Prepared by : Dr. Islam Sabry Abdelgwad**  
**lecturer of Critical Care Nursing**





# Learning Objectives

1

Define chest trauma, head injuries, and spinal cord injuries with their clinical significance.

2

Describe the etiology and pathophysiology of flail chest, pneumothorax, massive hemothorax, head injuries & spinal cord injuries.

3

Classify each injury type and differentiate between primary and secondary injuries.

4

Identify signs & symptoms, diagnostic procedures and assessment methods for each condition.

5

Discuss comprehensive medical, pharmacological, surgical & nursing management.

6

Apply nursing interventions and evaluate patient outcomes using evidence-based criteria.

7

Develop discharge plans and patient/family education appropriate for each condition.

# The Trauma Nurse's Algorithm

Assess → Intervene → Evaluate

## A - Airway



Ensure patent airway + cervical spine protection.

Interventions in #06B6D4

## B - Breathing



High-flow O<sub>2</sub> and ventilation.

Interventions in #06B6D4

## C - Circulation



IV access, hemorrhage control, hemodynamics.

Interventions in #06B6D4  
Trauma in #E11D48

## D - Disability



GCS, pupil reactivity, motor function.

Assessments in #D97706

## E - Exposure



Full body assessment while preventing hypothermia.

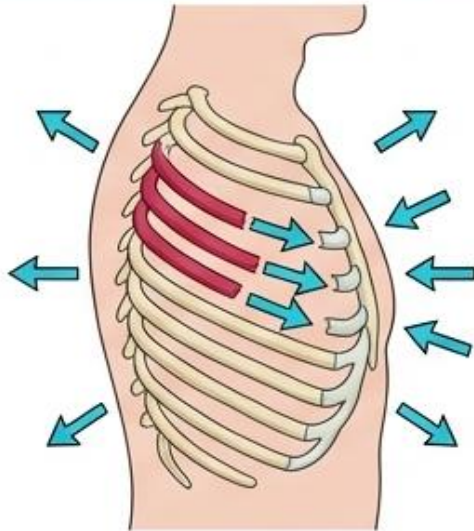
Assessments in #D97706

**Key Insight:** Do not progress to the next step until current life threats are managed.

# Chest Trauma: Flail Chest

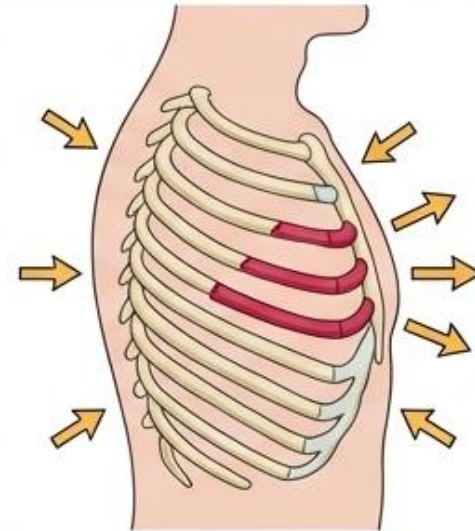
Definition: 2 or more adjacent ribs fractured in multiple places, creating a free-floating segment.

## Inspiration



Flail section moves paradoxically (inward)

## Expiration



Flail section moves paradoxically (outward)

## Pathophysiological Consequences



Paradoxical reversal reduces tidal volume



Hypoventilation



Hypoxia ( $\downarrow O_2$ ) & Hypercarbia ( $\uparrow CO_2$ )



# Flail Chest

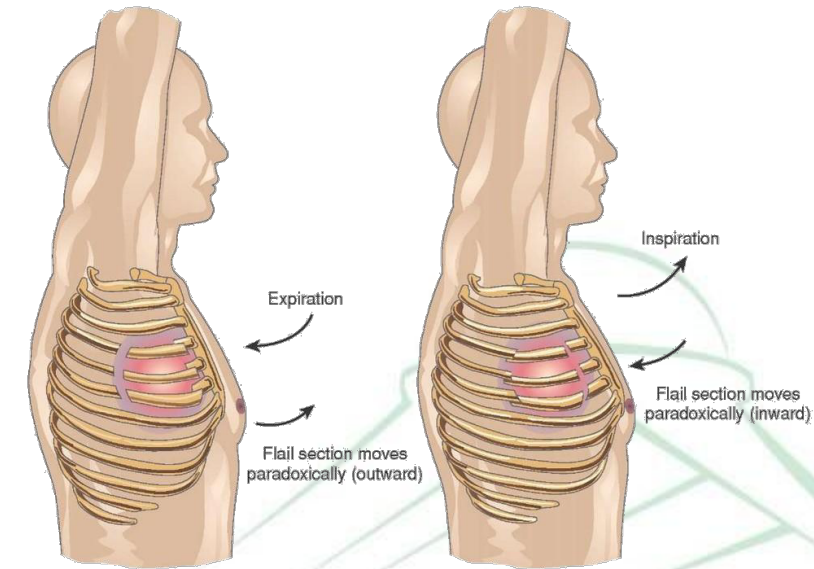
## Definition & Pathophysiology

**Flail chest** occurs when 2 or more adjacent ribs are fractured in multiple places, creating a free-floating ("flail") segment of the chest wall. The segment loses its bony connection to the rest of the thorax and moves independently paradoxically during respiration.

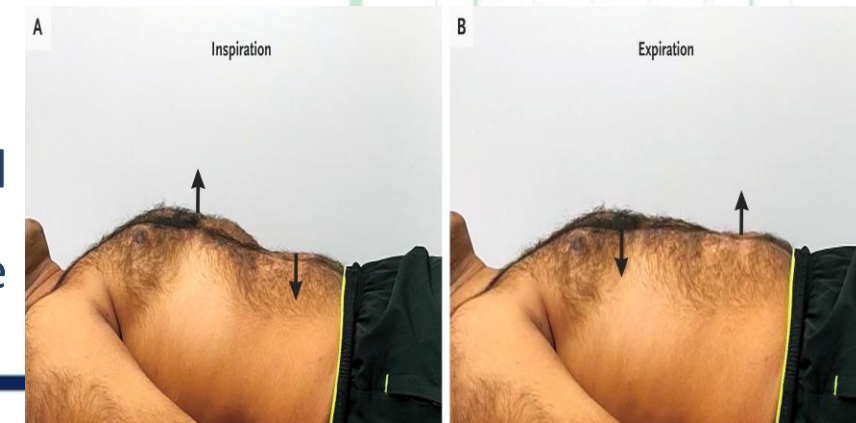
### Paradoxical Movement

**During INSPIRATION:** the diaphragm contracts and the chest wall expands outward, but the flail segment is sucked INWARD by the negative pressure.

**During EXPIRATION:** as the chest wall recoils inward, the flail segment is pushed OUTWARD. This reversal impairs effective ventilation.



*Paradoxical movement diagram*





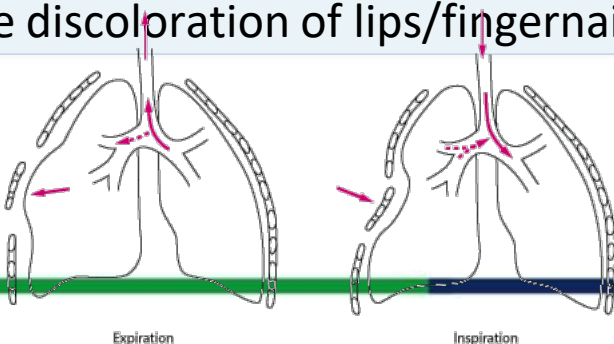
# Flail Chest – signs and symptoms & Diagnosis

## Respiratory Distress

- Dyspnea (shortness of breath) — rapidly progressive
- Rapid, shallow breathing (tachypnea  $> 20/\text{min}$ )
- Visible paradoxical wall movement on inspection
- Use of accessory muscles

## Pain & Cardiovascular Signs

- Severe localized chest pain — worsens with breathing
- Tachycardia (elevated heart rate)
- Hypotension if cardiac output is compromised
- Cyanosis: blue discoloration of lips/fingernails



## Diagnostic Procedures

### Physical Exam

Inspection: visible paradox, palpation: crepitus

### Chest X-Ray

Visualizes rib fractures; evaluate lung fields

### CT Thorax

Gold standard shows exact fracture location & lung contusion

### ABG Analysis

Confirms hypoxia ( $\downarrow \text{PaO}_2$ ) and hypercapnia ( $\uparrow \text{PaCO}_2$ )

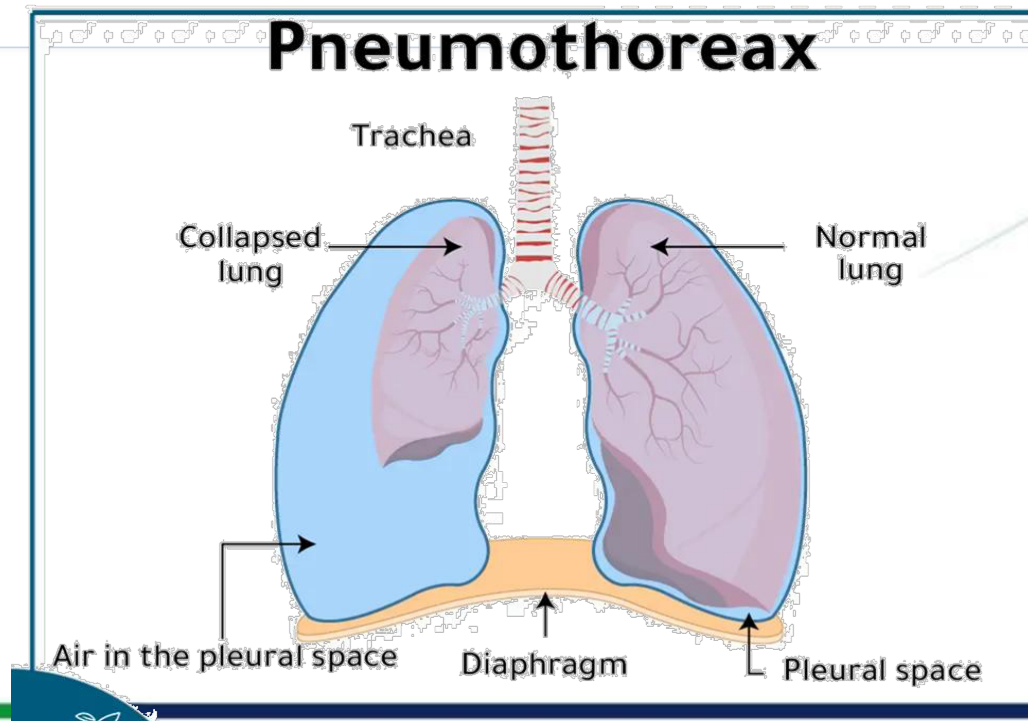
### ECG

Rules out associated cardiac contusion



## Pneumothorax – Definition & Pathophysiology

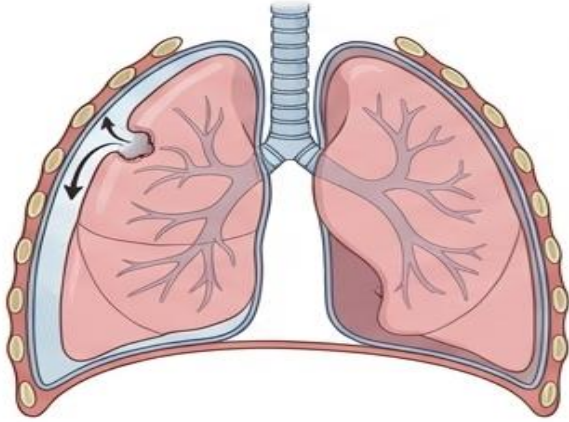
**Pneumothorax** is the accumulation of air in the pleural space. Normally, pleural pressure is sub-atmospheric (negative), keeping the lungs expanded. When air enters this space, pressure rises, compressing the lung → atelectasis (collapse).



# Type of Pneumothorax

## Chest Trauma: The Pneumothorax Spectrum

### Spontaneous

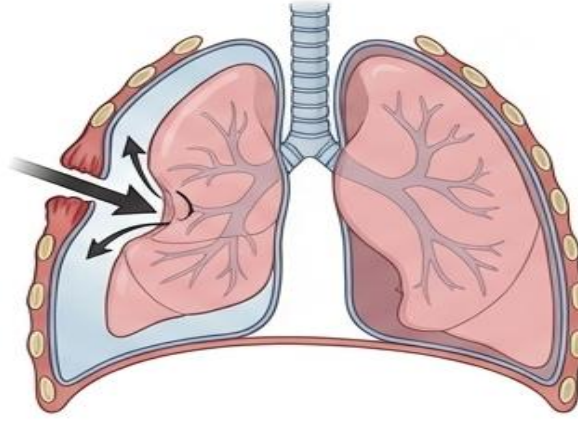


**Trigger:** Often occurs at rest in tall, thin males.

**Sign:** Mild to severe dyspnea.

Occurs without trauma. Caused by rupture of pulmonary blebs weakened, out-pouching alveolar clusters on the lung surface.

### Open

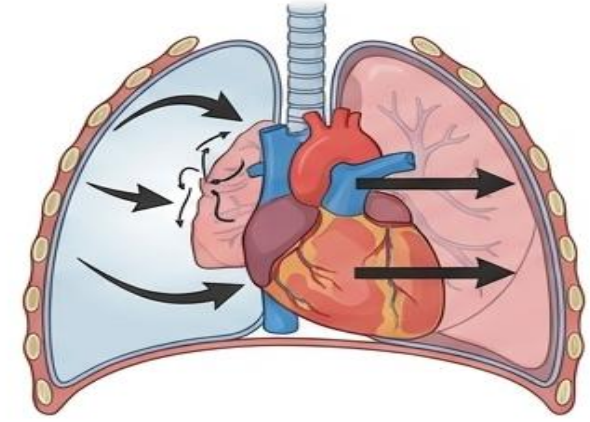


**Trigger:** Penetrating chest trauma.

**Intervention:** 3-sided occlusive dressing to prevent tension build-up.

Enters during inspiration (producing a "sucking" sound). Equalization of pleural and atmospheric pressure causes lung collapse.

### Tension (EMERGENCY)



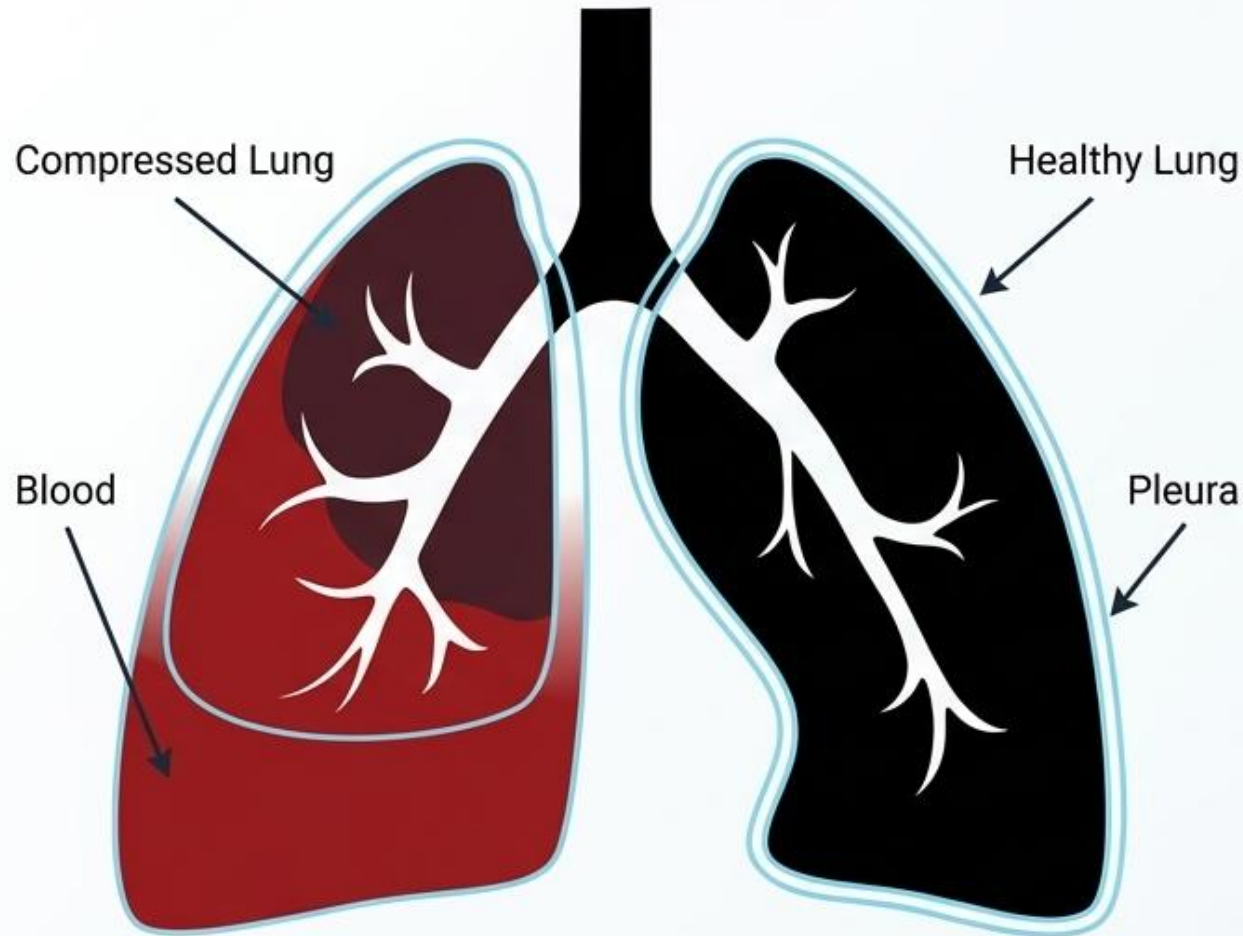
**Signs:** Tracheal deviation, JVD, hypotension.

**Intervention:** Immediate 14G needle decompression at 2nd ICS, Midclavicular line.

Air leaks into the pleural space through a one-way valve from a torn or ruptured lung.

# Chest Trauma: Massive Hemothorax

Definition: Accumulation of >1,500 mL of blood in the pleural cavity.



## The Respiratory Threat

Blood occupies pleural space → lung compression → severe hypoxia



## The Hemodynamic Threat

Massive internal blood loss → hypovolemia → hypovolemic shock



## Emergency Surgical Trigger

Emergency thoracotomy required if initial chest tube output is >1500 mL or ongoing drainage is >200 mL/hr.

# Synthesis Matrix: Differentiating Pleural Space Trauma

Crucial diagnostic differences between air and blood accumulation.

|                      | Air (Pneumothorax)                          | Blood<br>(Massive Hemothorax)                                                |
|----------------------|---------------------------------------------|------------------------------------------------------------------------------|
| Substance            | Air in pleural space                        | Blood in pleural space                                                       |
| Percussion           | <b>HYPER-RESONANT</b>                       | <b>DULL</b>                                                                  |
| Neck Veins           | Distended (JVD in tension)                  | Flat/Collapsed (hypovolemia)                                                 |
| Primary Intervention | Needle decompression /<br><b>Chest tube</b> | Large-bore Chest Tube (28-32 Fr) +<br><b>1:1:1 Massive Blood Transfusion</b> |

# Chest Trauma: Nursing Interventions

Consolidated care protocols for thoracic injuries



## Airway & O<sub>2</sub>

Apply high-flow O<sub>2</sub> (SpO<sub>2</sub> > 94%). Use PEEP during mechanical ventilation to internally stabilize flail segments and expand alveoli.



## Hemodynamics

Immediate placement of large-bore chest tube (28–32 Fr) connected to underwater-seal drainage. Record and replace all drainage. If initial output > 1500 mL or ongoing > 200 mL/hr → surgical intervention required. Provide chest tube care.



## Pain Management

IV/IM analgesia is critical. Effective pain relief directly improves respiratory effort, deep breathing, and tidal volume.

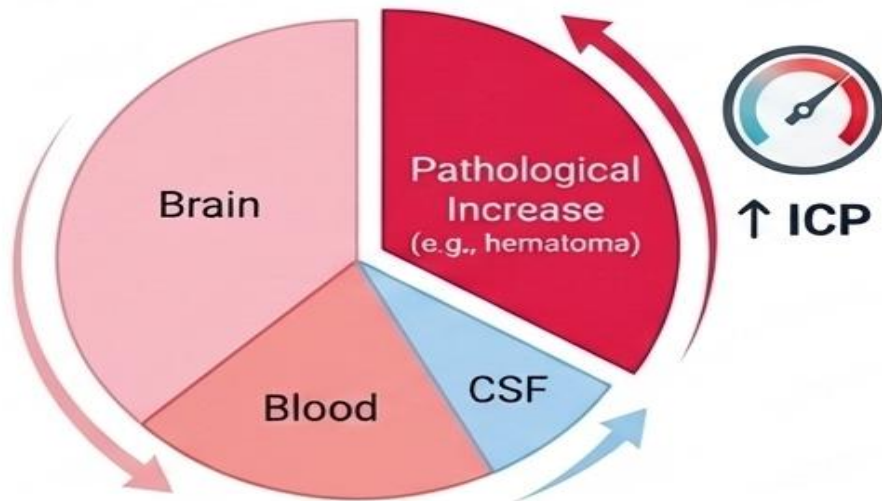


## Position & Comfort

Maintain Semi-Fowler's position (45°) to facilitate diaphragm drop and maximum lung expansion. Provide cough splinting education.

# Basal skull fracture

It is a break of a bone in the base of the skull, occurs in the floor of the skull: the areas around the eyes, ears, nose, or at the top of the neck, near the spine.



The skull is a fixed box. An increase in one component (e.g., hematoma, edema) must compress the others, spiking ICP.

## Primary Injury

Irreversible **mechanical trauma** at the moment of impact. Blunt or penetrating forces. Focus is on preventing further damage, as primary injury is irreversible.

## Secondary Injury

Develops hours later (Edema, Hypoxia, ↑ICP). This is the **direct target for nursing intervention**. Early intervention is crucial to interrupt this cascade and improve patient outcomes.



## Clinical Metric:

Normal ICP = 0–15 mmHg. Sustained >20 mmHg causes ischemia and necrosis. Sustained high ICP compromises cerebral perfusion pressure (CPP), leading to ischemia, brain herniation, and potential death.

# Head Injuries: Neurological Assessment *Sings and symptoms*

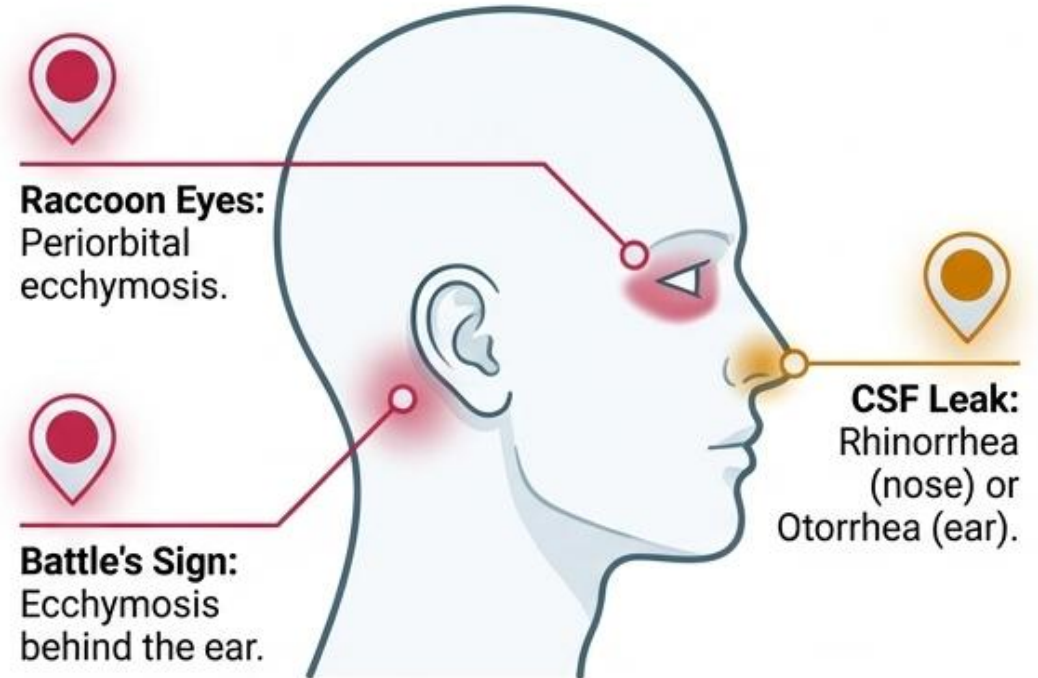
Key assessment tools and clinical indicators for head trauma.

## Glasgow Coma Scale (GCS)



Assessment combines Eye Opening, Verbal Response, and Motor Response.

## Basal Skull Fracture Red Flags



# *Nursing management*

## ICU Protocol (Preventing Secondary Injury)

- ✓ **Avoid the 3 Killers: Hypoxia, Hypotension, & Hypercapnia.** Maintain  $\text{SpO}_2 > 95\%$ ,  $\text{MAP} > 80$ ,  $\text{PaCO}_2$  35-45.
- ✓ **Positioning:** Head-of-bed 30–45° to promote cerebral venous drainage. Avoid neck flexion.
- ✓ **Medications:** Mannitol 0.25–1 g/kg IV (hyperosmolar therapy) and prophylactic anticonvulsants.

## Discharge Red Alerts



Instruct patient/family to return to ER immediately if:



**Repeated Vomiting:** Forceful or projectile vomiting (>2 episodes).

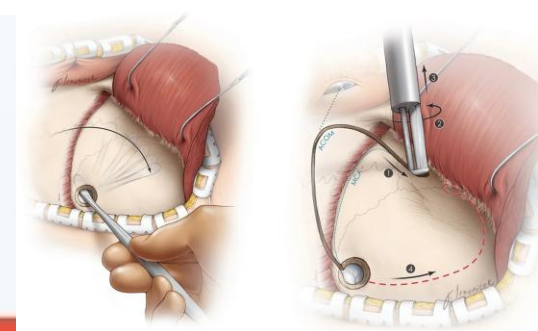


**Neurological Change:** New weakness of arms/legs, speech disturbance, or blurred vision.



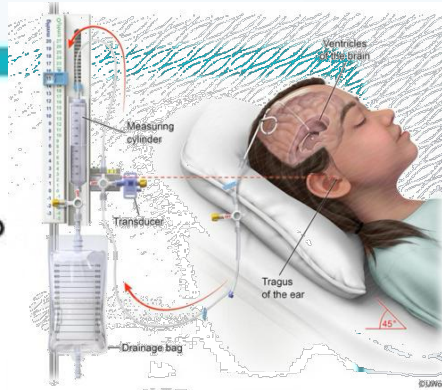
**Severe Headache:** Progressive headache totally unresponsive to paracetamol.

# Head Injuries: Continuum of Care



## Reduce ICP

- Head-of-bed elevation 30–45°
- Avoid neck flexion
- Control agitation
- Mannitol or hypertonic saline for hyperosmolar therapy
- Short-term hyperventilation only in herniation



## Surgical Management

- Craniotomy / burr holes for evacuating haematomas
- Decompressive craniectomy for refractory high ICP
- Depressed skull fractures may require elevation

## Neurological Monitoring

- Hourly GCS, pupil size & reactivity, limb power
- ICP monitoring via intraventricular catheter in severe TBI
- Continuous EEG if seizures suspected

## Pharmacological

- Prophylactic anticonvulsants (7-day phenytoin in severe TBI)
- Mannitol 0.25–1 g/kg IV
- Analgo-sedation to reduce cerebral metabolic rate
- Avoid steroids (increased mortality)

# Spinal Cord Injuries: Mechanisms & Severity

Spinal Cord Injury (SCI) is an acute insult to the spinal cord that disrupts sensory & motor communication within the CNS and between the CNS & peripheral nervous system. Consequences range from pain/weakness to complete paralysis depending on level and completeness.

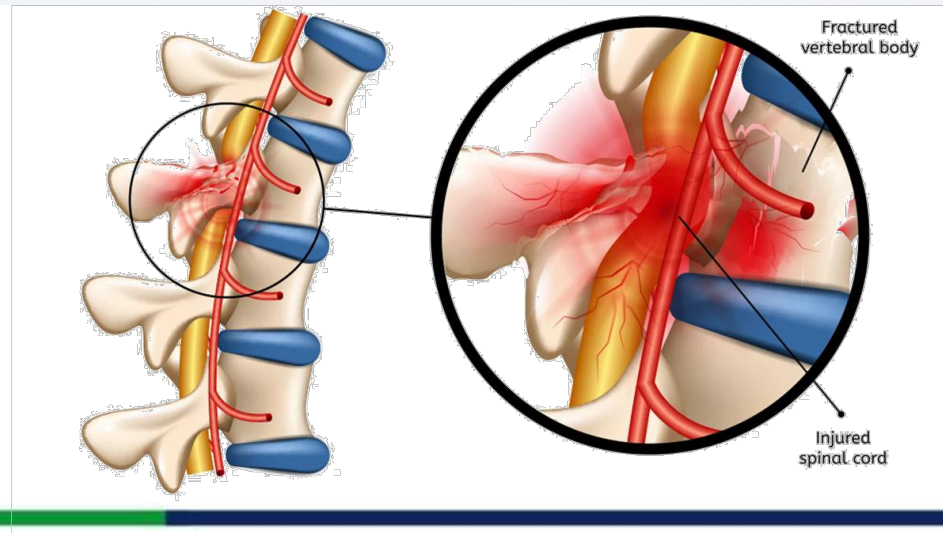
## Degree of Injury Or Type Spinal Cord Injuries

### Complete Injury

Permanent total loss of ALL motor and sensory function completely below the level of the injury.

### Incomplete Injury

Partial transection of the cord; resulting in mixed preservation of some motor and/or sensory function.



# Mechanism Spinal Cord Injuries

1. Flexion injury
2. Flexion rotation injury
3. Penetration injury
4. Hyperextension injury

## Mechanisms of Injury



**Flexion:** Forward bending force (e.g., diving).



**Flexion-Rotation:** Most unstable; dislocates facets (e.g., high-speed RTA).

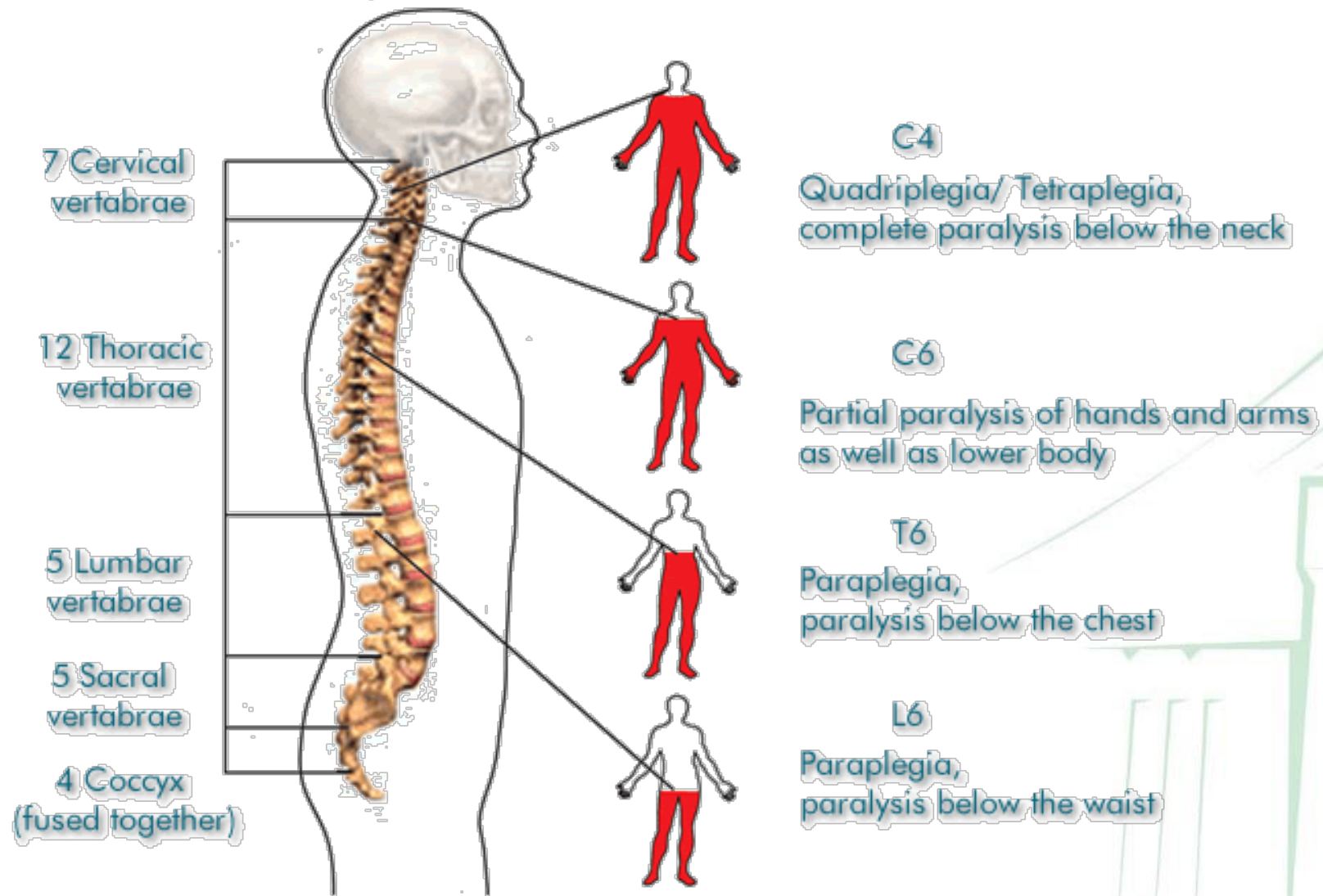


**Hyperextension:** Backward force (e.g., rear-end collision).



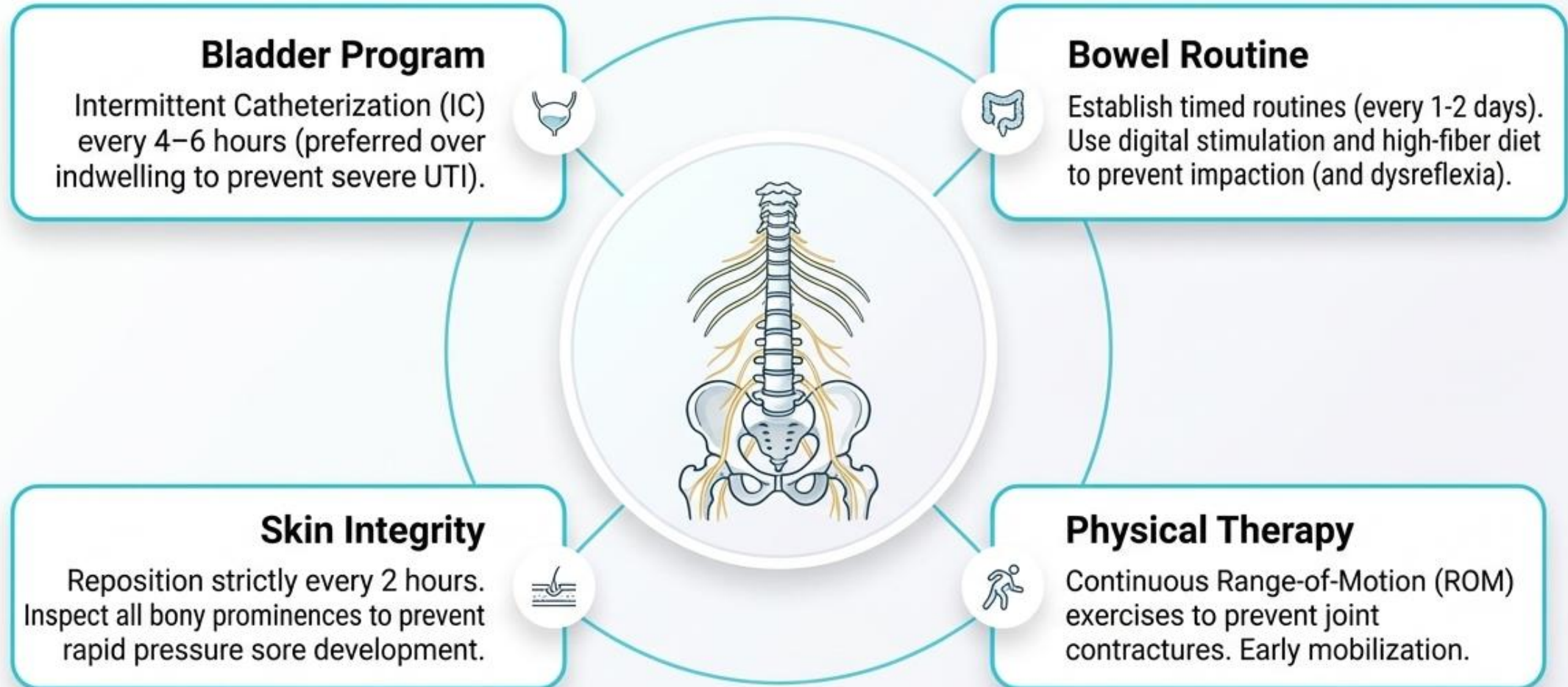
**Penetrating:** Direct laceration (e.g., gunshot).

# SCI: The spinal column



# Spinal Cord Injuries: Long-term Nursing Management

Holistic multidisciplinary care and rehabilitation.





# Comprehensive medical Management

## ❑ Medical Management

- Immobilise spine immediately  
rigid collar, logroll
- IV access + aggressive fluid  
resuscitation
- Vasopressors (norepinephrine) for  
neurogenic shock
- Maintain MAP 85–90 mmHg  
(spinal cord perfusion)
- Nasogastric tube  
(prevent aspiration in ileus)
- Urinary catheter (prevent retention)

## Pharmacological

- High-dose methylprednisolone (within  
8 hrs of injury — controversial)
- Vasopressors: norepinephrine for  
neurogenic shock
- Anticoagulation: LMWH for DVT  
prevention
- PPIs for stress ulcer prophylaxis
- Analgesics: IV morphine / paracetamol
- Anticonvulsants if neuropathic  
pain/seizures

## ❑ Surgical Management

- Decompression laminectomy
- Spinal stabilisation: rods, screws, fusion
- Penetrating injuries: fragment removal
- Timing: early surgery (< 24 hrs) →  
better outcomes
- Post-op: continued neurological  
monitoring



**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture 6**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**In- Hospital Emergency Life Support**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**

# Objectives

**By the end of this lecture the student will be able to:**

- Define emergency nursing care
- Indicate the role of emergency care nurse
- Indicate objective and categories of in –hospital triage system
- Define of triage system
- Mention roles and characteristics of triage nurse
- Enumerate objectives and categories of triage system
- Discuss triage levels
- Explain assessment of a seriously ill client
  - Primary survey
  - Secondary survey



# Outlines

- Definition emergency nursing care
- Roles of emergency care nurse
- Objectives and categories of in –hospital triage system
- Definition of triage system
- Roles and characteristics of triage nurse
- Objectives and categories of triage system
- Triage levels
- Assessment of a seriously ill client
  - Primary survey
  - Secondary survey



# Definition of Emergency Nursing Care

---

**Emergency nursing** is the specialized care for individuals of all ages experiencing acute physical or emotional health problems that require immediate intervention.



# Roles of Nurse in Emergency Department

**Triage:** Assess patients to determine the severity of their condition and prioritize them for immediate care.

**Patient assessment:** Quickly and thoroughly evaluate a patient's injuries or illness.

**Medical treatment:** Administer emergency medications, perform basic medical procedures, and provide wound care.

**Monitoring:** Closely monitor patients' vital signs and conditions, using advanced equipment as needed.

**Stabilization:** Implement life-saving interventions to stabilize patients, especially those with critical or life-threatening conditions.

**Collaboration:** Work with physicians, specialists, and other healthcare professionals

**Documentation:** Maintain accurate patient records, including medical history, treatment, and outcomes.

**Patient and family support:** Provide emotional support, education, and clear communication to patients and their families

# Definition of Triage System

The process by which patients are classified according to the type and urgency of their conditions to get the

**Right patient** to the

**Right place** at the

**Right time** with the

**Right care** provider



# Objectives of Triage System

- To specify a uniform method for initial evaluation of patients during multiple casualty incidence or disaster.
- Triage sorts patients to identify life-threatening conditions and ensure those who are most ill are seen first.
- It helps manage patient flow and ensures limited resources and personnel are used effectively and efficiently.
- Enabling prompt intervention for critical conditions, the system aims to enhance patient outcomes.
- It minimizes unnecessary waiting times for those who need immediate attention, which is crucial in emergency settings.
- Using a standardized system, it improves the overall quality of care provided in emergency departments

# Roles of Triage Nurse

---

- Maintain privacy and confidentiality
- Visualize all incoming patients even while interviewing others
- Use all resources to maintain high standard of care
- Crowd control
- Communicate with team leader and seek feed back of decisions

## Role of Triage Nurse



# Characteristics of triage nurse

- Extensive knowledge to emergency medical treatment
- Adequate training and competent skills, language, terminology
- Ability to use the critical thinker process
- Good decision maker



# Triage Categories

## 1. Non Disaster:

To provide the best care for each individual patient.

**The primary objectives of an ED triage are to:**

1. Identify patients requiring immediate care
2. Determine the appropriate area for treatment
3. Facilitate patient flow through the ED and avoid unnecessary congestion
4. Provide continual assessment and reassessment of arriving and waiting patients
5. Provide information and referrals to patients and families
6. Decrease patient and family anxiety

**N.B. Patient should wait no longer than 5 minutes for triage**

# Cont.....

## 2. Multi casualty/ disaster:

- To provide the most effective care for the greatest number of patients.
- Large or small number of patients needing services beyond immediately available resources.

# Triage Levels

---

1. Resuscitation
2. Emergent
3. Urgent
4. Less urgent
5. Non urgent

**Level I — Resuscitation**

**Level II — Emergent**

**Level III — Urgent**

**Level IV — Less Urgent**

**Level V — Non Urgent**

# 1. Resuscitation: threat to life

---

## Examples:

- Cardiac and respiratory arrest
- Major trauma
- Active seizure
- Shock
- Status asthmaticus



## 2. Emergent: Potential threat to life

### Examples:

- Decreased level of consciousness
- Severe respiratory distress
- Chest pain with cardiac suspicion
- Severe abdominal pain
- GI bleed with abnormal vital signs



### 3. Urgent: Condition with significant distress

#### Examples:

- Head injury: No LOC but with vomiting
- Mild to moderate respiratory distress
- Not actively G.I. bleed



## 4. Less urgent: Condition with mild to moderate discomfort

### Examples:

- Head injury: alert, no vomiting
- Chest pain: no distress, no cardiac suspicion



## 5. Non urgent: Condition can be delayed, no distress

### Examples:

- Minor trauma
- Sore throat with temperature  $<39$



# Triage level

|   |                                                         |
|---|---------------------------------------------------------|
| 1 | Level 1 – Immediate: life threatening                   |
| 2 | Level 2 – Emergency: could become life threatening      |
| 3 | Level 3 – Urgent: not life threatening                  |
| 4 | Level 4 – Semi Urgent: not life threatening             |
| 5 | Level 5 – Non-urgent: needs treatment when time permits |

# Triage Levels

|                  |                      |                                |
|------------------|----------------------|--------------------------------|
| <b>Level I</b>   | <b>Resuscitation</b> | <b>see patient immediately</b> |
| <b>Level II</b>  | <b>Emergency</b>     | <b>within 15 minutes</b>       |
| <b>Level III</b> | <b>Urgency</b>       | <b>within 30 minutes</b>       |
| <b>Level IV</b>  | <b>Less Urgency</b>  | <b>within 60 minutes</b>       |
| <b>Level V</b>   | <b>Non Urgency</b>   | <b>within 120 minutes</b>      |

# Assessment of a seriously ill client

- **Primary survey**
- **Secondary survey**



# Primary survey

- Subjective data

**SAMPLE**

- Objective data

**ACBDE**



# Subjective data. (SAMPLE )

**S**ymptoms

**A**llergies

**M**edication currently prescribed or using

**P**ast medical and surgical history

**L**ast meal

**E**vents preceding the emergency and care rendered

# Objective data (ACBD )

Airway

Circulation

Breathing

Disability



## - Secondary survey

### Subjective data:

1. History of present illness or injury (PQRST)

Precipitating factors

Quality of symptoms

Region/ radiation of pain

Severity of the symptoms

Time of onset

2. Mechanism of injury
3. Past medical history
4. Social & psychological history



# Cont.....

## Secondary survey

### Objective data

- Patient's general appearance
- Rapid head to toe examination
- Neurological examination



# Component of resuscitation phase

**1rst step:** Airway +cervical spine

**2nd step:** Circulation

**3rd step:** look, listen & feel Breathing

**4rth step:** Disability (AVPU)

**5th step:** Exposure

# Assessment neurological status (AVPU )

**A**lert

**V**erbal stimuli

**P**ainful stimuli

**U**nresponsive



# Question 1

1. which of the following conditions is considered the first element of the secondary assessment

A. Neurologic assessment

**B. Taking history of present illness**

C. Physical examination

D. Cardiac assessment



# Question 2

**2. which category should a patient with severe respiratory distress and cardiac arrest be placed?**

**A.Immediate**

B.Minor

C.Delayed

D.Dead / deceased



## Question 3

3. What is the time allocated for the emergent triage level?

- A. 15 minutes
- B. 30 minutes
- C. 60 minutes
- D. 120 minutes





**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture 7**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**Cardiac Emergencies**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**

# Objectives

**By the end of this lecture the student will be able to:**

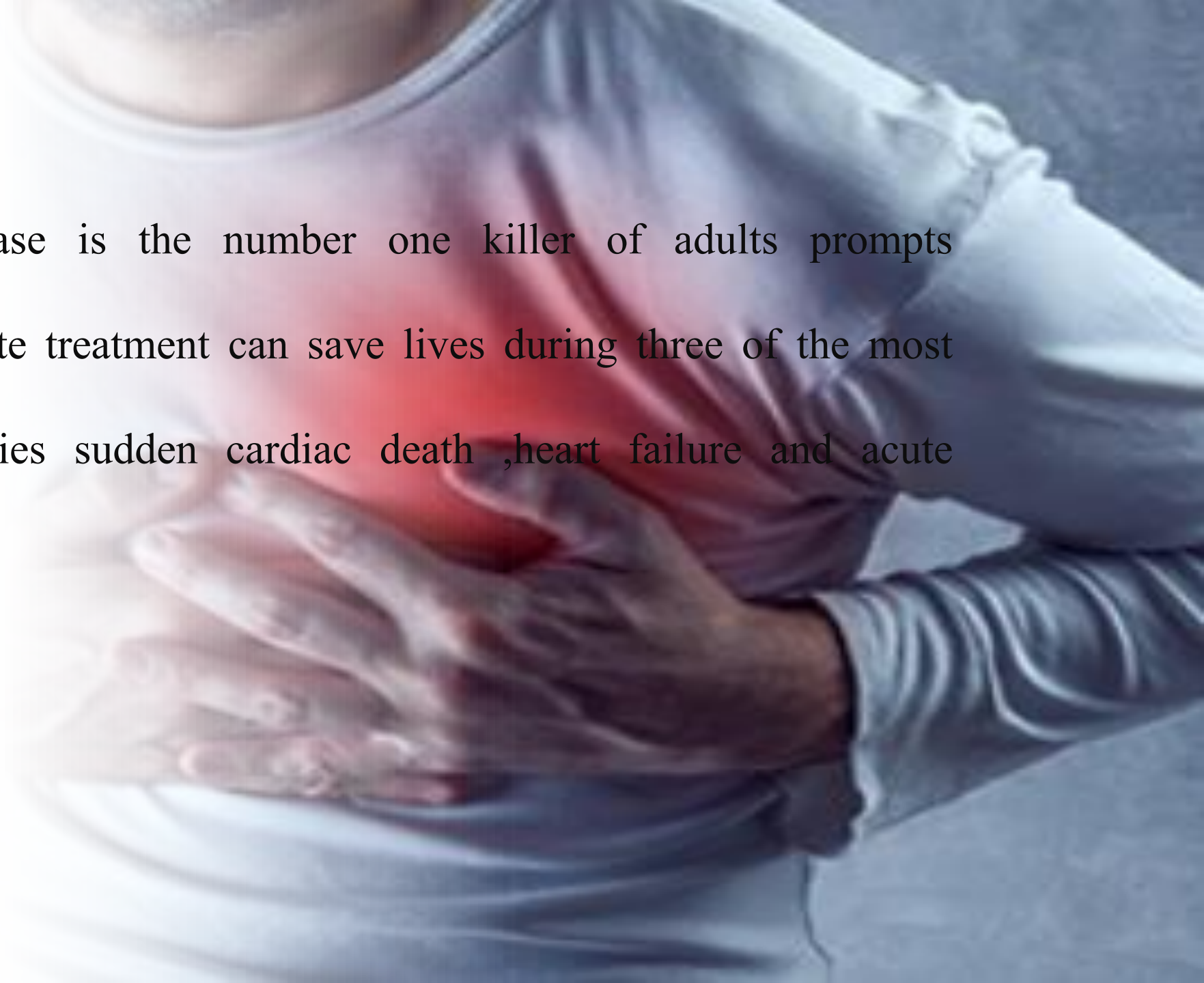
- ❑ Enumerate types of cardiac emergencies
- ❑ Mention definition, pathophysiology, risk factors, signs and symptoms, medical and surgical management and nursing management of angina pectoris, myocardial infarction, congestive heart failure ,cardiac tamponade and Cardiac arrest

# Outlines

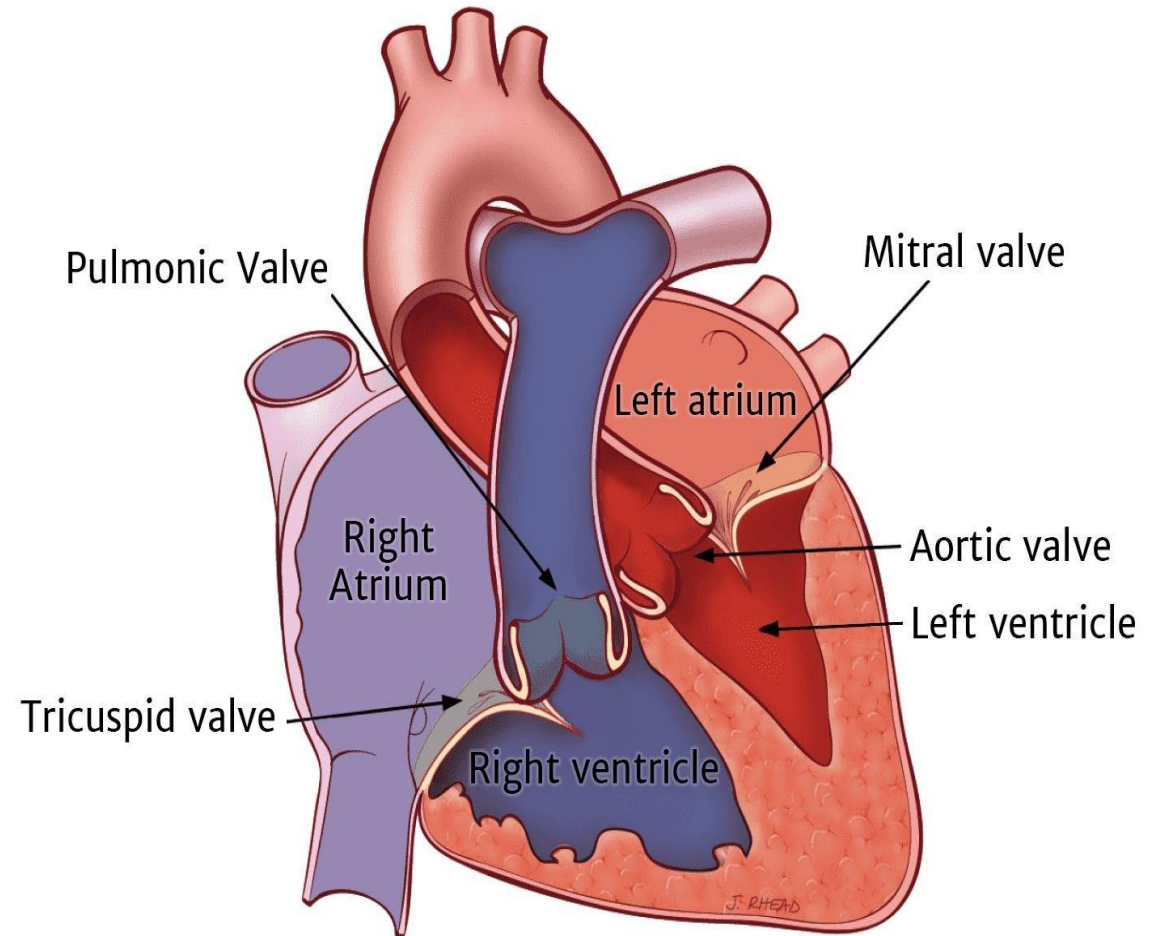
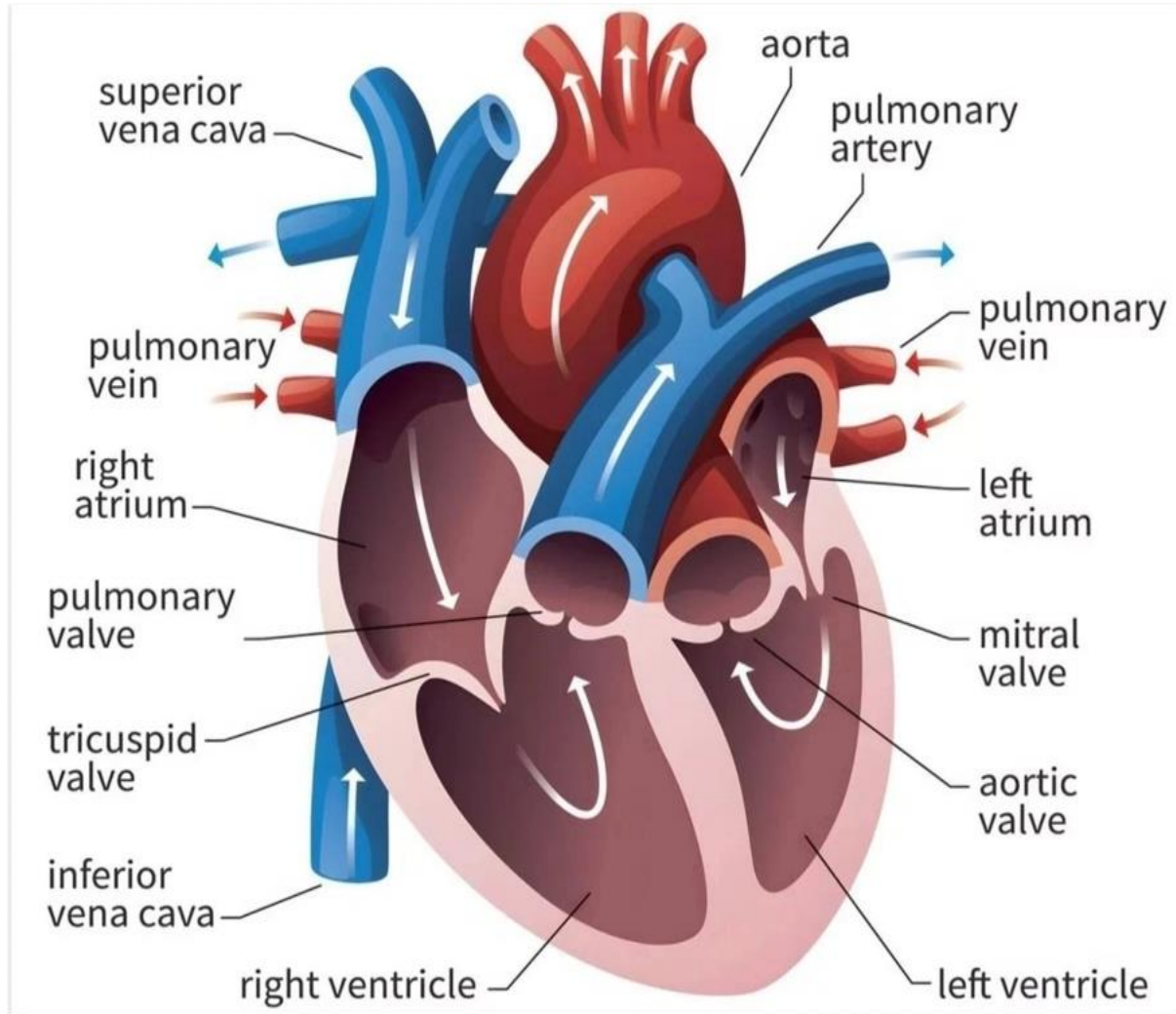
- ❑ Introduction
- ❑ Anatomy and physiology of the heart
- ❑ Types of cardiac emergencies
- ❑ Angina pectoris, myocardial infarction, congestive heart failure ,cardiac tamponade and cardiac arrest
  - Definition
  - Pathophysiology
  - Risk factors
  - Signs and symptoms
  - Medical and surgical management
  - Nursing management

# Introduction

Cardiac emergency disease is the number one killer of adults prompts recognition and appropriate treatment can save lives during three of the most deadly cardiac emergencies sudden cardiac death ,heart failure and acute pericarditis



# Anatomy and physiology of the heart



# Definition of cardiac emergency

**Cardiac emergencies** are life threatening conditions that must be recognized immediately to avoid delay in treatment and minimize morbidity and mortality.



# Types of cardiac emergencies

- **Angina pectoris**
- **Myocardial infarction**
- **Congestive heart failure**
- **Cardiac tamponade**
- **Cardiogenic shock**
- **Cardiac arrest**



# 1- Angina pectoris

- Is a clinical syndrome usually characterized by episodes of pain or pressure in the anterior chest.
- The cause is insufficient coronary blood flow resulting in a decreased oxygen supply where there is increased myocardial demand for oxygen.

# Pathophysiology

- Angina is usually caused by atherosclerotic disease.
- Angina is associated with significant obstruction of at least one major coronary artery.
- Oxygen demands not met. normally, the myocardium extracts a large amount of oxygen from the coronary circulation to meet its continuous demands.
- Increased demand. when there is an increase in demand, flow through the coronary arteries needs to be increased.
- Ischemia. when there is blockage in a coronary artery, flow cannot be increased, and ischemia results which may lead to necrosis or myocardial infarction.

# Types of angina

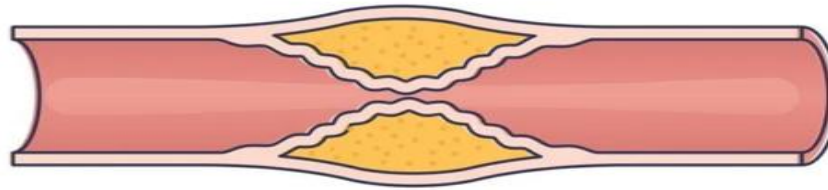
## TYPES OF ANGINA

**NORMAL**



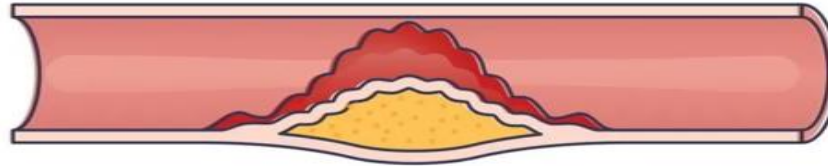
**Normal Coronary Artery**

**STABLE ANGINA**



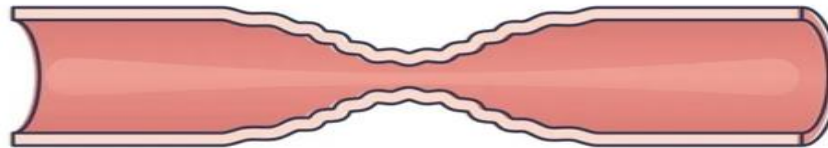
**Atherosclerosis**

**UNSTABLE ANGINA**



**Atherosclerosis with Blood Clot**

**VARIANT ANGINA**



**Coronary Spasm**

# Risk factors of angina

Increasing age.

Family history of heart disease.

Tobacco use.

Diabetes.

High blood pressure..

High cholesterol or triglycerides.

Not enough exercise.

Obesity.

Emotional stress. .

Drug misuse. Cocaine and other stimulants can cause blood vessel spasms and trigger angina.

# Signs and symptoms of angina

❑ Angina symptoms include chest pain and discomfort. The chest pain or discomfort may feel like:

- Burning.
- Fullness.
- Pressure.
- Squeezing.

Pain also may be felt in the arms, neck, jaw, shoulder or back.

❑ Other symptoms of angina include:

- Dizziness.
- Fatigue.
- Nausea.
- Shortness of breath.
- Sweating.

# Medical and surgical management

- Nitroglycerin
- Calcium channel blockers
- Anticoagulant
- Beta blockers



# Surgical management

- **Coronary Artery Bypass Grafting (CABG):** An open-heart procedure using a vein or artery from another body part to bypass a blocked heart artery.
- **Percutaneous Coronary Intervention (PCI):** Often called stenting or angioplasty, this procedure uses a balloon-tipped catheter to widen a blocked artery.

# Nursing management

## Immediate Nursing Interventions (Acute Episode)

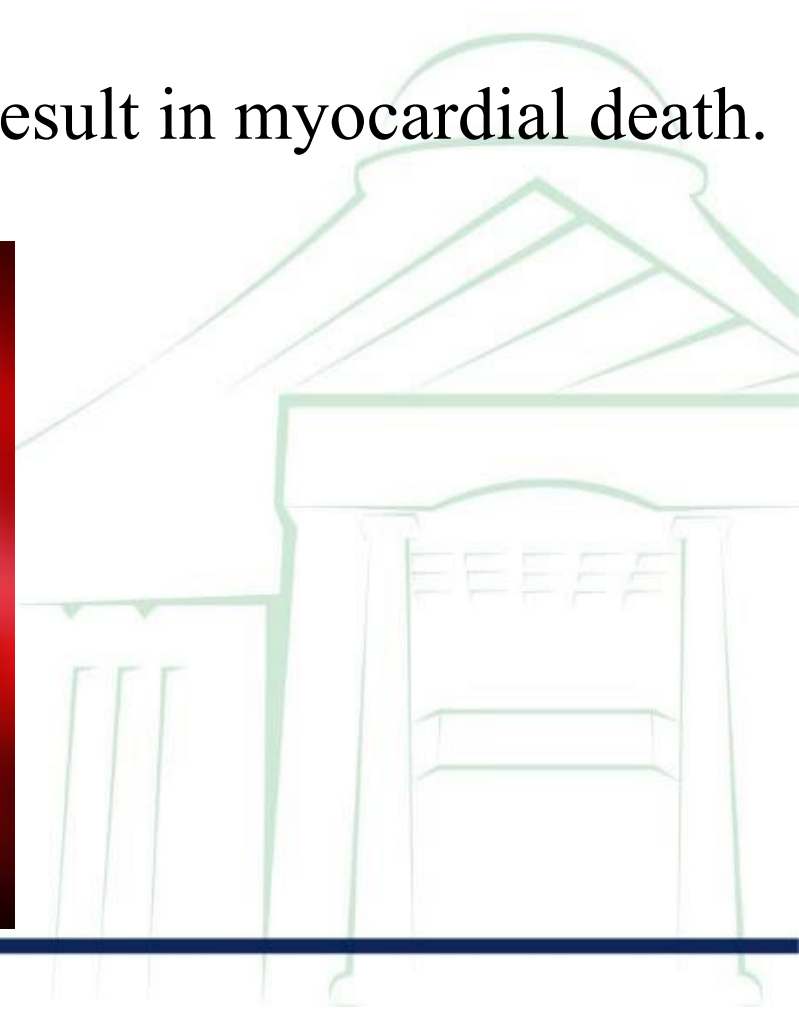
- **Stop Activity/Rest:** Immediately stop physical activity and place the patient in a semi-Fowler's position to reduce cardiac workload.
- **Assess Pain:** Evaluate the pain's location, quality, duration, and intensity using a 0–10 scale.
- **Oxygen Therapy:** Administer oxygen if the patient has respiratory distress or if oxygen saturation (Spa2 ) is less than 90%–94%.
- **Vital Signs/ECG:** Monitor blood pressure, heart rate, and obtain an ECG to monitor for ST-segment changes.

## Medication Administration:

- **Nitroglycerin:** Administer sublingual nitroglycerin to cause vasodilation, often every 5 minutes for up to three doses.
- **Aspirin:** Administer as ordered to reduce blood clotting.
- **Beta-blockers/Calcium Channel Blockers:** To reduce heart rate and contractility.

## 2- Myocardial Infarction (MI)

- Also known as acute coronary syndrome.
- Is an emergent situation characterized by ischemia result in myocardial death.



# Pathophysiology

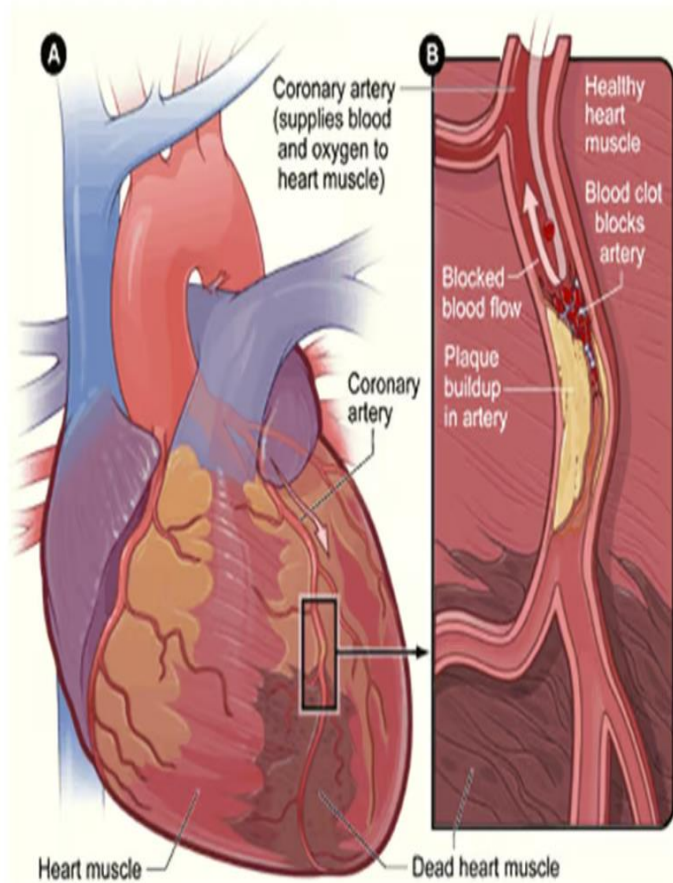


Image via: nhbl.nih.gov  
nurseslabs.com  
FOR ALL YOUR NURSING NEEDS

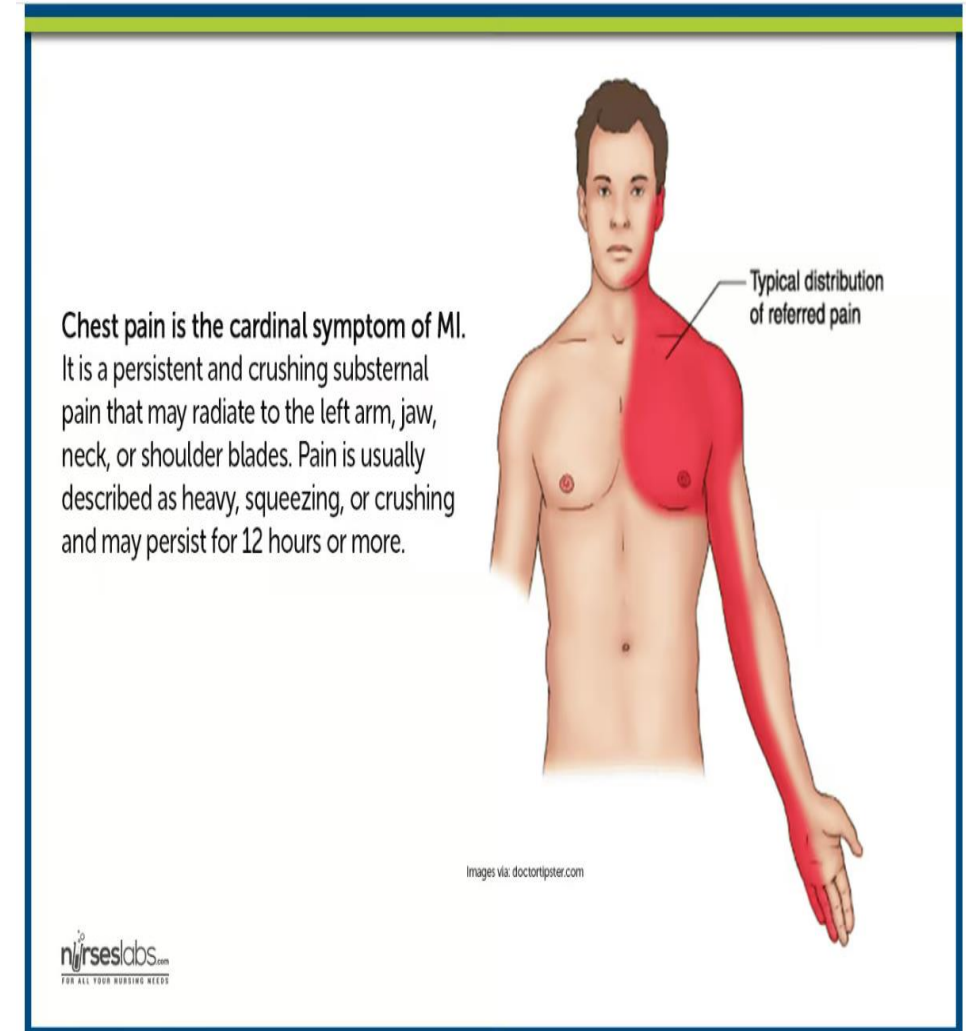
In each case of MI, a profound imbalance exists between myocardial oxygen supply and demand.

**Unstable angina.** There is reduced blood flow in a coronary artery, often due to rupture of an atherosclerotic plaque, but the artery is not completely occluded.

**Development of infarction.** As the cells are deprived of oxygen, ischemia develops, cellular injury occurs, and lack of oxygen leads to infarction or death of the cells.

# Signs and symptoms of myocardial Infarction

- **Chest pain.** Persistent and crushing substernal pain that may radiate to the left arm, jaw, neck, or shoulder blades. Pain is usually described as heavy, squeezing, or crushing and may persist for 12 hours or more.
- **Shortness of breath.**
- **Indigestion.**
- **Tachycardia and tachypnea.**



# Protocol approach of myocardial Infarction

- Immediate general treatment: (MONA)
- **M**orphine IV( if pain not relieved by nitroglycerine)
- O**xygen 4L/min.
- N**itroglycerine
- A**spirin 160 to 325 mg

# Surgical management

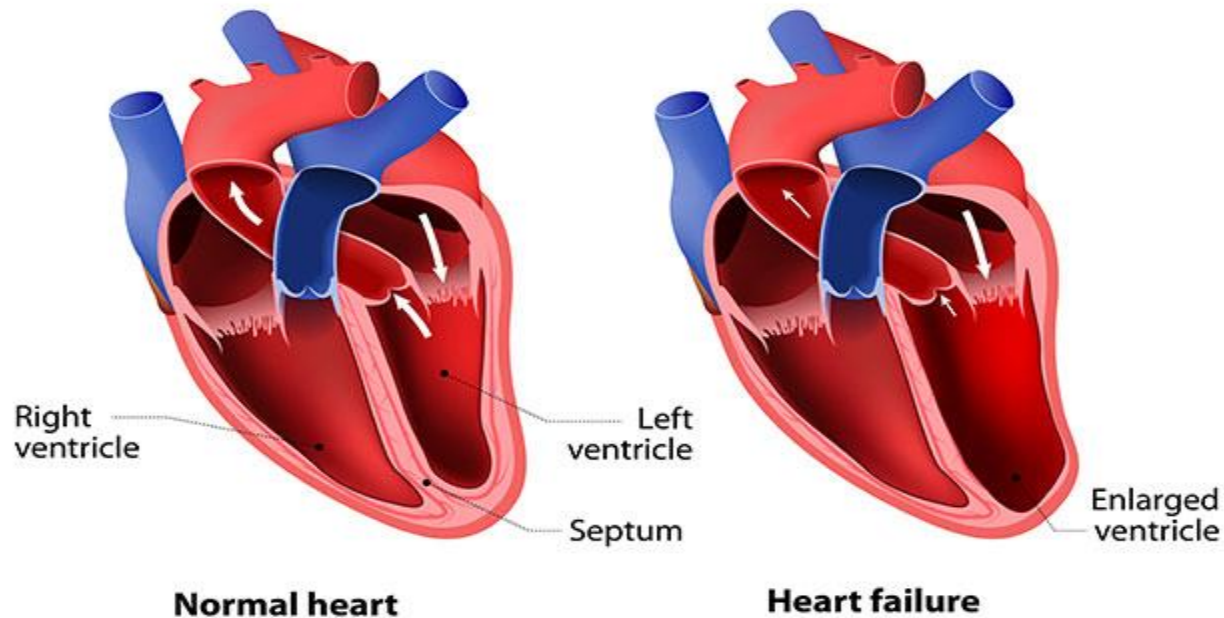
- **Coronary Artery Bypass Grafting (CABG):** An open-heart procedure using a vein or artery from another body part to bypass a blocked heart artery.
- **Percutaneous Coronary Intervention (PCI):** Often called stenting or angioplasty, this procedure uses a balloon-tipped catheter to widen a blocked artery.

# Nursing management

- Assess for chest pain not relieved by rest or medications.
- Monitor vital signs, especially the blood pressure and pulse rate.
- Assess for presence of shortness of breath, dyspnea, tachypnea, and crackles.
- Assess for nausea and vomiting.
- Assess for decreased urinary output.
- Perform a precise and complete physical assessment to detect complications and changes in the patient's status.
- Assess IV sites frequently.

### 3- Congestive heart failure (CHF)

Is inability of the heart to pump sufficient blood to meet the tissue for oxygen and nutrients in the past it is known as congenital heart diseases



# Pathophysiology of congestive heart failure

---



As Heart failure develops, the body activates neurohormonal compensatory mechanisms.



The sympathetic nervous system is then stimulated to release epinephrine and norepinephrine.



Decrease in renal perfusion causes renin release, and then promotes the formation of **angiotensin I**.

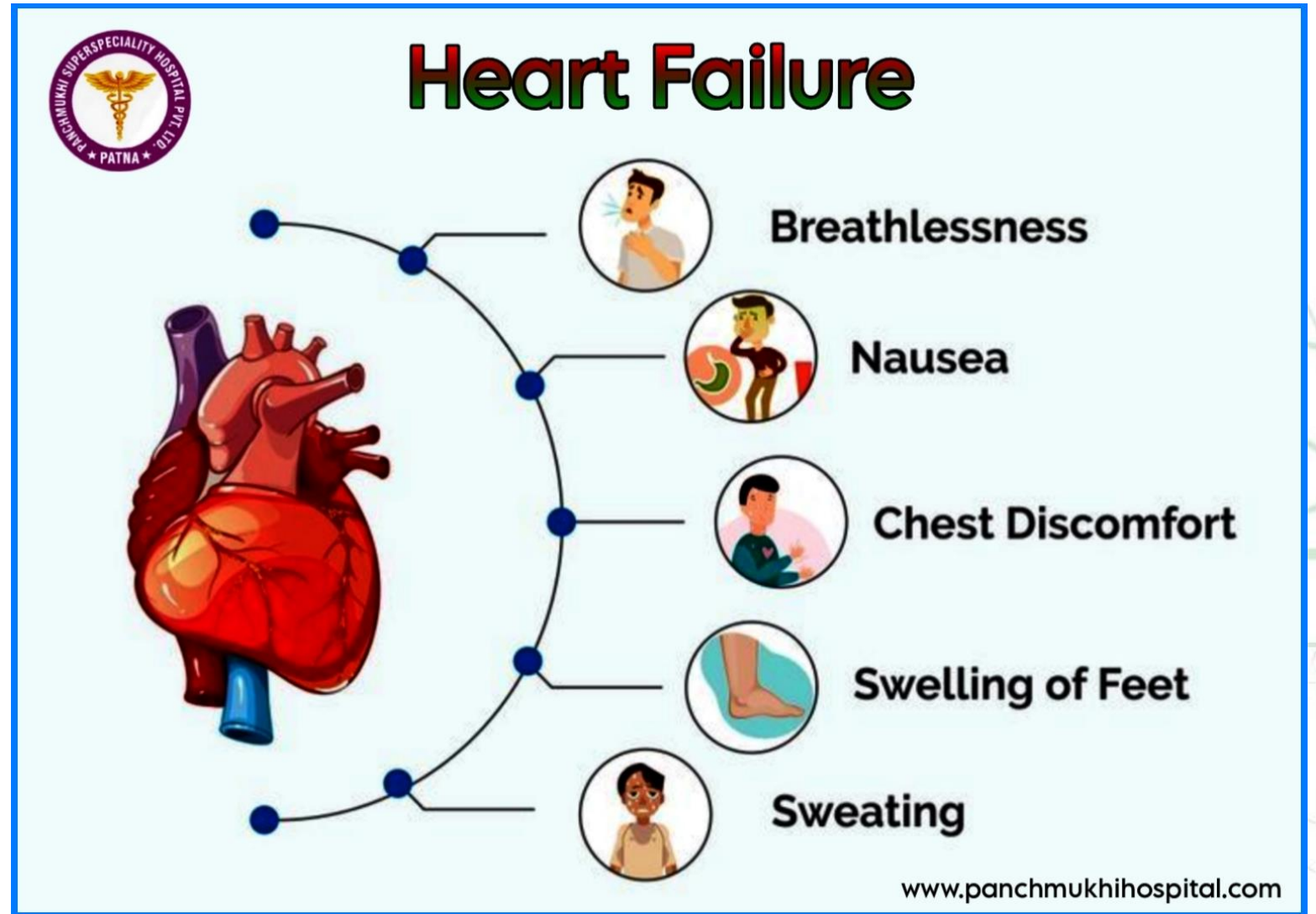


**Angiotensin I** is converted to **angiotensin II** by **Angiotensin-Converting Enzyme (ACE)** which constricts the blood vessels and stimulates aldosterone release that causes sodium and fluid retention.



There is a reduction in the contractility of the muscle fibers of the heart as the workload increases.

# Clinical manifestation of congestive heart failure



# Medical Management

- Calcium channel blockers
- Beta blockers
- Angiotensin receptors blockers
- Angiotensin-Converting Enzyme (ACE)



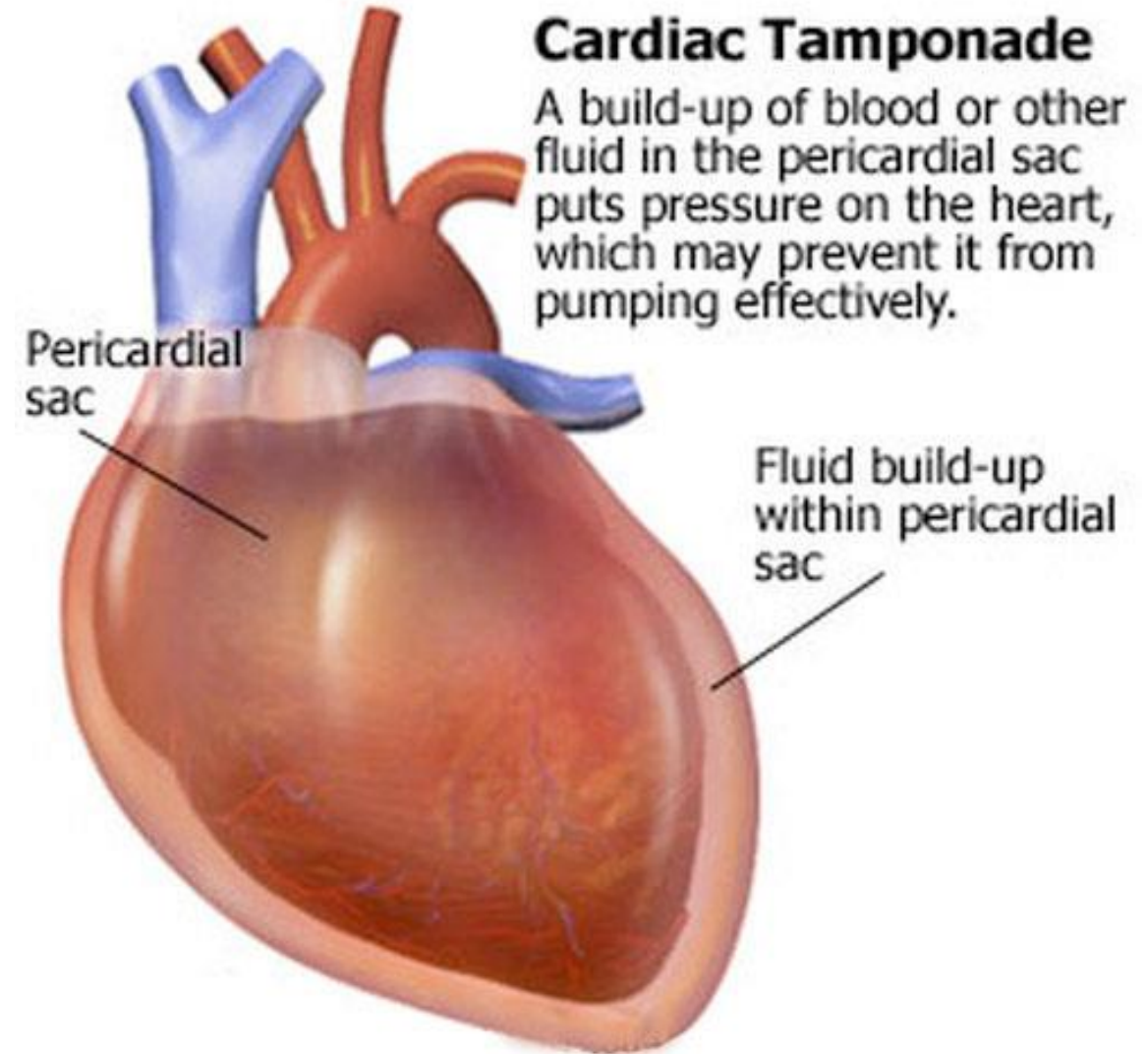
# Nursing Management

- Improve myocardial contractility and perfusion.
- **Manage fluid volume:** Monitor fluid balance
- **Promote activity tolerance:** Encourage 30 minutes of daily physical activity (as tolerated).
- Reduce anxiety. Provide comfort, psychological support.
- Provide disease information and prevention education.
- Educate patients about heart failure, its impact, prognosis, lifestyle modifications, medication adherence, and seeking timely care to prevent worsening of symptoms.

## 4- Cardiac Tamponade

---

Pericardial effusion (accumulation of excess in the pericardial sac may accompany advanced heart failure pericarditis cardiac surgery or trauma.



# **PATHOPHYSIOLOGY**

**Due to etiological factors**



**Accumulation of Fluid within Pericardium**



**pressure within the heart increases**



**Impairment in contraction and relaxation**

# Signs and symptoms of Cardiac Tamponade

**Sharp pain in the chest**

**Trouble breathing or breathing rapidly.**

**Fainting , dizziness or lightheadedness**

**Changes in skin color, especially pale, gray or blue-tinted skin.**

**Heart palpitations.**

**Fast pulse.**

# Surgical management

## To perform pericardiocentesis

Is an emergency or diagnostic procedure used to drain accumulated fluid from the pericardial sac (the membrane surrounding the heart) using a needle and catheter. It is primarily performed to treat cardiac tamponade, a life-threatening condition where fluid buildup restricts the heart's ability to pump.

## To perform pericardiotomy

Is a surgical incision into the pericardium the sac surrounding the heart. It is performed to drain excess fluid (pericardial effusion), alleviate pressure on the heart (cardiac tamponade)

# Nursing management

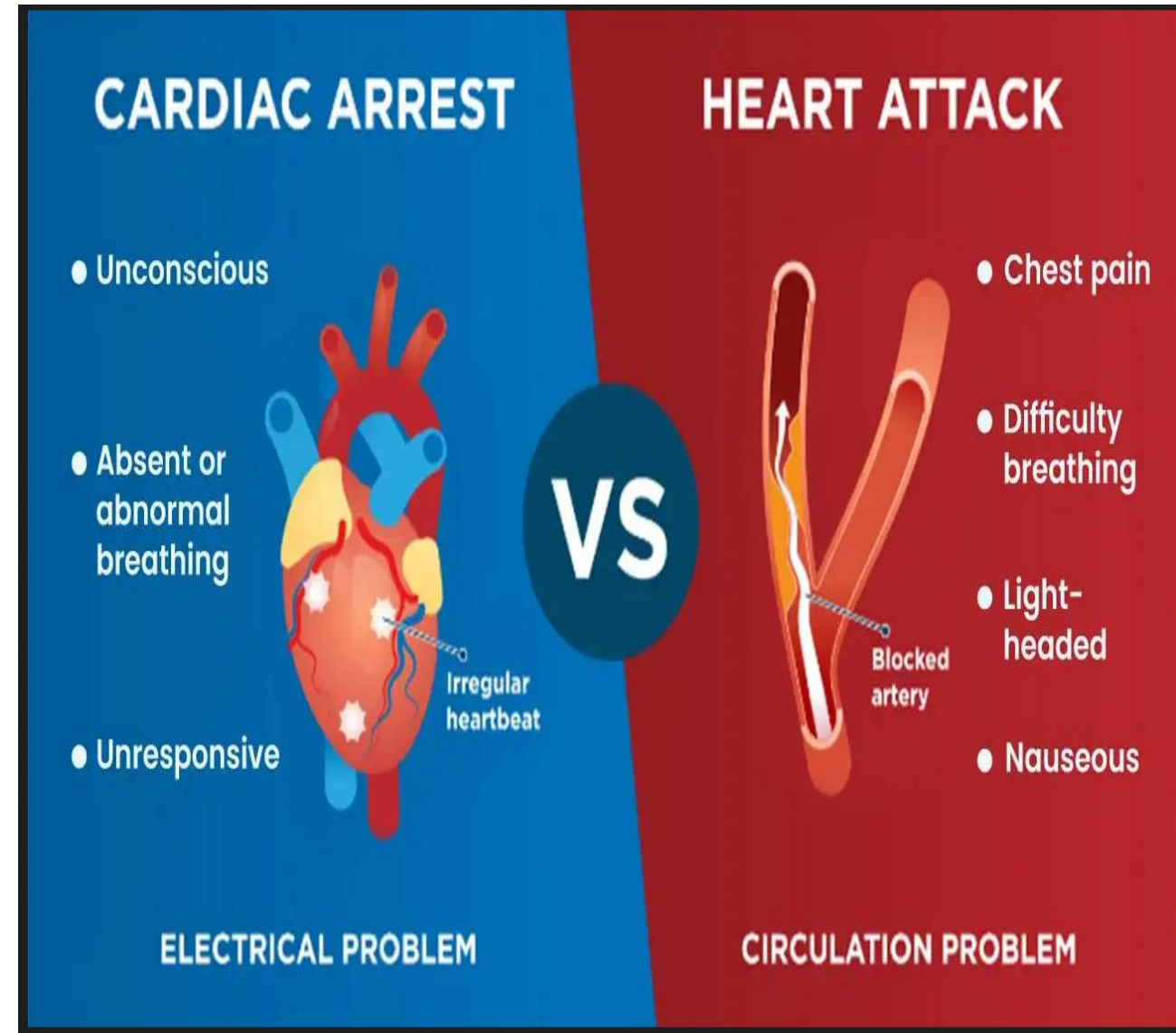
- **Monitor Vitals:** Check BP, heart rate, and rhythm every 5–15 minutes during the acute phase, looking for hypotension and tachycardia.
- **Assess Beck's Triad:** Monitor for Distant/muffled heart sounds, Jugular Venous Distension (JVD), and hypotension.
- **Electrocardiogram (ECG):** Monitor for electrical alternans (varying QRS amplitude).
- Cardiac Output Assessment:
- **Oxygen Therapy:** Administer high-flow oxygen (e.g., non-rebreather mask) as ordered to
- **Positioning:** Place the patient in high-Fowler's position to reduce venous return and relieve preload, unless they are severely hypotensive.
- **Pain/Anxiety Management:** Administer anxiety or pain meds cautiously, as opioids can exacerbate hypotension.

## 5- Cardiac arrest

---

Cardiac arrest is the sudden, unexpected cessation of effective heart function, causing immediate loss of consciousness, pulse, and breathing.

It occurs when the heart's electrical system fails, preventing blood from circulating to the brain and organs.



# Pathophysiology of Cardiac arrest

- Cardiac arrest causes global ischemia with consequences at the cellular level that adversely affect organ function even after resuscitation and restoration of perfusion.
- The main consequences involve direct cellular damage and edema formation.
- Edema is particularly harmful in the brain, which has minimal room to expand, and often results in increased intracranial pressure and corresponding decreased cerebral perfusion post resuscitation.

# Risk factors of cardiac arrest

- A family history of coronary artery disease.
- Smoking.
- High blood pressure.
- High blood cholesterol.
- Obesity.
- Diabetes.
- An inactive lifestyle.



# Signs and symptoms of cardiac arrest

## Signs and Symptoms of Sudden Cardiac Arrest



**Loss of Consciousness**



**Abnormal Pulse or Breathing**



**Chest Pain or Discomfort**



**Abnormal Heart Rhythms**



**Shortness of Breath**



**Fatigue and Weakness**



**Dizziness or Light-headedness**



**Nausea and Vomiting**

# Medical management

- CPR
- Calcium channel blockers
- Vasopressors



# Nursing management

- Assess vital signs
- Perform CPR
- Perform oxygen Therapy
- Assess patient medications



# Question 1

1- A patient presents with hypotension, distended neck veins, and muffled heart sounds.

**What is the priority intervention?**

- A. Administer diuretics
- B. Prepare for pericardiocentesis
- C. Give anticoagulants
- D. Encourage deep breathing exercises



## Question 2

2. A 58-year-old male presents with severe chest pain radiating to the left arm, sweating, and nausea. ECG shows ST-segment elevation.

**What is the priority nursing intervention?**

- A. Administer oral fluids
- B. Provide oxygen and prepare for reperfusion therapy
- C. Encourage ambulation
- D. Delay treatment until cardiac enzymes are confirmed



**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)

**Critical Care and Emergency Nursing  
Department**

**Lecture 11**



**MISR UNIVERSITY  
FOR SCIENCE & TECHNOLOGY**

**Respiratory Emergency**

**Prepared by : Dr. Ibrahim Atta Ashour**  
**lecturer of Critical Care Nursing**

# Objectives

**By the end of this lecture the student will be able to:**

- Indicate causes, intervention, and requirement of ventilation, diffusion and perfusion
- Indicate pathophysiology, etiology, clinical manifestation, assessment , diagnostic testes and intervention of pulmonary embolism (hidden threat)
- Indicate assessment, intervention and nursing diagnosis of pulmonary edema(flash flooding)

# Outlines

- Causes, intervention of ventilation, diffusion and perfusion
- Pathophysiology, etiology, clinical manifestation, assessment , diagnostic testes and intervention of:
  - pulmonary embolism(hidden threat)
  - pulmonary edema(flash flooding)



# Physiology of Respiration

## Define Respiration

- The exchange of gases between a living organism and the environment

## Define Ventilation

- Mechanical Process that moves air in and out of the lungs



# Causes of ventilation alteration

- Upper airway obstruction
- Lower airway obstruction
- Chest wall impairment
- Neurogenic dysfunction
- Foreign body, epiglottitis



# Ventilation Interventions

- Ensure that the upper and lower airways are open and unobstructed
- Provide assisted ventilations



# Diffusion

- Process of gas exchange between the air-filled alveoli and the pulmonary capillary bed
- Gas exchange is driven by simple diffusion in which gases move from areas of high concentration to low concentration until equal

# Causes of diffusion alteration

- Inadequate O<sub>2</sub>
- Alveolar pathology
- Fire, CO poisoning
- Lung disease, inhalation injury
- Pulmonary edema, drowning



# Diffusion Interventions

- Provide high flow O<sub>2</sub>
- Reduce inflammation of interstitial space



# Perfusion

The process of circulating blood through the pulmonary capillary bed

# Causes of perfusion alteration

- Inadequate blood volume
- Impaired circulation
- Capillary wall pathology
- Shock, anemia
- Pulmonary embolus



# Perfusion Interventions

- Ensure adequate circulating volume and Hgb levels
- Optimize left sided heart function



# Assessment Review of respiratory emergency

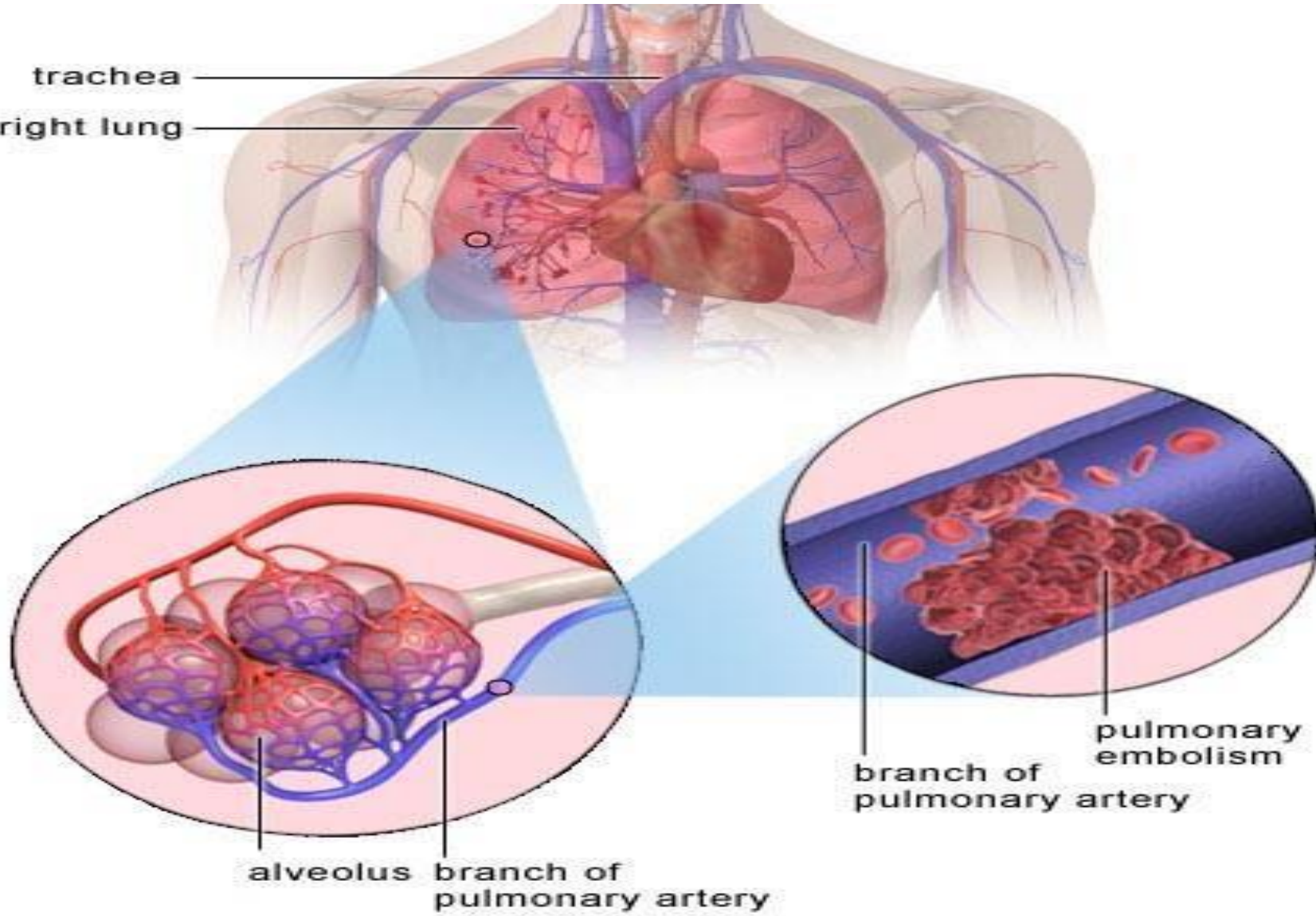
- Initial Assessment
- Recognition of signs of life threats
- Focused history
- Physical exam



# Pulmonary embolism: (A hidden threat)

- A pulmonary embolism (PE) is a blood clot, fat, amniotic fluid block a pulmonary vessels
- Most emboli (>90%) arise from deep venous thromboses (DVTs) in the femoral vein

trachea  
right lung



# Etiology & risk factors of pulmonary embolism

- Thrombosis of the leg veins ( DVT)
- Long bone fractures
- Prolonged immobilization
- Recent trauma or surgery
- Obesity
- Smoking
- A history of pulmonary or cardiac disease or coagulopathy increases the risk.

# Clinical manifestations of Pulmonary embolism

## **Respiratory distress:**

- Dyspnea
- Rales on auscultation.
- Tachycardia.
- Shallow respiration.
- Anxiety.
- Diaphoretic

## **Cough:**

- Blood-tinged sputum

## **Hypoxia:**

- Restless, headache, apprehension, pallor or cyanosis, behavioral changes, low- grade fever, engorged neck veins.

## **Dysrhythmias**

# Diagnostic studies of (PE)

- Computed tomography(CT) of the chest.
- Echocardiography
- pulmonary angiography.
- Chest X-ray
- Electrocardiograms (ECGs)
- Arterial blood gases



## How to assess (PE)

- It is difficult to diagnose clinically.
- **focused history** is crucial.

### Suspect PE if your patient has risk factors :

- Dyspnea and chest pain
- Respirations greater than 16
- Tachycardia
- Cough
- Anxiety



**If treatment is delayed or unsuccessful your patient may develop**

- Acute right ventricular ischemia that progresses to myocardial infarction,  
heart failure, cardiovascular collapse

# Treatment of PE

- Restoring blood flow through the pulmonary vasculature and preventing complications.
- Administer supplemental oxygen
- Monitor  $\text{paO}_2$ , cardiac rhythm, and vital signs.
- Antithrombotic medications
- Administer pain medication as ordered.



# **Pulmonary Edema** Flash flooding

Abnormal accumulation of fluid in the lung tissue, most commonly caused by left- ventricular failure.

# Causes of Pulmonary Edema

## Cardiogenic origin:

Increased pulmonary capillary filtrate pressure

Caused by:

- ✓ Acute left ventricular failure
- ✓ Acute myocardial infarction



# Causes of Pulmonary edema cont.....

## Non-cardiogenic:

Results from damage of the alveolar capillary membrane

### Caused by:

- ✓ Inhalation injury(aspiration, smoke, near- drowning)
- ✓ Direct trauma.
- ✓ Circulatory route (drug overdose, sepsis)
- ✓ Lymphatic obstruction (carcinoma)

# In cardiogenic pulmonary edema: clinical manifestations

## How to assess

Pulmonary edema causes dramatic respiratory symptoms:

- Tachypnea and tachycardia
- Labored breathing
- Intercostal and suprasternal retractions
- Bilateral inspiratory crackles
- Pink, frothy sputum
- Low  $paO_2$  level hypoxia
- Anxious , restless and decrease level of consciousness



# Investigations of pulmonary edema

- A chest X-ray
- CBC and electrolytes.
- ECG: tachycardia, dysrhythmias
- Arterial blood gases will show the extent of hypoxia

# Treatment of Pulmonary Edema

1. Elevate the head of your patient's bed and administer 100% oxygen by mask.
2. Monitor  $\text{paO}_2$  level, BP, and cardiac rate and rhythm.



**THANK YOU**

[www.must.edu.eg](http://www.must.edu.eg)